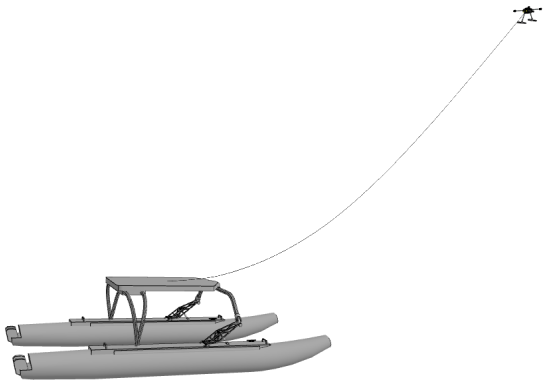
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Cooperative control of a tethered drone and surface vessel for sustained presence at sea

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Electrical and Computer Engineering

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Abstract

An unmanned surface vehicle and an unmanned aerial vehicle connected by a tether combine the endurance of the vessel with the view from above of the aircraft, which receives power and a reliable communication link through the tether. The tether, however, couples the vehicles, which must keep within its reach while each respects its own limits, under waves and wind. This thesis develops the cooperative control of such a system, formed by a WAM-V vessel and an x500 multirotor. Control-oriented models of both vehicles are derived and validated against a simulator, together with a closed-form model of the force of a slack tether that accounts for its weight. Each vehicle is driven by its own non-linear model predictive controller, and the two controllers exchange their predicted trajectories, so that each keeps the tether within its reach while anticipating the motion of the other. A disturbance observer compensates unmodelled forces, such as wind. The system is evaluated in a simulation at sea built on Gazebo, PX4 and MoorDyn. The aerial controller tracks a reference with less than a quarter of the error of the PX4 position controller, and both controllers plan, from simple or unreachable references, motions within the limits of the vehicles and of the tether. In missions of inspection of a tower, patrol with the pursuit of a target and recovery of the aircraft, the two vehicles work as one system, and the aircraft lands on the moving deck a few centimetres from the pad centre, under waves of $1.2\,$m and winds of up to $8\,$m/s.

Keywords: Tethered UAV, Unmanned Surface Vehicle, Cooperative Control, Non-linear Model Predictive Control, Maritime Robotics

Resumo

Um veículo de superfície não tripulado e um veículo aéreo não tripulado ligados por um cabo combinam a autonomia da embarcação com a visão aérea da aeronave, que recebe energia e comunicação fiável através do cabo. O cabo, no entanto, acopla os veículos, que têm de se manter ao seu alcance respeitando cada um os seus limites, sob ondas e vento. Esta tese desenvolve o controlo cooperativo deste sistema, formado por uma embarcação WAM-V e um multirotor x500. São derivados e validados num simulador modelos de ambos os veículos, bem como um modelo analítico da força de um cabo não esticado que considera o seu peso. Cada veículo é controlado pelo seu próprio controlador preditivo não linear, e os dois controladores trocam as trajetórias que preveem, mantendo cada um o cabo ao seu alcance e antecipando o movimento do outro. Um observador de perturbações compensa as forças não modeladas, como o vento. O sistema é avaliado numa simulação com Gazebo, PX4 e MoorDyn. O controlador aéreo segue uma referência com menos de um quarto do erro do controlador de posição do PX4, e ambos os controladores planeiam, a partir de referências simples ou inalcançáveis, movimentos dentro dos limites dos veículos e do cabo. Em missões de inspeção de uma torre, de patrulha com perseguição de um alvo e de recuperação da aeronave, os dois veículos funcionam como um só sistema, e a aeronave aterra no convés em movimento a poucos centímetros do centro da plataforma, sob ondas de $1{,}2$ m e ventos de até 8 m/s.

Palavras-chave: UAV com cabo, Veículo de Superfície Não Tripulado, Controlo Cooperativo, Controlo Preditivo Não Linear, Robótica Marítima

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Acronyms

CG	centre of gravity
DDS	Data Distribution Service
EMSA	European Maritime Safety Agency
HPIPM	High-Performance Interior Point Method
MPC	model predictive control
NMPC	non-linear model predictive control
NRMSE	normalised root-mean-square error
PID	proportional–integral–derivative
RMSE	root-mean-square error
ROS	Robot Operating System
RTI	real-time iteration
SITL	software-in-the-loop
SQP	sequential quadratic programming
UAV	unmanned aerial vehicle
uORB	micro Object Request Broker
USV	unmanned surface vehicle
VRX	Virtual RobotX
WAM-V	Wave Adaptive Modular Vessel

Chapter 1

Introduction

This chapter presents why a tethered unmanned surface vehicle (USV)–unmanned aerial vehicle (UAV) system is worth controlling as a single system, the problem addressed, how the proposed system is organised, the contributions of this work, and the structure of the thesis.

1.1 Motivation

Saving lives at sea, protecting marine ecosystems, and securing maritime infrastructure all rely on knowing what is happening, where, and when. In vast and changing maritime environments, this requires reliable and detailed awareness over long periods. The European Maritime Safety Agency (EMSA) recognises this need and highlights continuous monitoring and locally persistent operations as the basis of effective maritime surveillance and response [1].

Surface vessels provide a physical presence at sea, with long endurance and a large payload capacity [2], but they only see as far as the horizon, and operating them is expensive [3]. Aerial platforms, such as manned aircraft and UAVs, deploy quickly and sense from above [4]. Aircraft are costly, however, and UAVs suffer from short endurance and unreliable communication [5, 6].

Each of these assets compensates for what the other lacks. USVs keep a presence at sea for long periods at a much lower cost than crewed vessels, and UAVs extend their reach by sensing from the air, beyond the horizon of the vessel. Working as a team, a USV and a UAV can therefore keep a continuous and local awareness that neither achieves alone.

A tether between the two vehicles strengthens this cooperation, since it supplies the UAV with power and a reliable communication link from the USV [6]. With the energy and the communication on the USV, the two main limits of maritime UAVs, a short endurance and a weak long-range link, are overcome. Such systems already exist in industry. Exail and Elistair integrated a tethered UAV on the DriX USV for persistent maritime surveillance [7], and ACUA Ocean and Teledyne FLIR Defense demonstrated a tethered aerial-surface system for long-endurance maritime security [8]. Figure 1.1 illustrates the system considered in this work.

The tether turns the two vehicles into a single system. The distance between them must stay within

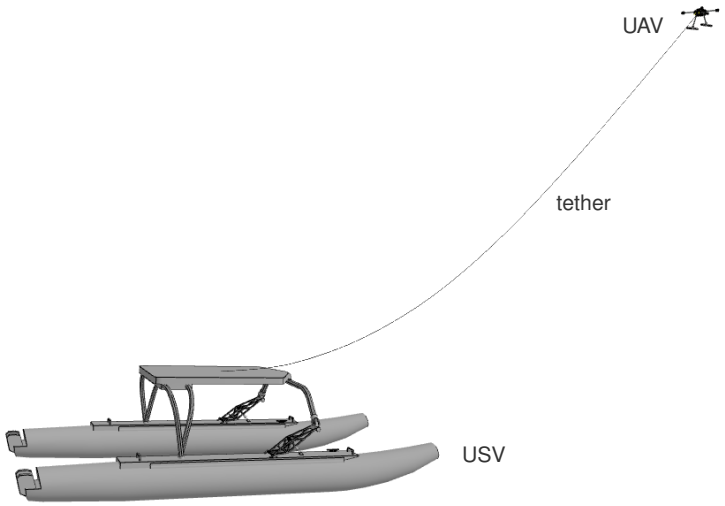


Figure 1.1: Tethered USV–UAV system.

the reach of the tether, so the motion of one restricts the motion of the other. The two vehicles also move on very different time scales, the USV slowly, dominated by its inertia and the drag of the water, and the UAV quickly. On top of this, wind and waves disturb both vehicles, and the UAV must take off from and land on a deck that moves with the waves.

Most existing tethered UAV systems focus on the supply of power and communication, while their operation remains manual or semi-autonomous. In research, the two vehicles are often controlled separately, the tether is simplified, or the surface vessel is assumed to be fixed. These assumptions simplify the design of the controllers, but few approaches coordinate both vehicles while accounting for the tether under realistic maritime conditions.

1.2 Problem Statement and Objectives

This work addresses the control of a USV, the Wave Adaptive Modular Vessel (WAM-V), and a UAV, the x500, connected by a tether, so that they carry out maritime missions autonomously as a single system. Each vehicle must follow its reference within its own limits, while the distance between them stays within the reach of the tether, also under waves and wind, and the UAV must take off from and land on the moving deck of the USV.

To address this problem, the work pursues four objectives. The first is to model both vehicles and the force of the tether on the UAV accurately enough to predict their motion over the horizon of a controller, and to validate these models. The second is to design a controller for each vehicle that respects its limits and anticipates the motion of the other vehicle, so that the two keep the tether within its reach together. The third is to develop the manoeuvres of both vehicles, including the takeoff from and the landing on the moving deck, together with missions that chain them into a complete operation, and the fourth is to evaluate the system in tracking and planning, in complete missions, and under waves and wind.

The system is studied in simulation, in an environment that reproduces the dynamics of both vehicles, the waves, the autopilot of the UAV and a tether that hangs and goes slack. The state of both vehicles is

taken from the simulator, so that the results measure the performance of the models and the controllers, and not that of the estimation of the state. The tests with the real vehicles are left for future work.

1.3 Proposed Approach

The two vehicles move on very different time scales, and a single controller for both would impose one horizon and one sampling time on them. Each vehicle is therefore controlled by its own non-linear model predictive control (NMPC), which plans its motion over a horizon of its own, within the limits of the vehicle. The two controllers exchange the trajectories they predict, so that each keeps the tether within its reach taking into account where the other vehicle is going, rather than where it is. Both rely on models derived from the physics of the vehicles and, for the UAV, on a model of the force that the hanging tether applies on it, while a disturbance observer compensates the forces that the models do not describe, such as the wind.

The controllers follow references, but do not decide where each vehicle should go. This is the role of the manoeuvres, such as a takeoff, a formation or a landing, which give the reference of each vehicle at every instant, and of the missions, which chain the manoeuvres of both vehicles into a complete operation. Since the controllers keep the vehicles within their limits whatever reference they are given, the manoeuvres can remain simple. Figure 1.2 shows how these parts fit together.

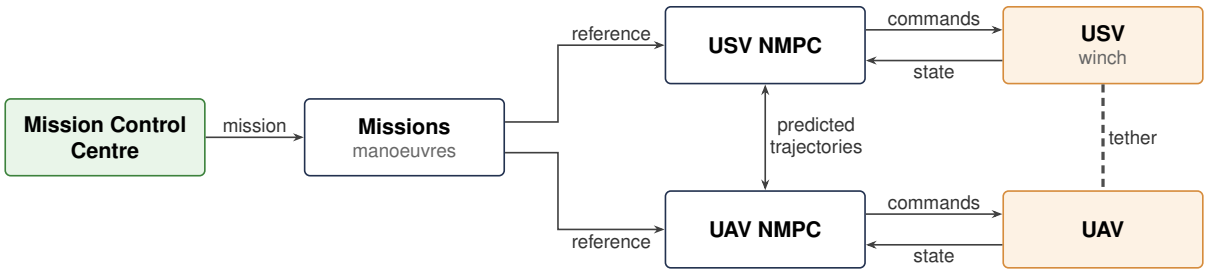


Figure 1.2: Overview of the proposed system.

1.4 Contributions

The main contributions of this work concern the models and the controllers of the tethered system. Control-oriented models of the WAM-V and of the x500 are derived, together with a closed-form model of the force of a slack tether, and all are validated against the simulator. These models support two cooperating NMPC controllers, one for each vehicle, which exchange their predicted trajectories to keep the tether within its reach while each vehicle respects its own limits. The same controllers plan the motion of both vehicles from simple or unreachable references.

The work further contributes a simulation environment of the tethered system at sea, with waves, the PX4 autopilot of the UAV, and a tether whose length is set by a winch. It also provides manoeuvres and missions for both vehicles, including the takeoff from and the landing on the moving deck, and the

support of the UAV by the USV around a structure. Finally, it presents an evaluation of the system in tracking and planning, in three missions, and under waves and wind.

1.5 Thesis Structure

The rest of this thesis is organised as follows.

- Chapter 2 reviews the modelling, the control, the cooperation and the planning of autonomous vehicles, with a focus on tethered and maritime systems.
- Chapter 3 derives the models of both vehicles and formulates their NMPC controllers.
- Chapter 4 presents the simulation environment, validates the models, and describes the implementation of the controllers, the manoeuvres and missions, and the software.
- Chapter 5 evaluates the system in tracking and planning, in the missions, and under disturbances.
- Chapter 6 summarises the achievements of this work and proposes future work.

Chapter 2

State of the Art and Related Work

This chapter is structured into four main sections, each representing a key element of cooperative system control. Each section presents the main techniques commonly used in control theory, together with subsections reviewing related work on topics closely related to this research.

The first section addresses *modelling and identification*, which provides the mathematical and experimental tools required to describe system dynamics and estimate relevant parameters, including tether physics. The second focuses on *control strategies*, ensuring stability, accurate tracking, and robustness to disturbances at the individual agent level, including predictive control and active winch architectures. The third section concerns *cooperative coordination*, which enables communication and shared decision-making so that multiple agents can operate as a unified system. Finally, *path and trajectory planning* deals with the generation of feasible and safe motion plans that connect mission objectives with low-level control actions.

2.1 Modelling and System Identification

Understanding how a system evolves is a prerequisite for any control design. Modelling provides this understanding by offering a mathematical description of the system dynamics, capturing the dominant behaviours while deliberately neglecting secondary effects that are not essential for analysis or control purposes [9]. The level of detail of such a model is inherently linked to the available physical insight, the amount and quality of experimental data, and the specific objectives for which the model is intended.

2.1.1 White-Box, Grey-Box, and Black-Box Modelling

Modelling methodologies are classified into three main groups, according to how much they rely on physical knowledge and on data. White-box models are built from first principles, expressing the dynamics through analytical formulations derived from Newtonian mechanics, hydrodynamics or aerodynamics. They provide clear insight into stability, controllability and energy transfer, and simplifying assumptions, such as linearisation or the neglect of higher-order effects, are commonly introduced to keep them tractable.

Grey-box models lie between purely analytical and purely data-driven approaches. Their structure follows from physical insight, while the parameters that are difficult to determine analytically, such as hydrodynamic or aerodynamic coefficients, are estimated from experimental data, which preserves a clear physical interpretation while improving accuracy. Black-box models, in contrast, rely entirely on measured input–output data and identify the mathematical relationships that best fit them, as in linear parametric models, state-space representations obtained from data, and learning-based methods.

2.1.2 Tether Dynamics Modelling

A flexible tether introduces highly non-linear forces and kinematic constraints on the vehicles, and three families of models describe it with increasing detail. Quasistatic catenary models describe the steady-state profile of a heavy tether suspended between two points, and resolve its sag and tension analytically by assuming static equilibrium at each instant. They are computationally efficient, but neglect dynamic effects such as transverse wave propagation and drag, which makes them unsuitable for rapid aerial manoeuvres.

Lumped-mass models discretise the tether into a chain of nodes connected by spring–damper segments. The segments model axial elasticity, damping and hydrodynamic drag, while the nodes carry the gravitational and buoyancy forces, so that this formulation, implemented in solvers such as MoorDyn [10], represents transient whipping, slackness and drag with high fidelity. Finite-segment and multibody models instead treat the tether as a chain of rigid cylinders connected by spherical joints, which captures bending and torsional moments accurately but requires solving high-index differential-algebraic equations, at a higher computational cost.

2.1.3 Optimisation-Based Estimation

System identification offers an alternative path by constructing models directly from measured input–output data when physical knowledge is incomplete. Optimisation-based estimation focuses on identifying model parameters offline by minimising the mismatch between measured and predicted system outputs. This is typically achieved through cost functions based on least-squares formulations, including weighted least-squares and prediction-error methods [11]. For non-linear or constrained problems, advanced optimisation schemes such as trust-region methods and constrained least-squares are often employed, allowing grey-box models to be calibrated while enforcing physical bounds and regularisation.

2.1.4 Related Work on Tethered Vehicle Modelling

In systems connected by a tether, the model of the overall system must account at least for the tether geometry to constrain the relative motion of the agents and avoid physically infeasible configurations that could lead to tether failure.

In [12], a buoy–UAV system is presented in which the tether is modelled explicitly as a geometric and dynamic constraint between the buoy and the aerial vehicle. The tether tension is computed from

the coupled dynamics and incorporated as an external force acting along the tether direction in the UAV model. This formulation relies on simplifying assumptions, namely that the tether is inextensible, massless, free of hydrodynamic effects, and remains taut during coupled operation.

An alternative modelling perspective is proposed in [13], where tether-induced forces are not included explicitly in the system dynamics. Instead, their effect is lumped into an external disturbance acting on the UAV. This disturbance is then handled at the control level through active disturbance rejection, avoiding the need for explicit tether modelling while still compensating for its influence in real time.

In [14], a cooperative UAV–USV system is modelled for the purpose of manipulating a floating object. Both vehicles are connected to the object through individual tethers, and the full system is described using simplified white-box models. The formulation assumes inextensible and massless tethers that remain permanently taut.

In [15], a tether is not included, and both the UAV and the USV are described using simplified white-box models. The USV is represented by a planar marine vehicle model commonly used for path-following applications, while the UAV is modelled independently using a reduced-order dynamic model.

2.2 Control Strategies for Autonomous Vehicles

Control theory provides the mathematical framework for designing systems that regulate their behaviour through feedback, ensuring stability, accuracy, and robustness in the presence of uncertainty and disturbances.

2.2.1 Classical and Modern Control

Classical control is based on the representation of systems through transfer functions in the Laplace or frequency domain, under the assumption of linear time-invariant dynamics. Within this framework, feedback laws are typically implemented through proportional–integral–derivative (PID) controllers and frequency-domain compensators designed to shape the closed-loop response.

Modern control theory generalises these principles to systems that are multivariable, time-varying, or subject to strong coupling between states and inputs. It is formulated in the *state-space* domain, where the internal dynamics of a system are described by sets of first-order differential equations. This representation enables systematic analysis of controllability, observability, and stability. Within this framework, optimal control theory defines feedback laws that minimise a performance index, such as the linear quadratic regulator or robust control methods such as H_∞ [16, 17].

2.2.2 Predictive and Optimisation-Based Control

Predictive and optimisation-based control methods determine control actions by solving an optimisation problem that predicts the system behaviour over a finite horizon. At each sampling instant, the controller minimises a cost function subject to system dynamics and operational constraints, typically

penalising deviations from reference trajectories and excessive control effort. Only the first control action is applied, and the process repeats at the next time step, forming the receding-horizon principle that defines model predictive control (MPC) [18].

Extensions to non-linear systems led to the development of NMPC, which replaces linear models with non-linear formulations and uses numerical optimisation techniques to compute control inputs in real time [19, 20]. To enable real-time embedded execution on high-rate systems (e.g. UAV flight control), modern applications rely on symbolic framework code generators (such as CasADi) that export native C solvers. These solvers utilise highly efficient quadratic programming kernels such as High-Performance Interior Point Method (HPIPM) and execution frameworks like the sequential quadratic programming (SQP)–real-time iteration (RTI) scheme, which performs a single optimisation iteration per time step to minimise solver latency.

2.2.3 Active Winch and Tension Control

In tethered UAV operations, the winch that manages the tether is a core element, and three designs are common. Passive tension tethers rely on spring-loaded spools or constant-torque slip rings. They are simple and require no electrical control, but cannot adapt to fast USV manoeuvres, which leads to large tension spikes or excessive slack.

Active spooling winches use motorised actuators regulated by microcontrollers, and adjust the length of the tether from the distance between the platforms or from force feedback. In tension-feedback designs, a load cell measures the force and a closed-loop proportional–derivative controller regulates the spooling speed to maintain a target tension. At sea, this scheme acts as active heave compensation, isolating the UAV in flight from the heave of the USV caused by the waves.

2.2.4 Related Work on Tethered Vehicle Control

In tethered UAV systems, the tether introduces additional forces and coupling effects that influence control design. In [12], this coupling is addressed through a non-linear backstepping controller derived directly from the white-box dynamics of the UAV–buoy system. The work in [13] adopts an alternative approach based on active disturbance rejection, where the tether influence is not explicitly modelled but treated as an external disturbance estimated in real time by a reduced-order observer.

In [14], cooperative manipulation with two tethered agents is controlled using a MPC formulation based on simplified white-box models of the UAV, the USV, and the floating object. A similar predictive strategy is presented in [21], where both the UAV and the USV are coordinated using NMPC, relying on known dynamic models.

In [15], the USV employs a line-of-sight guidance law combined with PID control for heading and speed regulation. The UAV motion is controlled using an NMPC formulation, which generates reference accelerations to track the desired relative trajectory. No joint or centralised control law is employed, and each vehicle is controlled using its own dedicated controller.

2.3 Cooperative Coordination and Systems

Cooperative coordination refers to the control and organisation of multiple agents working towards achieving goals through communication and joint decision-making.

2.3.1 Centralised vs. Distributed Coordination

Depending on how information is exchanged and how the control authority is structured, cooperative architectures are broadly classified as centralised or distributed. In centralised coordination, a single decision node aggregates global information and computes the commands of all agents. Since it has access to the full state and constraints of the system, it facilitates global optimisation and keeps the mission-level objectives consistent, but it scales poorly with the size of the team and is vulnerable to communication delays and single-point failures.

Distributed coordination removes the central planner, and each agent decides from locally available information and limited communication, exchanging its state or intent with the others to achieve group-level objectives. This provides scalability and robustness to communication losses and agent failures, at the cost of challenges related to synchronisation and network stability.

2.3.2 UAV–USV Cooperative Systems

Cooperative deployments of UAVs and USVs combine the speed and altitude of aerial sensing with the load-carrying capability and range of a hull, and three applications are common. The first is the autonomous landing of the UAV on the moving deck of the USV in waves. It requires a precise estimate of the relative state, from visual markers or real-time kinematic satellite positioning, and predictive control to synchronise the landing velocity of the UAV with the pitch and roll of the deck.

In the second application, the USV acts as a mobile recharging station. The UAVs fly reconnaissance sweeps and return to its deck to swap or recharge their batteries, which extends their persistence at sea. In the third, cooperative target tracking, path-following algorithms steer the USV to intercept a target, while the UAV orbits above it to provide continuous visual tracking.

2.3.3 Related Work on Maritime Cooperative Robotics

In [21], cooperation is based on a tracking strategy in which the UAV is controlled to follow and land on a moving USV. Only the UAV is actively controlled, while the USV moves independently and provides its state as a reference. The interaction is therefore unidirectional, with the UAV adapting its motion to the USV.

In contrast, [14] proposes a centralised MPC approach in which the UAV and the USV are controlled jointly. A single optimisation problem computes the control inputs for both vehicles, explicitly accounting for the coupling introduced by the tether. This approach focuses on cooperative object manipulation rather than on mission-level coordination between the vehicles.

In [15], coordination is achieved through a leader–follower structure. The USV acts as the leader and follows a predefined path, while the UAV adjusts its motion to maintain a desired relative configuration with respect to the USV. Coordination is therefore based on reference sharing and relative motion constraints, without joint optimisation or centralised decision-making.

2.4 Path and Trajectory Planning

Path and trajectory planning address the problem of generating feasible, collision-free, and dynamically consistent motions for autonomous systems operating in structured or unstructured environments, translating abstract goals into trackable trajectories.

2.4.1 Offline and Online Planning

Planning methods are broadly divided into offline and online approaches, depending on when the plan is computed and how information about the environment is handled. Offline planning is performed before execution and assumes full or near-complete knowledge of the environment. Classical approaches include graph-search methods, such as Dijkstra's algorithm and A*, sampling-based methods, such as rapidly-exploring random trees and probabilistic roadmaps, and trajectory optimisation. They generate globally optimal paths, but are computationally demanding and cannot adapt to changes in the environment.

Online planning generates and adapts the trajectories in real time, under dynamic or uncertain conditions. Reactive techniques, such as potential fields and dynamic window approaches, avoid obstacles quickly but can be trapped in local minima. Optimisation-based MPC frameworks instead integrate the dynamics of the vehicle into the planning loop, and generate dynamically consistent, collision-free paths online.

2.4.2 Related Work on Trajectory Planning

A centralised predictive planning approach is presented in [14], where a single MPC formulation jointly optimises the motion of both the UAV and the USV. In this case, planning is achieved online through joint trajectory optimisation that explicitly accounts for the coupling introduced by tethers.

Planning is addressed from a different perspective in [15], where cooperative behaviour is achieved through coordinated path following. A predefined path is assigned to the USV, while the UAV generates its motion online as part of the control process to maintain coordination. In this framework, planning is limited to the online generation of the UAV motion, and no cooperative path planning between the two vehicles is performed.

Overall, existing works typically adopt one of two approaches to planning in UAV–USV systems. Either a predefined path is assigned to one vehicle while the other generates its motion online to follow or support it, or both vehicles generate their motion online as part of a predictive control framework.

Chapter 3

Proposed Methodology

This chapter presents the methodology developed for the cooperative control of the tethered USV–UAV system. Section 3.1 derives the models of both vehicles and of the force that the tether applies on the UAV, and Section 3.2 formulates the two cooperating NMPC controllers that use these models to drive each vehicle within its limits while keeping the tether within its reach.

3.1 System Modelling

Real-time control needs models that are simple enough to compute quickly, yet detailed enough to capture how each vehicle actually moves and interacts with the other. The kinematics, dynamics, and physical coupling between the two vehicles let the controller predict how the system evolves and compute feasible, optimal trajectories.

The reference frames and notation are fixed first, followed by the kinematic and dynamic models of the USV and the UAV.

3.1.1 Reference Coordinate Frames and Notation

Deriving the equations of motion requires specifying the frames against which positions, orientations, and velocities are measured. Since the USV and the UAV move both with respect to the world and with respect to each other, a common frame of reference is required to describe their motion. Figure 3.1 illustrates these frames.

Inertial Reference Frame $\{\mathcal{I}\}$ This frame establishes the global coordinate system following the East-North-Up convention, aligned with the standard Robot Operating System (ROS) convention. Its origin $\mathcal{O}_I$ is fixed at an arbitrary point on the water surface within the operating area, following the usual flat-Earth convention. The axis $\mathbf{x}^{\mathcal{I}}$ points East, $\mathbf{y}^{\mathcal{I}}$ points North, and $\mathbf{z}^{\mathcal{I}}$ points upward, opposite to gravity.

USV Body-Fixed Reference Frame $\{\mathcal{B}_v\}$ This frame is rigidly attached to the USV, with origin $\mathcal{O}_v$ at its centre of gravity (CG). This choice decouples the linear and rotational momentum equations. The

forward-pointing axis $\mathbf{x}^{\mathcal{B}_v}$ defines surge, $\mathbf{y}^{\mathcal{B}_v}$ points to port (sway), and $\mathbf{z}^{\mathcal{B}_v}$ points upward (heave).

UAV Body-Fixed Reference Frame $\{\mathcal{B}_a\}$ Similarly, this frame attaches to the UAV, with origin $\mathcal{O}_a$ at its CG, decoupling the translational and rotational dynamics in the same way. The longitudinal axis $\mathbf{x}^{\mathcal{B}_a}$ points forward, the lateral axis $\mathbf{y}^{\mathcal{B}_a}$ points to the left, and the vertical axis $\mathbf{z}^{\mathcal{B}_a}$ points upward, aligning with the collective thrust direction.

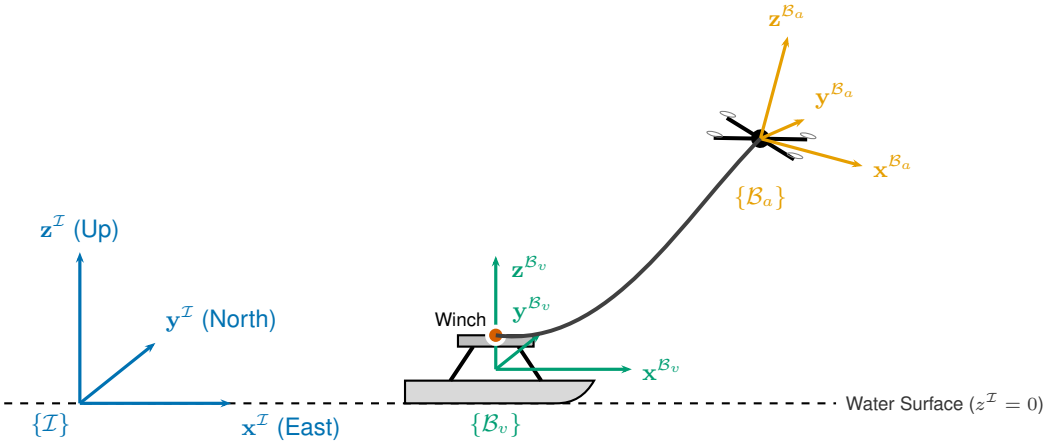


Figure 3.1: Reference coordinate frames $\{\mathcal{I}\}$, $\{\mathcal{B}_v\}$, and $\{\mathcal{B}_a\}$.

3.1.2 WAM-V Model

The mathematical model of the WAM-V catamaran, covering its horizontal manoeuvring dynamics and thruster allocation, forms a control-oriented state-space model suitable for trajectory tracking and predictive control.

Although a rigid body moving freely in the marine environment inherently has six degrees of freedom, the WAM-V is a USV that moves on the surface, which allows the motion to be simplified to the horizontal plane. Under this assumption, the model is reduced to three degrees of freedom.

Predicting the USV's motion, as needed for trajectory tracking and predictive control, requires knowing how its state changes over time for a given input. This relationship is captured by the non-linear state-space model

$$\dot{\mathbf{x}}_v = f(\mathbf{x}_v, \mathbf{u}_v), \tag{3.1}$$

The state vector $\mathbf{x}_v \in \mathbb{R}^6$ is given by

$$\mathbf{x}_v = [x, y, \psi, u, v, r]^\top, \tag{3.2}$$

where:

- x, y – position of the origin $\mathcal{O}_v$ of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{I}\}$;
- ψ – heading of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$;
- u, v – surge and sway linear velocities of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{B}_v\}$;
- r – angular velocity of $\{\mathcal{B}_v\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{B}_v\}$;

The control input vector $\mathbf{u}_v \in \mathbb{R}^3$ is given by

$$\mathbf{u}_v = [X, Y, N]^\top, \tag{3.3}$$

where:

- X – surge force generated by the propulsion system, applied to $\{\mathcal{B}_v\}$, expressed in $\{\mathcal{B}_v\}$;
- Y – sway force generated by the propulsion system, applied to $\{\mathcal{B}_v\}$, expressed in $\{\mathcal{B}_v\}$;
- N – yaw moment generated by the propulsion system, applied to $\{\mathcal{B}_v\}$, expressed in $\{\mathcal{B}_v\}$;

The vector field $f(\cdot)$ splits into a kinematic block and a dynamic block, derived in turn in the following subsections.

Kinematic Model

The kinematics describe how the USV's inertial position and heading evolve as a function of its body-fixed velocities, forming the first half of (3.1).

Since $\{\mathcal{B}_v\}$ is rotated relative to $\{\mathcal{I}\}$ by the heading ψ, the body-fixed velocities u, v give the inertial velocity of $\mathcal{O}_v$ through the planar rotation matrix $\mathbf{R}(\psi)$, while the heading rate is the angular velocity r,

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \mathbf{R}(\psi) \begin{bmatrix} u \\ v \end{bmatrix}, \qquad \mathbf{R}(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}, \qquad \dot{\psi} = r. \tag{3.4}$$

Dynamic Model

The dynamics describe how the USV's body-fixed velocities evolve as a function of the applied forces and moments, forming the second half of (3.1).

The body-fixed basis vectors, denoted $\mathbf{x}^{\mathcal{B}}, \mathbf{y}^{\mathcal{B}}$ for brevity, rotate with the vehicle at the rate r,

$$\dot{\mathbf{x}}^{\mathcal{B}} = r\,\mathbf{y}^{\mathcal{B}}, \qquad \dot{\mathbf{y}}^{\mathcal{B}} = -r\,\mathbf{x}^{\mathcal{B}}. \tag{3.5}$$

The inertial velocity $\dot{\mathbf{p}}$ of $\mathcal{O}_v$, expressed in the body-fixed frame, is $\dot{\mathbf{p}} = u\,\mathbf{x}^{\mathcal{B}} + v\,\mathbf{y}^{\mathcal{B}}$. Differentiating once more and substituting (3.5) yields the inertial acceleration

$$\ddot{\mathbf{p}} = (\dot{u} - vr)\,\mathbf{x}^{\mathcal{B}} + (\dot{v} + ur)\,\mathbf{y}^{\mathcal{B}}, \tag{3.6}$$

where the cross terms $-vr$ and ur are Coriolis/centripetal terms arising because $\{\mathcal{B}_v\}$ is a rotating, non-inertial frame.

Let F_x, F_y denote the resultant surge and sway forces applied at $\mathcal{O}_v$, so that $\mathbf{F} = F_x\,\mathbf{x}^{\mathcal{B}} + F_y\,\mathbf{y}^{\mathcal{B}}$. By Newton's second law, $\mathbf{F} = m\,\ddot{\mathbf{p}}$, where m is the mass of the USV, and comparing with (3.6),

$$F_x = m(\dot{u} - vr), \qquad F_y = m(\dot{v} + ur). \tag{3.7}$$

Since the USV's motion is planar, Euler's rotational equation about $\mathbf{z}^{\mathcal{B}}$ takes the form

$$M_z = I_z\,\dot{r}, \tag{3.8}$$

where M_z is the resultant yaw moment about $\mathcal{O}_v$ and I_z the yaw moment of inertia. Accelerating the hull also accelerates the surrounding water, whose inertia acts on the USV as an added mass. For a hull symmetric about its centreline, it is described by the diagonal coefficients $X_{\dot{u}}, Y_{\dot{v}}, N_{\dot{r}} \leq 0$ [22], which add inertia and Coriolis terms to (3.7) and (3.8),

$$(m-X_{\dot{u}})\dot{u}-(m-Y_{\dot{v}})vr = F_x, \qquad (m-Y_{\dot{v}})\dot{v}+(m-X_{\dot{u}})ur = F_y, \qquad (I_z-N_{\dot{r}})\dot{r}+(X_{\dot{u}}-Y_{\dot{v}})uv = M_z. \tag{3.9}$$

The resultant force and moment F_x, F_y, M_z combine two contributions: the thrust generated by the propulsion system, X, Y, N, and the hydrodynamic damping opposing the USV's motion. This damping follows the standard linear-plus-quadratic model for USVs [22]. The quadratic surge term is further split into forward and backward coefficients, to capture the asymmetric drag between the two directions:

$$F_x = X - \big(X_u + X^{\pm}_{uu}|u|\big)u, \qquad X^{\pm}_{uu} = \begin{cases} X^{+}_{uu}, & u \geq 0 \\ X^{-}_{uu}, & u < 0 \end{cases} \tag{3.10}$$

$$F_y = Y - \big(Y_v + Y_{vv}|v|\big)v, \qquad M_z = N - \big(N_r + N_{rr}|r|\big)r. \tag{3.11}$$

Substituting (3.10)–(3.11) into (3.9), and solving for the accelerations, gives the dynamic block of (3.1):

$$\dot{u} = \frac{1}{m - X_{\dot{u}}}\Big[X - \big(X_u + X^{\pm}_{uu}|u|\big)u + (m - Y_{\dot{v}})vr\Big], \tag{3.12}$$

$$\dot{v} = \frac{1}{m - Y_{\dot{v}}}\Big[Y - \big(Y_v + Y_{vv}|v|\big)v - (m - X_{\dot{u}})ur\Big], \tag{3.13}$$

$$\dot{r} = \frac{1}{I_z - N_{\dot{r}}}\Big[N - \big(N_r + N_{rr}|r|\big)r - (X_{\dot{u}} - Y_{\dot{v}})uv\Big]. \tag{3.14}$$

Thruster Allocation

The thruster allocation maps the physical actuator commands, the left and right thrust magnitudes T_L, T_R and azimuth angles α_L, α_R, to the resultant force and moment X, Y, N required by the dynamics of Section 3.1.2, according to the geometry shown in Figure 3.2.

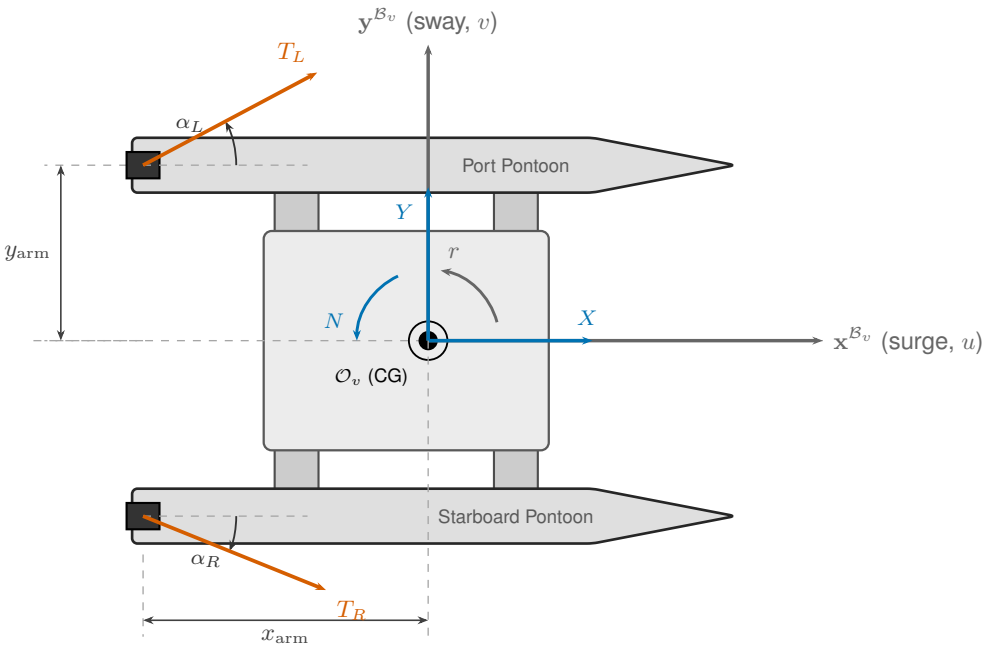


Figure 3.2: Planar USV schematic and thruster allocation geometry.

Each azimuth angle is measured relative to $\mathbf{x}^{\mathcal{B}}$, so that each thruster produces a planar force

$$\mathbf{f}_L = T_L \cos\alpha_L \, \mathbf{x}^{\mathcal{B}} + T_L \sin\alpha_L \, \mathbf{y}^{\mathcal{B}}, \qquad \mathbf{f}_R = T_R \cos\alpha_R \, \mathbf{x}^{\mathcal{B}} + T_R \sin\alpha_R \, \mathbf{y}^{\mathcal{B}}. \tag{3.15}$$

Summing the components of both thrusters along $\mathbf{x}^{\mathcal{B}}$ and $\mathbf{y}^{\mathcal{B}}$ gives the resultant force applied at $\mathcal{O}_v$:

$$X = T_L \cos\alpha_L + T_R \cos\alpha_R, \qquad Y = T_L \sin\alpha_L + T_R \sin\alpha_R. \tag{3.16}$$

Both thrusters are mounted behind $\mathcal{O}_v$, at the same longitudinal distance x_{arm}, and symmetrically on either side of the centreline, at a lateral distance y_{arm}. Expressed in $\{\mathcal{B}_v\}$, the left (port) thruster is located at $(-x_{\text{arm}}, y_{\text{arm}})$ and the right (starboard) thruster at $(-x_{\text{arm}}, -y_{\text{arm}})$.

The yaw moment produced by a planar force (f_x, f_y) applied at (r_x, r_y) is given by the two-dimensional cross product $r_x f_y - r_y f_x$. Applying this to each thruster and summing gives the resultant yaw moment about $\mathcal{O}_v$:

$$N = -x_{\text{arm}}\big(T_L \sin\alpha_L + T_R \sin\alpha_R\big) - y_{\text{arm}}\big(T_L \cos\alpha_L - T_R \cos\alpha_R\big). \tag{3.17}$$

Equations (3.16) and (3.17) complete the mapping from the four physical actuator commands T_L, T_R, α_L, α_R to the control input $\mathbf{u}_v = [X, Y, N]^\top$ used by the dynamic model.

3.1.3 x500 UAV Model

The mathematical model of the x500 UAV, covering its translational dynamics in three-dimensional space, forms a control-oriented state-space model suitable for trajectory tracking and predictive control. The vehicle's attitude is stabilised by an inner-loop attitude controller, to which the attitude used in the

translational dynamics is commanded.

Predicting the vehicle's motion, as needed for trajectory tracking and predictive control, requires knowing how its state changes over time for a given input. This relationship is captured by the non-linear state-space model

$$\dot{\mathbf{x}}_a = f(\mathbf{x}_a, \mathbf{u}_a), \tag{3.18}$$

The state vector $\mathbf{x}_a \in \mathbb{R}^6$ is given by

$$\mathbf{x}_a = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top, \tag{3.19}$$

where:

- x, y, z – position of the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{I}\}$;
- $\dot{x}, \dot{y}, \dot{z}$ – linear velocity of $\mathcal{O}_a$ relative to $\{\mathcal{I}\}$, expressed in $\{\mathcal{I}\}$;

The control input vector $\mathbf{u}_a \in \mathbb{R}^4$ is given by

$$\mathbf{u}_a = [T, \phi, \theta, \psi]^\top, \tag{3.20}$$

where:

- T – total thrust force, applied to $\{\mathcal{B}_a\}$, expressed in $\{\mathcal{B}_a\}$;
- ϕ, θ, ψ – roll, pitch, and yaw angles of $\{\mathcal{B}_a\}$ relative to $\{\mathcal{I}\}$;

The vector field $f(\cdot)$ splits into a kinematic block, relating the inertial position and velocity, and a dynamic block, mapping the thrust, attitude, aerodynamic drag, and tether force to the inertial translational acceleration.

Kinematic Model

The kinematics describe how the vehicle's inertial position evolves as a function of its inertial velocity, forming the first half of (3.18).

Let $\mathbf{p} = x\,\mathbf{x}^{\mathcal{I}} + y\,\mathbf{y}^{\mathcal{I}} + z\,\mathbf{z}^{\mathcal{I}}$ denote the inertial position of the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$. Since $\mathbf{x}^{\mathcal{I}}, \mathbf{y}^{\mathcal{I}}, \mathbf{z}^{\mathcal{I}}$ are constant, differentiating yields

$$\dot{\mathbf{p}} = \dot{x}\,\mathbf{x}^{\mathcal{I}} + \dot{y}\,\mathbf{y}^{\mathcal{I}} + \dot{z}\,\mathbf{z}^{\mathcal{I}}, \tag{3.21}$$

i.e. $\dot{x}, \dot{y}, \dot{z}$ are, by definition, the components of $\dot{\mathbf{p}}$ in the inertial frame, which are precisely the linear velocity components in $\mathbf{x}_a$. The kinematic block of (3.18) is therefore given directly by (3.21), without any change of basis. This is in contrast to a USV such as the WAM-V (Section 3.1.2), where the linear velocities are instead expressed in $\{\mathcal{B}_v\}$, requiring an additional change of basis via $\mathbf{R}(\psi)$ to relate them to $\dot{\mathbf{p}}$. Here, this rotation is required instead to express the thrust force, applied along $\mathbf{z}^{\mathcal{B}_a}$, in $\{\mathcal{I}\}$. It is therefore deferred to the dynamic block, derived next.

Dynamic Model

The dynamics describe how the vehicle's inertial velocity evolves as a function of the applied thrust, attitude, aerodynamic drag, and tether force, forming the second half of (3.18).

The thrust generated by the propulsion system acts along the vertical axis of $\{\mathcal{B}_a\}$, so that the resultant thrust force is

$$\mathbf{F}_{\text{thr}} = T\,\mathbf{z}^{\mathcal{B}}, \tag{3.22}$$

where $\mathbf{z}^{\mathcal{B}}$ denotes the vertical basis vector of $\{\mathcal{B}_a\}$, i.e. $\mathbf{z}^{\mathcal{B}_a}$.

Since (3.21) relates the inertial position and velocity directly, without an intermediate change of basis, Newton's second law is more conveniently applied directly in $\{\mathcal{I}\}$, requiring $\mathbf{F}_{\text{thr}}$ to instead be expressed in the inertial frame.

Since $\{\mathcal{B}_a\}$ is rotated relative to $\{\mathcal{I}\}$ by the roll, pitch, and yaw angles ϕ, θ, ψ, the inertial unit vectors are related to the body-fixed ones by the rotation matrix $\mathbf{R}(\phi, \theta, \psi) \in SO(3)$,

$$\begin{bmatrix} \mathbf{x}^{\mathcal{I}} \\ \mathbf{y}^{\mathcal{I}} \\ \mathbf{z}^{\mathcal{I}} \end{bmatrix} = \mathbf{R}(\phi, \theta, \psi) \begin{bmatrix} \mathbf{x}^{\mathcal{B}} \\ \mathbf{y}^{\mathcal{B}} \\ \mathbf{z}^{\mathcal{B}} \end{bmatrix}, \tag{3.23}$$

following the ZYX (yaw-pitch-roll) Euler angle convention,

$$\mathbf{R}(\phi, \theta, \psi) = \begin{bmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{bmatrix}, \tag{3.24}$$

where $s_\bullet = \sin(\bullet)$ and $c_\bullet = \cos(\bullet)$.

Projecting $\mathbf{F}_{\text{thr}}$ onto $\{\mathcal{I}\}$ then gives

$$\mathbf{F}_{\text{thr}}^{\mathcal{I}} = \mathbf{R}(\phi, \theta, \psi) \begin{bmatrix} 0 \\ 0 \\ T \end{bmatrix}. \tag{3.25}$$

Figure 3.3 illustrates this projection in the longitudinal plane, showing how the body tilt θ splits the thrust force T into vertical and horizontal inertial components.

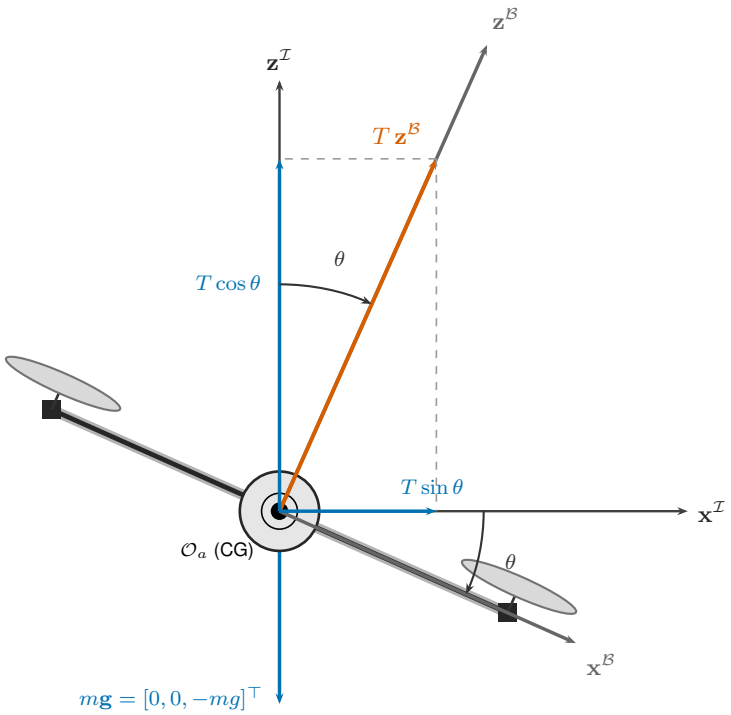


Figure 3.3: UAV thrust vectoring in the longitudinal plane.

Gravity, unlike the thrust, is already naturally expressed in $\{\mathcal{I}\}$:

$$\mathbf{F}^{\mathcal{I}}_{\text{grav}} = \begin{bmatrix} 0 \\ 0 \\ -mg \end{bmatrix}. \tag{3.26}$$

where m is the mass of the UAV and g the gravitational acceleration.

During translational flight, the vehicle is also subject to aerodynamic drag opposing its planar motion, arising primarily from the relative airflow across the spinning rotor disks (rotor drag). This force is naturally expressed in $\{\mathcal{B}_a\}$ as a linear damping $\mathbf{D} = \text{diag}(d_r, d_r, 0)$ of the velocity along $\mathbf{x}^{\mathcal{B}}$ and $\mathbf{y}^{\mathcal{B}}$, which in $\{\mathcal{I}\}$ gives

$$\mathbf{F}^{\mathcal{I}}_{\text{drag}} = -\mathbf{R}(\phi, \theta, \psi)\,\mathbf{D}\,\mathbf{R}(\phi, \theta, \psi)^\top \dot{\mathbf{p}}. \tag{3.27}$$

For small roll and pitch angles, the horizontal axes of the body remain approximately horizontal in $\{\mathcal{I}\}$, so the drag simplifies to

$$\mathbf{F}^{\mathcal{I}}_{\text{drag}} \approx \begin{bmatrix} -d_r\,\dot{x} \\ -d_r\,\dot{y} \\ 0 \end{bmatrix}, \tag{3.28}$$

where the small coupling between horizontal and vertical velocities caused by the tilt of the vehicle is neglected.

In addition to thrust, gravity, and aerodynamic drag, the vehicle is subject to a force transmitted through a tether attached at the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$, denoted $\mathbf{F}^{\mathcal{I}}_{\text{tether}}$ and derived separately in Section 3.1.3.

Since $\{\mathcal{I}\}$ is non-rotating, Newton's second law applies directly to the inertial acceleration, without the transport terms that arise when applying it in a rotating frame, such as $\{\mathcal{B}_v\}$ in the WAM-V model (Section 3.1.2):

$$m\,\ddot{\mathbf{p}} = \mathbf{F}^{\mathcal{I}}_{\text{thr}} + \mathbf{F}^{\mathcal{I}}_{\text{grav}} + \mathbf{F}^{\mathcal{I}}_{\text{drag}} + \mathbf{F}^{\mathcal{I}}_{\text{tether}}. \tag{3.29}$$

Substituting (3.25), (3.26), and (3.28) into (3.29), and solving for the acceleration, gives the dynamic block of (3.18):

$$\begin{bmatrix}\ddot{x}\\ \ddot{y}\\ \ddot{z}\end{bmatrix} = \frac{T}{m}\,\mathbf{R}(\phi,\theta,\psi)\begin{bmatrix}0\\0\\1\end{bmatrix} + \begin{bmatrix}0\\0\\-g\end{bmatrix} - \frac{d_r}{m}\begin{bmatrix}\dot{x}\\ \dot{y}\\ 0\end{bmatrix} + \frac{1}{m}\mathbf{F}^{\mathcal{I}}_{\text{tether}}. \tag{3.30}$$

Tether Force Model

The tether connects the origin $\mathcal{O}_a$ of $\{\mathcal{B}_a\}$ to the winch on the deck of the WAM-V. The winch is vertically aligned with $\mathcal{O}_v$, at a fixed height z_t, so that its position in $\{\mathcal{I}\}$ is $\mathbf{p}_w = [x_v, y_v, z_t]^\top$, where x_v, y_v denote the horizontal position of the USV. It is treated here only as a known point, without modelling the (negligible) reaction force exerted on the USV.

The relevant geometric quantity is the chord, the straight-line vector between the UAV and the winch,

$$\mathbf{r} = \mathbf{p}_w - \mathbf{p}, \qquad d = \|\mathbf{r}\|, \qquad \hat{\mathbf{d}}_0 = \frac{\mathbf{r}}{d}, \tag{3.31}$$

where d is the length of the chord, i.e. the straight-line distance between the UAV and the USV, and $\hat{\mathbf{d}}_0$ is its direction, which the tether would have if taut.

The tether lies in the vertical plane that contains the chord. Within this plane, the horizontal direction towards the winch is the unit vector $\hat{\mathbf{e}}_h$ along the horizontal component of the chord, and the depression angle α of the chord below it follows from

$$\hat{\mathbf{d}}_0 = \cos\alpha\,\hat{\mathbf{e}}_h - \sin\alpha\,\mathbf{z}^{\mathcal{I}}. \tag{3.32}$$

Denoting by $L \geq d$ the tether length at a given instant, its slack is conveniently quantified by the dimensionless ratio

$$\varepsilon = \frac{L-d}{L}, \tag{3.33}$$

with $\varepsilon = 0$ corresponding to a taut tether, and ε increasing as more slack is introduced.

Under its own weight, a tether of linear mass density μ suspended between its two ends takes the shape of a catenary, contained in the vertical plane defined by the two endpoints. Due to the geometric self-similarity of the catenary curve at a given slack ratio ε and chord inclination, the tension magnitude T_{tether} at $\mathcal{O}_a$ scales proportionally with the length d of the chord (or with the unstretched length L), while the tension direction is governed by the angular deflection γ by which the tension vector at $\mathcal{O}_a$ deviates from $\hat{\mathbf{d}}_0$ towards the vertical.

The exact relations governing the deflection angle $\gamma_{\text{exact}}(\varepsilon,\alpha)$ and the normalised tension coefficient $c_d^{\text{exact}}(\varepsilon,\alpha) = T_{\text{tether}}/d$ do not admit a closed-form analytical solution, requiring numerical root-finding

algorithms at every control cycle.

The tether control scheme considered in this work maintains ε close to a constant operating point ε_0. Under this operational regime, the tension magnitude is approximated as scaling linearly with the length of the chord d:

$$T_{\text{tether}}(d) = c_d(\varepsilon_0)\, d, \tag{3.34}$$

where the nominal coefficient at ε_0 is obtained by approximating the tension at the UAV by the weight of the tether, $T_{\text{tether}}(L) \approx \mu\, g\, L$, with $L = d/(1-\varepsilon_0)$:

$$c_d(\varepsilon_0) = \frac{\mu\, g}{1-\varepsilon_0}. \tag{3.35}$$

This neglects the dependence of the tension on the inclination of the chord, which for the operating slack changes it by about 12% at most. The deflection angle also depends on the inclination of the chord, given by the depression angle α in (3.32). Only the component of the weight of the tether normal to the chord, $\mu\, g\, L \cos\alpha$, bends it away from the straight line, so that, for the small slack of the operating point, the deflection approximately scales with $\cos\alpha$. It is maximal for a horizontal chord and vanishes for a vertical one, where the tether hangs straight. The deflection angle is therefore given by its nominal value at ε_0 for a horizontal chord, $\gamma_0 = \gamma_{\text{exact}}(\varepsilon_0, 0)$, scaled by $\cos\alpha$:

$$\gamma(\alpha) = \gamma_0 \cos\alpha, \tag{3.36}$$

which replaces the transcendental relation with a closed-form expression. Since ε_0 is fixed, γ_0 is a constant, obtained once from the exact relation in Section 4.2.3.

Figure 3.4 illustrates this construction within the catenary plane, showing the local horizontal $\hat{\mathbf{e}}_h$, the depression angle α, and the tension direction $\hat{\mathbf{d}}$, tilted further from $\hat{\mathbf{d}}_0$ by the angle γ.

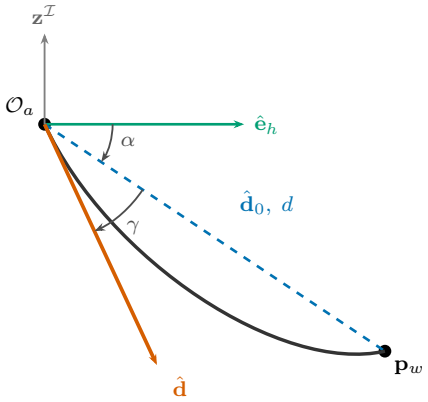


Figure 3.4: Tether catenary plane.

The tension direction at $\mathcal{O}_a$ then follows by tilting $\hat{\mathbf{d}}_0$ further towards the vertical by the angle $\gamma(\alpha)$, within the same plane,

$$\hat{\mathbf{d}} = \cos(\alpha+\gamma)\, \hat{\mathbf{e}}_h - \sin(\alpha+\gamma)\, \mathbf{z}^{\mathcal{I}}, \tag{3.37}$$

and the resulting tether force, applied at $\mathcal{O}_a$ and expressed in $\{\mathcal{I}\}$, is

$$\mathbf{F}^{\mathcal{I}}_{\text{tether}} = T_{\text{tether}}(d)\,\hat{\mathbf{d}} = c_d(\varepsilon_0)\,d\,\hat{\mathbf{d}}, \tag{3.38}$$

with d evaluated at each instant from (3.31). Equation (3.38) is the term $\mathbf{F}^{\mathcal{I}}_{\text{tether}}$ introduced in (3.29). For a vertical chord, $\hat{\mathbf{e}}_h$ is undefined, but then $\alpha = \pi/2$ and $\gamma = 0$, so that $\cos(\alpha+\gamma) = 0$ and the direction is simply $-\mathbf{z}^{\mathcal{I}}$.

3.2 System Control

Having established, in Section 3.1, the kinematic and dynamic models governing the USV and UAV, the natural next step is to determine how these models can be exploited to compute the control inputs that drive each vehicle along a desired trajectory. Because the two vehicles are physically coupled through the tether, their motion cannot be planned independently without accounting for this interaction, motivating a cooperative control strategy.

The overall control approach is therefore established first, followed by the design of the controllers for each vehicle.

3.2.1 Cooperative Control Design

Before formulating either controller, the strategies available are weighed against the system's dynamics and its objectives, so that the method chosen suits both vehicles.

Control Method

Keeping the tether within safe operating conditions depends on how the distance between the two vehicles evolves over time. A controller that only reacts to the current state can correct a violation only after it happens. Anticipating where the system is headed allows corrective action before reaching the safety limits.

A predictive controller does this by repeatedly solving, at each control instant, an optimisation problem over a finite prediction horizon. Because the optimisation runs over the predicted trajectory rather than the current state alone, requirements such as keeping the tether within safe conditions can be stated as explicit constraints in the problem itself, enforcing them explicitly over the prediction horizon.

The predictions rely on the model of the system dynamics, so their accuracy is tied to how well that model represents the real behaviour. The USV and UAV models derived in Section 3.1 include non-linear terms. The predictive controller uses these non-linear models directly, making NMPC the method used for both vehicles.

Cooperative Architecture

The USV and the UAV operate on different timescales. The USV moves in the horizontal plane with slow motion, dominated by its inertia and by hydrodynamic forces. The UAV's dynamics evolve much faster and demand a finer sampling time. A single NMPC built around the combined state of both vehicles would force one prediction horizon and one sampling time onto both, compromising one vehicle's timescale to accommodate the other.

Combining both vehicles into one NMPC would also enlarge the state, input, and constraint sets a single solver must handle, making a problem that already runs under real-time constraints harder to solve within the available computation time. Keeping the two problems separate matches each optimisation's size to its own vehicle and keeps the computation more manageable in real time.

Each vehicle is therefore controlled by its own NMPC. This separation isolates the two controllers from each other's failures: if one solver fails to find a feasible solution, the other vehicle's NMPC continues operating on its own well-posed problem.

The tether's safety limits depend on the relative position between the USV and the UAV, so each vehicle's NMPC accounts for the other vehicle's position. At every control instant, each controller receives the other vehicle's most recently computed predicted trajectory, and uses it to compute the predicted inter-vehicle distance at the nodes of its own horizon covered by that trajectory. Because the two controllers operate at different sampling intervals, the received predicted trajectory is temporally interpolated and resampled at the receiver's own prediction nodes.

Without this exchange, a vehicle would have to treat the other as stationary or extrapolate its motion from the last known state. Either assumption misrepresents the tether coupling and can drive the system into a constraint violation.

Because the UAV operates with a finer sampling time and a possibly shorter prediction horizon than the USV, its predicted trajectory may cover only part of the USV's horizon. The USV therefore enforces the tether constraint only at the nodes covered by the received trajectory, and leaves it inactive at the remaining ones, where no information about the UAV is available. This avoids constraining the USV with an assumed UAV position that has no basis in the prediction. Since the problem is solved again at every instant, the covered window slides forward with time, and any potential violation is accounted for as soon as it enters it. In the opposite direction no such restriction is needed, since the USV's horizon fully covers that of the UAV.

Figure 3.5 shows this architecture, with two independent NMPC loops, each driving its own vehicle, exchanging predicted horizons to enforce the tether constraint.

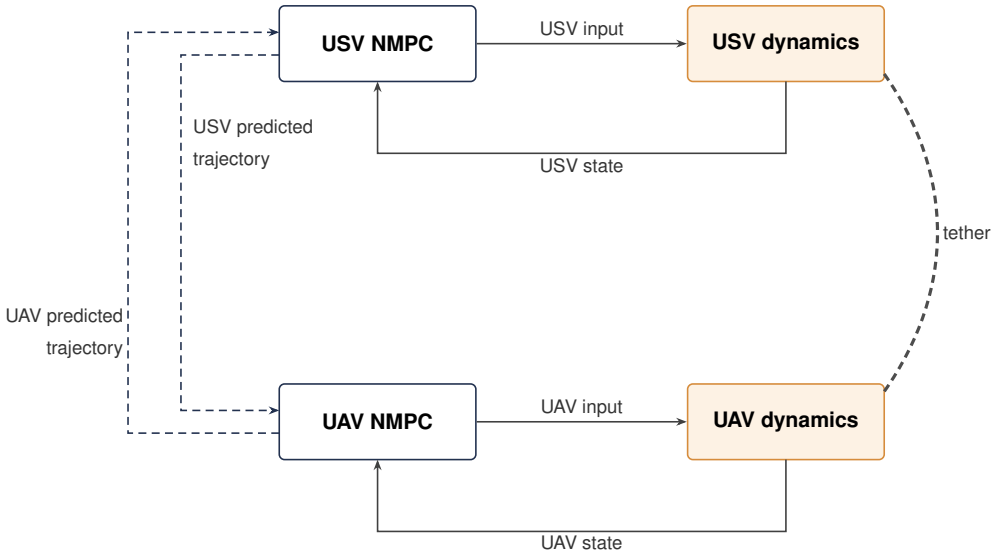


Figure 3.5: Cooperative control architecture diagram.

3.2.2 Non-linear Model Predictive Control

The vehicle models of Section 3.1 are continuous-time systems of the form

$$\dot{\mathbf{x}} = \mathbf{f}\left(\mathbf{x}, \mathbf{u}\right), \tag{3.39}$$

where $\mathbf{x} \in \mathbb{R}^{n_x}$ is the state vector and $\mathbf{u} \in \mathbb{R}^{n_u}$ is the control input vector.

However, the controller acts at discrete sampling instants $t_k = kT_s$, with the input held constant over each interval $[t_k, t_{k+1})$. Under this assumption, the state at the next instant follows from integrating the dynamics over one sampling interval,

$$\mathbf{x}(t_{k+1}) = \mathbf{x}(t_k) + \int_{t_k}^{t_{k+1}} \mathbf{f}\left(\mathbf{x}(\tau), \mathbf{u}_k\right) d\tau. \tag{3.40}$$

Since this integral generally has no closed-form solution for non-linear dynamics, it is approximated by applying a numerical integration scheme to $\mathbf{f}$ over one sampling interval. This defines a one-step discrete map $\mathbf{\Phi}$,

$$\mathbf{x}_{k+1} = \mathbf{\Phi}\left(\mathbf{x}_k, \mathbf{u}_k\right), \tag{3.41}$$

where $\mathbf{x}_k \approx \mathbf{x}(t_k)$ is the system state at instant k, so that $\mathbf{\Phi}$ returns the approximate state one sampling interval ahead, given the current state and input.

At every instant k, NMPC uses this map to predict the future behaviour of the system over a finite prediction horizon of N steps, and computes the input that steers it towards a reference. Denote by $\hat{\mathbf{x}}_{k+i|k}$ and $\hat{\mathbf{u}}_{k+i|k}$ the state and input predicted i steps ahead at instant k, for $i = 0, \ldots, N$ and $i = 0, \ldots, N-1$, respectively. The predicted trajectory starts from the estimated current state,

$$\hat{\mathbf{x}}_{k|k} = \mathbf{x}_k, \tag{3.42}$$

and evolves according to the system model,

$$\hat{\mathbf{x}}_{k+i+1|k} = \boldsymbol{\Phi}\left(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}\right), \quad i = 0, \ldots, N-1. \tag{3.43}$$

Each predicted step is then scored against a reference trajectory. A stage cost $\ell(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k})$, typically quadratic, penalises the tracking error and the control effort at every step of the horizon, and a terminal cost $\ell_f(\hat{\mathbf{x}}_{k+N|k})$ penalises the tracking error at its final step. Summing both over the horizon gives the total predicted cost,

$$J = \sum_{i=0}^{N-1} \ell\left(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}\right) + \ell_f\left(\hat{\mathbf{x}}_{k+N|k}\right). \tag{3.44}$$

Minimising J alone ignores the physical limits of the system, so the prediction is also constrained. The predicted states and inputs must remain within the admissible sets $\mathcal{X}$ and $\mathcal{U}$ at every step of the horizon,

$$\hat{\mathbf{x}}_{k+i|k} \in \mathcal{X}, \quad i = 0, \ldots, N, \tag{3.45}$$

$$\hat{\mathbf{u}}_{k+i|k} \in \mathcal{U}, \quad i = 0, \ldots, N-1, \tag{3.46}$$

where $\mathcal{X}$ and $\mathcal{U}$ collect the admissible states and inputs. These sets may be defined by simple bounds or by general non-linear inequality constraints of the form $h(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}) \leq 0$.

Enforcing these sets as hard constraints can make the problem infeasible. If a disturbance pushes the true state outside the admissible set, no predicted trajectory satisfies the constraints, and the solver has no solution to return even though one is required at every sampling instant. To avoid this, a constraint $h(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}) \leq 0$ can be softened by a slack variable $\sigma \geq 0$,

$$h(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}) \leq \sigma, \tag{3.47}$$

which allows the constraint to be violated by an amount σ. The slack becomes an additional decision variable at each step where the constraint is softened, and a penalty

$$\rho(\sigma) = \mu_1 \sigma + \mu_2 \sigma^2, \qquad \mu_1, \mu_2 \geq 0, \tag{3.48}$$

is added to the cost J for each slack [23]. The linear term gives the penalty a positive slope at $\sigma = 0$, so that, for a sufficiently large μ_1, the constraint is violated only when it cannot be met, which a quadratic term alone, with zero slope at $\sigma = 0$, would not ensure. The quadratic term penalises large violations more heavily and makes the penalty strictly convex in σ.

Assembling the cost, the prediction model, and the constraints defines the optimal control problem

solved at every sampling instant k,

$$\min_{\hat{\mathbf{x}}_{k+i|k},\, \hat{\mathbf{u}}_{k+i|k},\, \hat{\sigma}_{k+i|k}} \quad J + \sum_{i=0}^{N-1} \rho\left(\hat{\sigma}_{k+i|k}\right) \tag{3.49}$$

$$\begin{aligned}
\text{s.t.} \quad & \hat{\mathbf{x}}_{k|k} = \mathbf{x}_k, \\
& \hat{\mathbf{x}}_{k+i+1|k} = \boldsymbol{\Phi}\left(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}\right), \quad i = 0, \ldots, N-1, \\
& \hat{\mathbf{x}}_{k+i|k} \in \mathcal{X}, \quad i = 0, \ldots, N, \\
& \hat{\mathbf{u}}_{k+i|k} \in \mathcal{U}, \quad i = 0, \ldots, N-1, \\
& h\left(\hat{\mathbf{x}}_{k+i|k}, \hat{\mathbf{u}}_{k+i|k}\right) \leq \hat{\sigma}_{k+i|k}, \quad \hat{\sigma}_{k+i|k} \geq 0, \quad i = 0, \ldots, N-1.
\end{aligned}$$

Solving this problem returns a full predicted input sequence, $\hat{\mathbf{u}}_{k|k}, \ldots, \hat{\mathbf{u}}_{k+N-1|k}$, but only its first element, $\hat{\mathbf{u}}_{k|k}$, is applied to the system. At the next instant, $k+1$, the state is estimated again and the problem is solved again from $\hat{\mathbf{x}}_{k+1|k+1} = \mathbf{x}_{k+1}$, with the horizon shifted one step forward. This strategy, known as the receding horizon principle, computes every input from the latest state estimate, which closes the control loop and allows the controller to correct for disturbances and model errors that the previous prediction did not anticipate.

3.2.3 USV NMPC

The USV NMPC computes, at every sampling instant, the thruster commands that best track a reference trajectory over a finite prediction horizon, subject to the USV's dynamics, the physical limits of its actuators, and the tether workspace. It predicts the USV's motion with the model derived in Section 3.1, and solves a constrained optimisation problem at each step to obtain the commands applied to the thrusters.

Predictive Model

The USV model of Section 3.1 is driven by the resultant thrust force and moment $\boldsymbol{\tau} = [X, Y, N]^\top$, which are obtained from the physical actuator commands $T_L, T_R, \alpha_L, \alpha_R$ through the allocation

$$X = T_L \cos\alpha_L + T_R \cos\alpha_R, \qquad Y = T_L \sin\alpha_L + T_R \sin\alpha_R, \tag{3.50}$$

$$N = -x_{\mathrm{arm}}\big(T_L \sin\alpha_L + T_R \sin\alpha_R\big) - y_{\mathrm{arm}}\big(T_L \cos\alpha_L - T_R \cos\alpha_R\big). \tag{3.51}$$

Since the controller ultimately has to command $T_L, T_R, \alpha_L, \alpha_R$, the NMPC is formulated directly in terms of these variables, rather than choosing $\boldsymbol{\tau}$ and allocating it separately. This way, the limits of the physical actuators can be imposed directly in the optimisation problem.

Each thruster is limited both in the thrust and azimuth it can deliver,

$$T_{L,min} \leq T_L \leq T_{L,max}, \quad T_{R,min} \leq T_R \leq T_{R,max}, \quad \alpha_{L,min} \leq \alpha_L \leq \alpha_{L,max}, \quad \alpha_{R,min} \leq \alpha_R \leq \alpha_{R,max}, \tag{3.52}$$

and in how fast it can change them,

$$|\dot{T}_L| \le \dot{T}_{L,max}, \quad |\dot{T}_R| \le \dot{T}_{R,max}, \quad |\dot{\alpha}_L| \le \dot{\alpha}_{L,max}, \quad |\dot{\alpha}_R| \le \dot{\alpha}_{R,max}. \tag{3.53}$$

The rate limits act on the time derivatives of the commands, so imposing them directly as bounds requires these derivatives to become the control input, replacing the input of (3.3),

$$\mathbf{u}_v = [\dot{T}_L, \dot{T}_R, \dot{\alpha}_L, \dot{\alpha}_R]^\top \in \mathbb{R}^4, \tag{3.54}$$

which in turn makes $T_L, T_R, \alpha_L, \alpha_R$ part of the state vector, extending that of (3.2),

$$\mathbf{x}_v = [x, y, \psi, u, v, r,\ T_L, T_R, \alpha_L, \alpha_R]^\top \in \mathbb{R}^{10}. \tag{3.55}$$

With this choice, the range limits act on the state, and the rate limits act directly on the input.

With $\mathbf{x}_v$ and $\mathbf{u}_v$ defined, the prediction model $\dot{\mathbf{x}}_v = \mathbf{f}_v(\mathbf{x}_v, \mathbf{u}_v)$ is assembled from the USV kinematics and dynamics.

The kinematics are given by (3.4),

$$\dot{x} = u\cos\psi - v\sin\psi, \qquad \dot{y} = u\sin\psi + v\cos\psi, \qquad \dot{\psi} = r. \tag{3.56}$$

The dynamics follow (3.12)–(3.14), with X, Y, N given by (3.50)–(3.51) in terms of the thruster state,

$$\dot{u} = \frac{1}{m - X_{\dot{u}}}\Big[X - \big(X_u + X^{\pm}_{uu}|u|\big)u + (m - Y_{\dot{v}})vr\Big], \tag{3.57}$$

$$\dot{v} = \frac{1}{m - Y_{\dot{v}}}\Big[Y - \big(Y_v + Y_{vv}|v|\big)v - (m - X_{\dot{u}})ur\Big], \tag{3.58}$$

$$\dot{r} = \frac{1}{I_z - N_{\dot{r}}}\Big[N - \big(N_r + N_{rr}|r|\big)r - (X_{\dot{u}} - Y_{\dot{v}})uv\Big]. \tag{3.59}$$

Finally, the derivatives of the thruster states $T_L, T_R, \alpha_L, \alpha_R$ are directly the components of $\mathbf{u}_v$.

Discretisation over a sampling interval T_s, following Section 3.2.2, gives the one-step map $\mathbf{\Phi}_v$ such that $\hat{\mathbf{x}}_{v,k+i+1|k} = \mathbf{\Phi}_v\big(\hat{\mathbf{x}}_{v,k+i|k}, \hat{\mathbf{u}}_{v,k+i|k}\big)$ approximates the solution of $\dot{\mathbf{x}}_v = \mathbf{f}_v(\mathbf{x}_v, \mathbf{u}_v)$ over one sampling interval.

Cost and Constraints

Let $\mathbf{p} = [x, y]^\top$ denote the position, ψ the heading, and $\boldsymbol{\nu} = [u, v, r]^\top$ the body-fixed velocities, with $\mathbf{p}_{ref}$, ψ_{ref}, and $\boldsymbol{\nu}_{ref}$ the corresponding references from a given trajectory.

Tracking the reference position is penalised quadratically as

$$\|\mathbf{p} - \mathbf{p}_{ref}\|^2_{Q_p}, \tag{3.60}$$

where $\|\boldsymbol{\xi}\|^2_W \coloneqq \boldsymbol{\xi}^\top W \boldsymbol{\xi}$ for a positive semi-definite weighting matrix W, which sets the relative importance

of each component of $\boldsymbol{\xi}$ in the cost.

Tracking the heading reference is penalised as

$$q_\psi\big(1-\cos(\psi-\psi_{ref})\big), \tag{3.61}$$

where $q_\psi \geq 0$ is a scalar weight setting the relative importance of heading tracking.

Both ψ and ψ_{ref} are angles represented within an interval of width 2π, such as $(-\pi, \pi]$. When they lie near opposite ends of this interval, their raw difference $\psi - \psi_{ref}$ approaches $\pm 2\pi$, even though the two headings are then almost identical. As $\cos(\cdot)$ has period 2π, its value approaches one whenever $\psi - \psi_{ref}$ is near zero or $\pm 2\pi$, so the cost (3.61) is near zero in both cases, correctly reflecting that the two headings are almost identical.

Tracking the reference velocity is penalised in the same way as the position,

$$\|\boldsymbol{\nu} - \boldsymbol{\nu}_{ref}\|^2_{Q_\nu}. \tag{3.62}$$

The thruster commands $T_L, T_R, \alpha_L, \alpha_R$ are regularised directly as

$$\big\|[T_L, T_R, \alpha_L, \alpha_R]^\top\big\|^2_{Q_\tau}, \tag{3.63}$$

discouraging thrust and azimuth commands larger than necessary to track the reference.

The last term penalises $\mathbf{u}_v$, the rate of change of the thruster commands,

$$\big\|[\dot{T}_L, \dot{T}_R, \dot{\alpha}_L, \dot{\alpha}_R]^\top\big\|^2_{R}, \tag{3.64}$$

discouraging abrupt changes in thrust and azimuth between consecutive instants.

Summing (3.60)–(3.64) gives the stage cost

$$\begin{aligned}\ell(\mathbf{x}_v, \mathbf{u}_v) = \|\mathbf{p} - \mathbf{p}_{ref}\|^2_{Q_p} + q_\psi\big(1-\cos(\psi-\psi_{ref})\big) + \|\boldsymbol{\nu} - \boldsymbol{\nu}_{ref}\|^2_{Q_\nu} \\ + \big\|[T_L, T_R, \alpha_L, \alpha_R]^\top\big\|^2_{Q_\tau} + \big\|[\dot{T}_L, \dot{T}_R, \dot{\alpha}_L, \dot{\alpha}_R]^\top\big\|^2_{R}.\end{aligned} \tag{3.65}$$

The terminal cost ℓ_f penalises the last predicted state, so that the optimiser does not disregard where the USV ends up at the end of the horizon. It equals the stage cost (3.65) without the input rate term, since no control input is chosen at the final node:

$$\ell_f(\mathbf{x}_v) = \|\mathbf{p} - \mathbf{p}_{ref}\|^2_{Q_p} + q_\psi\big(1-\cos(\psi-\psi_{ref})\big) + \|\boldsymbol{\nu} - \boldsymbol{\nu}_{ref}\|^2_{Q_\nu} + \big\|[T_L, T_R, \alpha_L, \alpha_R]^\top\big\|^2_{Q_\tau}. \tag{3.66}$$

The thruster limits introduced in the predictive model enter the optimisation problem as constraints. Since thrust and azimuth are components of the state, their range limits define the admissible state set

$$\mathcal{X} = \left\{ \mathbf{x}_v : \begin{array}{ll} T_{L,min} \leq T_L \leq T_{L,max}, & T_{R,min} \leq T_R \leq T_{R,max}, \\ \alpha_{L,min} \leq \alpha_L \leq \alpha_{L,max}, & \alpha_{R,min} \leq \alpha_R \leq \alpha_{R,max} \end{array} \right\}, \tag{3.67}$$

while the limits on their rates of change, which act on the input, define the admissible input set

$$\mathcal{U} = \left\{ \mathbf{u}_v : \begin{array}{ll} |\dot{T}_L| \le \dot{T}_{L,max}, & |\dot{T}_R| \le \dot{T}_{R,max}, \\ |\dot{\alpha}_L| \le \dot{\alpha}_{L,max}, & |\dot{\alpha}_R| \le \dot{\alpha}_{R,max} \end{array} \right\}. \tag{3.68}$$

The surge speed is bounded from below, so that the USV does not sail astern: without this bound, when the reference changes direction the tracking cost can be reduced faster by turning the stern towards it and backing than by turning the bow, which is not how the USV is meant to be operated. A small negative bound $u_{min} \le 0$ still allows the slow backing needed to hold a position or to brake. Since the speed is a state that a disturbance can push below the bound, it is enforced as a soft constraint through a slack variable $\sigma_u \ge 0$,

$$u_{min} - u \le \sigma_u, \tag{3.69}$$

penalised by $\rho_u(\sigma_u)$ of the form (3.48), with its own weights $\mu_{u,1}$ and $\mu_{u,2}$.

The last constraint couples the USV to the UAV through the tether, and is the only one that depends on the other vehicle. The tether limits the distance between the winch on the USV deck and the UAV to a maximum admissible separation d_{max}. The winch is vertically aligned with the origin of the USV's body frame, at a fixed height z_t in the inertial frame, so its position is (x, y, z_t). Along the horizon, the USV controller uses the UAV's most recently computed predicted trajectory $\hat{\mathbf{p}}_a = [\hat{x}_a, \hat{y}_a, \hat{z}_a]^\top$, temporally resampled at the USV's prediction nodes covered by it, to compute the squared distance between the winch and the UAV,

$$d^2 = (x - \hat{x}_a)^2 + (y - \hat{y}_a)^2 + (\hat{z}_a - z_t)^2. \tag{3.70}$$

The requirement $d^2 \le d_{max}^2$ is enforced as a soft constraint through a slack variable $\sigma \ge 0$,

$$\underbrace{d^2 - d_{max}^2}_{h(\mathbf{x}_v)} \le \sigma, \tag{3.71}$$

which is penalised by $\rho(\sigma)$ as in (3.48), with large weights μ_1 and μ_2 so that violations remain small whenever geometrically feasible.

Optimal Control Problem

At every sampling instant k, the kinematic and dynamic components of the initial state come from the USV's current estimate, while the thruster components use the last command actually applied:

$$\mathbf{x}_{v,k} = \left[x_k, y_k, \psi_k, u_k, v_k, r_k,\ T_{L,k-1}, T_{R,k-1}, \alpha_{L,k-1}, \alpha_{R,k-1}\right]^\top. \tag{3.72}$$

Starting from the last applied command ensures that the rate limits in (3.53) are enforced relative to the actual thruster values, so the first input chosen is always attainable by the thrusters.

The references and the UAV's predicted trajectory are given data at each instant. Since the UAV's predicted trajectory can be shorter than the USV's horizon, let $N_c \le N$ denote the last node of the USV's

horizon covered by it. The tether constraint, and its slack, are enforced only for $i = 0, \dots, N_c$.

Assembling the cost, the model, and the constraints gives the optimal control problem the USV solves at every sampling instant:

$$\begin{aligned}
\min_{\hat{\mathbf{x}}_v,\, \hat{\mathbf{u}}_v,\, \hat{\sigma},\, \hat{\sigma}_u} \quad & \sum_{i=0}^{N-1} \Big[\ell\big(\hat{\mathbf{x}}_{v,k+i|k}, \hat{\mathbf{u}}_{v,k+i|k}\big) + \rho_u\big(\hat{\sigma}_{u,k+i|k}\big) \Big] + \ell_f\big(\hat{\mathbf{x}}_{v,k+N|k}\big) + \sum_{i=0}^{N_c} \rho\big(\hat{\sigma}_{k+i|k}\big) \\
\text{s.t.} \quad & \hat{\mathbf{x}}_{v,k|k} = \mathbf{x}_{v,k}, \\
& \hat{\mathbf{x}}_{v,k+i+1|k} = \boldsymbol{\Phi}_v\big(\hat{\mathbf{x}}_{v,k+i|k}, \hat{\mathbf{u}}_{v,k+i|k}\big), \quad i = 0, \dots, N-1, \\
& \hat{\mathbf{x}}_{v,k+i|k} \in \mathcal{X}, \quad i = 0, \dots, N, \\
& \hat{\mathbf{u}}_{v,k+i|k} \in \mathcal{U}, \quad i = 0, \dots, N-1, \\
& u_{min} - \hat{u}_{k+i|k} \le \hat{\sigma}_{u,k+i|k}, \quad \hat{\sigma}_{u,k+i|k} \ge 0, \quad i = 1, \dots, N-1, \\
& h\big(\hat{\mathbf{x}}_{v,k+i|k}\big) \le \hat{\sigma}_{k+i|k}, \quad \hat{\sigma}_{k+i|k} \ge 0, \quad i = 0, \dots, N_c.
\end{aligned} \tag{3.73}$$

The USV's low-level actuators receive the integrated thruster commands $T_{L,k}, T_{R,k}, \alpha_{L,k}, \alpha_{R,k}$, obtained by integrating the first optimal input rate $\hat{\mathbf{u}}_{v,k|k}$ over the sampling interval T_s. These become the thruster components of the initial state at the subsequent sampling instant.

3.2.4 UAV NMPC

The UAV NMPC computes, at every sampling instant, the thrust and attitude commands that best track a reference trajectory over a finite prediction horizon, subject to the vehicle's dynamics, its actuation limits, and the tether workspace. It predicts the UAV's motion with the model derived in Section 3.1, and solves a constrained optimisation problem at each step to obtain the commands applied to the UAV.

Predictive Model

The UAV model defined in Section 3.1 assumed that the commanded attitude is tracked instantaneously by the low-level flight controller. In reality, the autopilot tracks the commanded attitude with a delay, and it does so by rotating the vehicle so that the direction of the thrust approaches the commanded one, while the yaw is controlled separately. The commanded attitude is given by the setpoints $\phi_{\mathrm{cmd}}, \theta_{\mathrm{cmd}}, \psi_{\mathrm{cmd}}$, and, following (3.25), the commanded direction of the thrust is $\mathbf{R}(\phi_{\mathrm{cmd}}, \theta_{\mathrm{cmd}}, \psi_{\mathrm{cmd}})$ $[0,0,1]^\top$. The setpoints reach the autopilot after a transport delay, τ_d for roll and pitch and $\tau_{d,\psi}$ for yaw, and the closed-loop response of the autopilot can be approximated by a first-order lag, with a time constant $\tau > 0$ for the direction of the thrust and $\tau_\psi > 0$ for the yaw:

$$\begin{aligned}
\dot{\mathbf{z}}^{\mathcal{B}_a} &= \frac{1}{\tau}\Big(\mathbf{R}\big(\phi_{\mathrm{cmd}}(t-\tau_d), \theta_{\mathrm{cmd}}(t-\tau_d), \psi_{\mathrm{cmd}}(t-\tau_d)\big)[0,0,1]^\top - \mathbf{z}^{\mathcal{B}_a}\Big), \\
\dot{\psi} &= \frac{1}{\tau_\psi}\big(\psi_{\mathrm{cmd}}(t-\tau_{d,\psi}) - \psi\big),
\end{aligned} \tag{3.74}$$

where $\mathbf{z}^{\mathcal{B}_a}$, the direction of the thrust, is expressed in $\{\mathcal{I}\}$, and ψ is the actual yaw of the UAV.

Since the yaw only rotates the direction of the thrust about the vertical, this direction is described by two tilt angles ϕ, θ, measured relative to the commanded yaw,

$$\begin{aligned} \mathbf{z}^{\mathcal{B}_a} &= \mathbf{R}(\phi, \theta, \psi_{\mathrm{cmd}})[0,0,1]^\top = \mathbf{R}(0,0,\psi_{\mathrm{cmd}})\,\mathbf{n}(\phi,\theta), \\ \mathbf{n}(\phi,\theta) &= \mathbf{R}(\phi,\theta,0)[0,0,1]^\top = [\sin\theta\cos\phi,\ -\sin\phi,\ \cos\theta\cos\phi]^\top, \end{aligned} \tag{3.75}$$

which coincide with the roll and pitch angles of the UAV when its yaw equals the commanded yaw. The setpoints $\phi_{\mathrm{cmd}}, \theta_{\mathrm{cmd}}$ are, in the same way, the tilt angles of the commanded direction of the thrust.

Furthermore, similarly to the thrusters of the USV, the attitude commands and the thrust cannot change arbitrarily fast, and their rates of change are subject to physical slew-rate limits:

$$|\dot{\phi}_{\mathrm{cmd}}| \le \dot{\phi}_{max}, \quad |\dot{\theta}_{\mathrm{cmd}}| \le \dot{\theta}_{max}, \quad |\dot{\psi}_{\mathrm{cmd}}| \le \dot{\psi}_{max}, \quad |\dot{T}| \le \dot{T}_{max}. \tag{3.76}$$

Imposing these limits directly as bounds in the optimisation problem requires the rates of the commanded angles and of the thrust to form the control input vector, replacing that of (3.20),

$$\mathbf{u}_a = [\dot{\phi}_{\mathrm{cmd}}, \dot{\theta}_{\mathrm{cmd}}, \dot{\psi}_{\mathrm{cmd}}, \dot{T}]^\top \in \mathbb{R}^4, \tag{3.77}$$

which in turn incorporates the commanded attitude and the thrust into the state vector, extending that of (3.19),

$$\mathbf{x}_a = [x, y, z, \dot{x}, \dot{y}, \dot{z},\ \phi, \theta, \psi,\ \phi_{\mathrm{cmd}}, \theta_{\mathrm{cmd}}, \psi_{\mathrm{cmd}},\ T]^\top \in \mathbb{R}^{13}. \tag{3.78}$$

With this state augmentation, range limits act on the state and rate limits act directly on the input.

With $\mathbf{x}_a$ and $\mathbf{u}_a$ defined, the prediction model $\dot{\mathbf{x}}_a = \mathbf{f}_a(\mathbf{x}_a, \mathbf{u}_a)$ is assembled from kinematics, translational dynamics, attitude lag, and command integrators.

The kinematics follow (3.21), and the translational dynamics follow (3.30), with the thrust term expanded using the direction of the thrust (3.75), including the linear horizontal rotor drag opposing translational motion, the tether force $\mathbf{F}^{\mathcal{I}}_{\mathrm{tether}} = [F^{\mathcal{I}}_{\mathrm{tether},x}, F^{\mathcal{I}}_{\mathrm{tether},y}, F^{\mathcal{I}}_{\mathrm{tether},z}]^\top$ given by (3.31)–(3.38), and an estimated external force $\hat{\mathbf{F}}_{\mathrm{ext}} = [\hat{F}_{\mathrm{ext},x}, \hat{F}_{\mathrm{ext},y}, \hat{F}_{\mathrm{ext},z}]^\top$ that accounts for the forces the model does not describe, such as wind, and is held constant along the horizon (its estimation is given with the optimal control problem):

$$\ddot{x} = \frac{T}{m}\big(\cos\psi_{\mathrm{cmd}}\sin\theta\cos\phi + \sin\psi_{\mathrm{cmd}}\sin\phi\big) - \frac{d_r}{m}\dot{x} + \frac{1}{m}\big(F^{\mathcal{I}}_{\mathrm{tether},x} + \hat{F}_{\mathrm{ext},x}\big), \tag{3.79}$$

$$\ddot{y} = \frac{T}{m}\big(\sin\psi_{\mathrm{cmd}}\sin\theta\cos\phi - \cos\psi_{\mathrm{cmd}}\sin\phi\big) - \frac{d_r}{m}\dot{y} + \frac{1}{m}\big(F^{\mathcal{I}}_{\mathrm{tether},y} + \hat{F}_{\mathrm{ext},y}\big), \tag{3.80}$$

$$\ddot{z} = \frac{T}{m}\cos\theta\cos\phi - g + \frac{1}{m}\big(F^{\mathcal{I}}_{\mathrm{tether},z} + \hat{F}_{\mathrm{ext},z}\big). \tag{3.81}$$

The attitude dynamics follow from the response (3.74). Since the delayed setpoints are not part of the state, they are approximated to first order with their rates, which are inputs of the model, $\phi_{\mathrm{cmd}}(t-\tau_d) \approx \phi_{\mathrm{cmd}} - \tau_d\,\dot{\phi}_{\mathrm{cmd}}$, and likewise for θ_{cmd} and ψ_{cmd}. Expressing the response of the direction of the thrust in

the frame rotated by ψ_{cmd}, in which this direction is $\mathbf{n}(\phi,\theta)$, gives

$$\dot{\mathbf{n}} = \frac{1}{\tau}\Big(\mathbf{R}\big(0,0,-\tau_d\dot{\psi}_{\mathrm{cmd}}\big)\,\mathbf{n}\big(\phi_{\mathrm{cmd}}-\tau_d\dot{\phi}_{\mathrm{cmd}},\ \theta_{\mathrm{cmd}}-\tau_d\dot{\theta}_{\mathrm{cmd}}\big)-\mathbf{n}\Big)-\dot{\psi}_{\mathrm{cmd}}\,[0,0,1]^\top\times\mathbf{n}, \tag{3.82}$$

where the rotation inside the brackets accounts for the change of the commanded yaw during the delay, and the last term for the rotation of the frame itself. The partial derivatives of $\mathbf{n}$ with respect to ϕ and θ are orthogonal, with norms 1 and $\cos\phi$, so the rates of the tilt angles are the projections of $\dot{\mathbf{n}}$ onto them. Together with the response of the yaw, the attitude dynamics are

$$\dot{\phi} = \Big(\frac{\partial\mathbf{n}}{\partial\phi}\Big)^\top\dot{\mathbf{n}}, \qquad \dot{\theta} = \frac{1}{\cos^2\phi}\Big(\frac{\partial\mathbf{n}}{\partial\theta}\Big)^\top\dot{\mathbf{n}}, \qquad \dot{\psi} = \frac{1}{\tau_\psi}\big(\psi_{\mathrm{cmd}}-\tau_{d,\psi}\dot{\psi}_{\mathrm{cmd}}-\psi\big), \tag{3.83}$$

and they are completed by the commanded attitude and thrust, whose derivatives are directly the components of $\mathbf{u}_a$.

Discretisation over a sampling interval T_s, following Section 3.2.2, gives the one-step map $\mathbf{\Phi}_a$ such that $\hat{\mathbf{x}}_{a,k+i+1|k} = \mathbf{\Phi}_a\big(\hat{\mathbf{x}}_{a,k+i|k},\hat{\mathbf{u}}_{a,k+i|k}\big)$ approximates the solution of $\dot{\mathbf{x}}_a = \mathbf{f}_a(\mathbf{x}_a,\mathbf{u}_a)$ over one sampling interval.

Cost and Constraints

Let $\mathbf{p} = [x,y,z]^\top$ and $\mathbf{v} = [\dot{x},\dot{y},\dot{z}]^\top$ denote the position and velocity, with $\mathbf{p}_{ref}$ and $\mathbf{v}_{ref}$ the corresponding references.

Tracking the reference position and velocity is penalised quadratically as

$$\|\mathbf{p}-\mathbf{p}_{ref}\|^2_{Q_p} + \|\mathbf{v}-\mathbf{v}_{ref}\|^2_{Q_v}. \tag{3.84}$$

The next term penalises ϕ,θ directly, rather than comparing them against any reference,

$$\big\|[\phi,\theta]^\top\big\|^2_{Q_{tilt}}, \tag{3.85}$$

discouraging attitude larger than necessary to track the reference.

The next term penalises the yaw error, but rather than tracking a fixed yaw angle, ψ is aligned with the direction to a reference point $\mathbf{p}_{b,ref} = [x_{b,ref},y_{b,ref}]^\top$, itself possibly varying along the horizon. The yaw reference ψ_{ref} is the direction of the bearing vector

$$\mathbf{b} = \mathbf{p}_{b,ref}-\mathbf{p}_{xy} = \|\mathbf{b}\|\,[\cos\psi_{ref},\ \sin\psi_{ref}]^\top, \tag{3.86}$$

where $\mathbf{p}_{xy} = [x,y]^\top$, and the yaw error is penalised as the heading of the USV in (3.61), which avoids the same periodicity issue, but scaled by a weight w_b,

$$q_\psi\big(1-w_b\cos(\psi-\psi_{ref})\big), \qquad w_b = \frac{\|\mathbf{b}\|}{\sqrt{\|\mathbf{b}\|^2+\delta}}, \tag{3.87}$$

where $q_\psi \geq 0$ is a scalar weight setting the relative importance of yaw tracking, and $\delta > 0$ is a small

regularisation constant. Far from the reference point, where $\|\mathbf{b}\|^2 \gg \delta$, $w_b \approx 1$, while within a horizontal distance of the order of $\sqrt{\delta}$ it fades to zero. The term thus remains well defined above the reference point, where ψ_{ref} is undefined, and does not drive the UAV to rotate when it passes close to it.

The thrust is penalised relative to the hover thrust, as penalising it towards zero opposes gravity,

$$Q_T\,(T - mg)^2, \tag{3.88}$$

where $Q_T \geq 0$ is a scalar weight.

The last term penalises the input $\mathbf{u}_a$, the rates of change of the attitude and thrust commands,

$$\|\mathbf{u}_a\|_R^2, \tag{3.89}$$

discouraging abrupt changes of the commands between consecutive instants.

Summing the penalties (3.84)–(3.89) gives the stage cost

$$\begin{aligned}\ell(\mathbf{x}_a, \mathbf{u}_a) = \|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2 + \|\mathbf{v} - \mathbf{v}_{ref}\|_{Q_v}^2 + \left\|[\phi, \theta]^\top\right\|_{Q_{tilt}}^2 \\ + q_\psi\big(1 - w_b \cos(\psi - \psi_{ref})\big) + Q_T\,(T - mg)^2 + \|\mathbf{u}_a\|_R^2.\end{aligned} \tag{3.90}$$

The terminal cost ℓ_f penalises the last predicted state, so that the optimiser does not disregard where the UAV ends up at the end of the horizon. It equals the stage cost (3.90) without the input term, since no input is chosen at the final node:

$$\begin{aligned}\ell_f(\mathbf{x}_a) = \|\mathbf{p} - \mathbf{p}_{ref}\|_{Q_p}^2 + \|\mathbf{v} - \mathbf{v}_{ref}\|_{Q_v}^2 + \left\|[\phi, \theta]^\top\right\|_{Q_{tilt}}^2 \\ + q_\psi\big(1 - w_b \cos(\psi - \psi_{ref})\big) + Q_T\,(T - mg)^2.\end{aligned} \tag{3.91}$$

The UAV's operating envelope bounds the altitude, translational velocity, and attitude. To guarantee safe clearance above the water surface and USV deck, the UAV is restricted to $z \geq z_{min}$. Furthermore, because horizontal and vertical velocities are limited by distinct physical factors, the UAV's horizontal and vertical speeds are bounded separately. The attitude is bounded through the tilt, the angle between the direction of the thrust and the vertical, whose cosine is the vertical component of $\mathbf{n}$ in (3.75), $\cos\phi\cos\theta$. Limiting the tilt as a whole, rather than the roll and pitch about each axis, bounds the inclination of the UAV equally in every direction, independently of its yaw, up to a maximum tilt ϑ_{max}:

$$z \geq z_{min}, \quad \left\|[\dot{x}, \dot{y}]^\top\right\| \leq v_{xy,max}, \quad |\dot{z}| \leq v_{z,max}, \quad \cos\phi\cos\theta \geq \cos\vartheta_{max}. \tag{3.92}$$

Unlike the thruster commands of the USV, the velocity and attitude are subject to external disturbances such as wind, which can push the true state past these bounds regardless of what the NMPC has commanded. The bounds of (3.92) are therefore softened as in (3.47), each with its own slack variable

$\sigma_j \geq 0$, $j = 1, \ldots, 5$:

$$\underbrace{z_{min} - z}_{h_1(\mathbf{x}_a)} \leq \sigma_1, \tag{3.93}$$

$$\underbrace{\left\| [\dot{x}, \dot{y}]^\top \right\| - v_{xy,max}}_{h_2(\mathbf{x}_a)} \leq \sigma_2, \tag{3.94}$$

$$\underbrace{|\dot{z}| - v_{z,max}}_{h_3(\mathbf{x}_a)} \leq \sigma_3, \tag{3.95}$$

$$\underbrace{\cos\vartheta_{max} - \cos\phi\cos\theta}_{h_4(\mathbf{x}_a)} \leq \sigma_4, \tag{3.96}$$

$$\underbrace{\cos\vartheta_{max} - \cos\phi_{\mathrm{cmd}}\cos\theta_{\mathrm{cmd}}}_{h_5(\mathbf{x}_a)} \leq \sigma_5. \tag{3.97}$$

The tilt limit applies both to the attitude, h_4, and to its command, h_5. The attitude follows the command with a delay and a time constant, so that, with a bound on the attitude alone, the NMPC could command a tilt beyond the limit to reach the limit sooner, and the real attitude, if it responds faster than predicted, would follow the command beyond it. Each slack is penalised by $\rho_j(\sigma_j)$ as in (3.48), with weights $\mu_{1,j}$ and $\mu_{2,j}$.

The thrust range acts on the state, since the thrust is a component of $\mathbf{x}_a$, defining the admissible state set

$$\mathcal{X} = \{\mathbf{x}_a : \; T_{min} \leq T \leq T_{max}\}, \tag{3.98}$$

while the command rate limits of the predictive model (3.76) act on the input, defining the admissible input set

$$\mathcal{U} = \left\{ \mathbf{u}_a : \begin{array}{ll} |\dot{\phi}_{\mathrm{cmd}}| \leq \dot{\phi}_{max}, & |\dot{\theta}_{\mathrm{cmd}}| \leq \dot{\theta}_{max}, \\ |\dot{\psi}_{\mathrm{cmd}}| \leq \dot{\psi}_{max}, & |\dot{T}| \leq \dot{T}_{max} \end{array} \right\}. \tag{3.99}$$

The remaining constraints couple the UAV to the USV through the tether, and are the only ones that depend on the other vehicle. First, as for the USV, the tether limits the distance between the UAV and the winch, described in Section 3.2.3, to the maximum admissible separation d_{max}. The winch is located at $\mathbf{p}_w = [\hat{x}_v, \hat{y}_v, z_t]^\top$, as in Section 3.1.3, where the horizontal position $\hat{\mathbf{p}}_v = [\hat{x}_v, \hat{y}_v]^\top$ at each node is taken from the USV's most recently computed predicted trajectory, temporally resampled at the UAV's prediction nodes. Since the USV's horizon covers that of the UAV, the constraint is enforced at every node. With the chord $\mathbf{r} = \mathbf{p}_w - \mathbf{p}$ between the UAV and the winch, as in (3.31), the squared separation distance is

$$d^2 = \|\mathbf{r}\|^2 = (x - \hat{x}_v)^2 + (y - \hat{y}_v)^2 + (z - z_t)^2. \tag{3.100}$$

The requirement $d^2 \leq d_{max}^2$ is enforced as a soft constraint through a slack variable $\sigma_6 \geq 0$,

$$\underbrace{d^2 - d_{max}^2}_{h_6(\mathbf{x}_a)} \leq \sigma_6, \tag{3.101}$$

which is penalised by $\rho_6(\sigma_6)$ as in (3.48), with large weights $\mu_{1,6}$ and $\mu_{2,6}$ so that violations remain small.

Similarly, the length of the tether can only follow the distance between the vehicles as fast as the winch spools it, in either direction. If the distance grows faster than the winch pays out the tether, the tether is stretched and its tension rises sharply. If it shrinks faster than the winch reels the tether in, the tether is left with more slack than the operating point ε_0 of its model, and sags further, towards the water. The rate of change of the distance must therefore not exceed the maximum speed of the winch, $|\dot{d}| \leq \dot{d}_{max}$. Differentiating $d = \|\mathbf{r}\|$ with respect to time yields the projection of the relative velocity $\dot{\mathbf{r}} = [\hat{\dot{x}}_v - \dot{x},\, \hat{\dot{y}}_v - \dot{y},\, -\dot{z}]^\top$ onto the direction $\hat{\mathbf{d}}_0$ of the chord:

$$\dot{d} = \frac{\mathrm{d}}{\mathrm{d}t}\|\mathbf{r}\| = \frac{\mathbf{r}^\top \dot{\mathbf{r}}}{d} = \hat{\mathbf{d}}_0^\top \dot{\mathbf{r}}. \tag{3.102}$$

This formulation restricts only the radial component of the relative motion, leaving the motion around the winch, at constant distance, unconstrained. Enforcing $|\dot{d}| \leq \dot{d}_{max}$ with the regularisation δ to avoid numerical singularity when $d \to 0$ gives the softened constraint:

$$\underbrace{\frac{|\mathbf{r}^\top \dot{\mathbf{r}}|}{\sqrt{d^2 + \delta}} - \dot{d}_{max}}_{h_7(\mathbf{x}_a)} \leq \sigma_7, \tag{3.103}$$

which is penalised by $\rho_7(\sigma_7)$ as in (3.48).

Together, the minimum altitude z_{min} and the maximum tether separation d_{max} delimit the workspace of the UAV, illustrated in Figure 3.6.

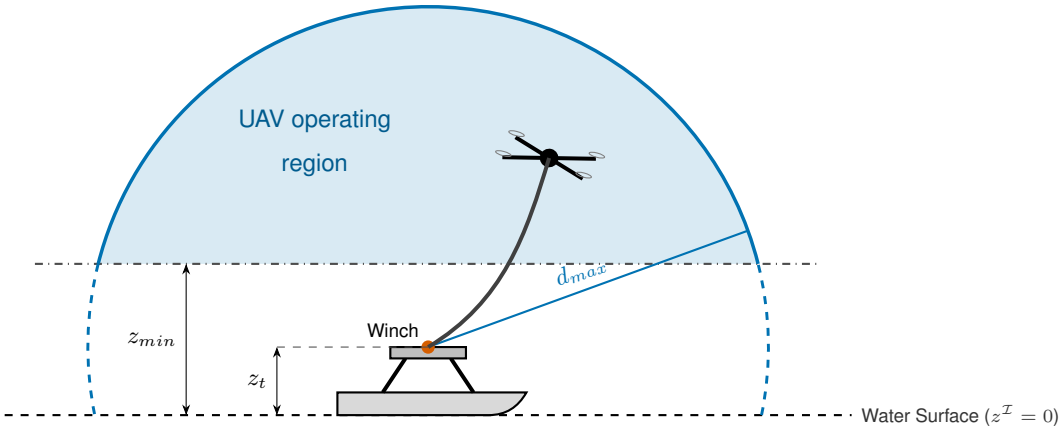


Figure 3.6: Operating region of the UAV.

All seven of these state constraints have the form $h_j(\mathbf{x}_a) \leq \sigma_j$ and are softened, so that the only hard constraint on the state is the thrust range (3.98). With the slacks stacked as $\boldsymbol{\sigma} = [\sigma_1, \ldots, \sigma_7]^\top$, the combined penalty is

$$P(\boldsymbol{\sigma}) = \sum_{j=1}^{7} \rho_j(\sigma_j). \tag{3.104}$$

Optimal Control Problem

At every sampling instant k, the translational and physical attitude components of the initial state come from the UAV's current state estimate, while the commanded attitude and thrust components use the last commands actually applied:

$$\mathbf{x}_{a,k} = \big[x_k, y_k, z_k, \dot{x}_k, \dot{y}_k, \dot{z}_k,\ \phi_k, \theta_k, \psi_k,\ \phi_{\mathrm{cmd},k-1}, \theta_{\mathrm{cmd},k-1}, \psi_{\mathrm{cmd},k-1},\ T_{k-1}\big]^\top. \tag{3.105}$$

The tilt angles ϕ_k, θ_k are obtained from the estimated direction of the thrust, expressed in the frame rotated by the last commanded yaw as in (3.75), $\mathbf{R}(0,0,\psi_{\mathrm{cmd},k-1})^\top \mathbf{z}^{\mathcal{B}_a} = \mathbf{n}(\phi_k, \theta_k)$. Since the state estimate is not synchronised with the sampling instants, the position is also propagated with the estimated velocity from the instant t_e of the most recent estimate to the sampling instant t_k, $\mathbf{p}_k = \mathbf{p}(t_e) + \mathbf{v}(t_e)\,(t_k - t_e)$.

A force that the model does not describe makes the prediction drift from the true motion, and without a model of it the controller settles with a steady offset. Following the disturbance-model approach to offset-free predictive control [24, 25], the external force $\hat{\mathbf{F}}_{\mathrm{ext}}$ in (3.79)–(3.81) is estimated at every sampling instant by a disturbance observer, from the one-step prediction error of the velocity. The velocity $\hat{\mathbf{v}}_{k|k-1}$ predicted for the current instant by the solution of the previous instant is compared with the estimated velocity $\mathbf{v}_k$, and the force left unexplained over the last interval corrects the estimate,

$$\hat{\mathbf{F}}_{\mathrm{ext},k} = \mathrm{sat}_{\bar{F}}\Big(\hat{\mathbf{F}}_{\mathrm{ext},k-1} + \beta\,\frac{m}{T_s}\big(\mathbf{v}_k - \hat{\mathbf{v}}_{k|k-1}\big)\Big), \tag{3.106}$$

where $0 < \beta < 1$ sets the speed of the estimate, trading the rejection of the disturbance against the transmission of measurement noise, and $\mathrm{sat}_{\bar{F}}$ bounds each component to $\pm\bar{F}$. For a constant force the update vanishes only when the prediction matches the motion, so the steady offset is removed.

The references and the USV's latest predicted trajectory are given data at each instant.

Assembling the cost, the model, and the constraints gives the optimal control problem the UAV solves at every sampling instant, with the slacks $\hat{\sigma}_{j,k+i|k}$ entering as additional decision variables:

$$\begin{aligned}
\min_{\hat{\mathbf{x}}_a,\,\hat{\mathbf{u}}_a,\,\hat{\boldsymbol{\sigma}}} \quad & \sum_{i=0}^{N-1} \ell\big(\hat{\mathbf{x}}_{a,k+i|k}, \hat{\mathbf{u}}_{a,k+i|k}\big) + \ell_f\big(\hat{\mathbf{x}}_{a,k+N|k}\big) + \sum_{i=0}^{N} P\big(\hat{\boldsymbol{\sigma}}_{k+i|k}\big) \\
\text{s.t.} \quad & \hat{\mathbf{x}}_{a,k|k} = \mathbf{x}_{a,k}, \\
& \hat{\mathbf{x}}_{a,k+i+1|k} = \boldsymbol{\Phi}_a\big(\hat{\mathbf{x}}_{a,k+i|k}, \hat{\mathbf{u}}_{a,k+i|k}\big), \quad i = 0, \dots, N-1, \\
& \hat{\mathbf{x}}_{a,k+i|k} \in \mathcal{X}, \quad i = 0, \dots, N, \\
& \hat{\mathbf{u}}_{a,k+i|k} \in \mathcal{U}, \quad i = 0, \dots, N-1, \\
& h_j\big(\hat{\mathbf{x}}_{a,k+i|k}\big) \le \hat{\sigma}_{j,k+i|k}, \quad \hat{\sigma}_{j,k+i|k} \ge 0, \quad j = 1, \dots, 7, \quad i = 0, \dots, N.
\end{aligned} \tag{3.107}$$

The UAV's low-level autopilot receives the attitude setpoints $\phi_{\mathrm{cmd},k}, \theta_{\mathrm{cmd},k}, \psi_{\mathrm{cmd},k}$, and the thrust command T_k, obtained by integrating the first optimal input $\hat{\mathbf{u}}_{a,k|k}$ over T_s. These become the commanded attitude and thrust components of the initial state at the subsequent sampling instant.

Chapter 4

Implementation

Chapter 3 defines the models and the controllers of the tethered USV–UAV system, but their behaviour and performance can only be assessed once they are implemented and executed. This depends on how the controllers interact with the vehicles, the tether, and the marine environment, which cannot be studied analytically.

For this reason, a realistic simulation environment, integrated with ROS 2, is developed to support the design of the controllers. It represents the vehicles and the tether, and provides controlled and repeatable conditions under which the models of Chapter 3 are first validated and the controllers are tested. The controllers are then implemented to run in real time and adapted to the vehicles used in simulation. Finally, manoeuvres and missions are defined to evaluate the performance of the controllers in representative operations, and all the components are integrated in a software architecture operated from a single interface.

4.1 Simulation Environment

Simulation is the natural first step in the development of the controllers, since it allows them to be tested and adjusted before being used on the real vehicles. The simulations in this work are run in Gazebo [26], a robotics simulator. A simulator is a program that creates a virtual environment, places robots in it, and calculates how they would behave in reality. At each step, it receives the commands sent to the robots, computes the forces acting on them, updates their positions accordingly, and returns the measurements that their sensors would produce. To the controller, the simulated robot behaves like a real one.

This allows controllers to be developed through many fast iterations, and different scenarios to be repeated under the same conditions, which makes the comparison between controllers and parameter choices fair. Sensor noise and estimation error can also be included in simulation, but they can be left out during development, so that the behaviour of the controllers is first analysed on its own.

4.1.1 World

The space where the simulation takes place is called the world. It contains the environment around the robots, such as the terrain, obstacles, and light, and defines the rules that apply inside it, such as the way time advances, the physical effects that are calculated, and the sensors that can be simulated. The robots are not part of the world. They are added to it afterwards, and what happens to them results from the interaction between their own models and the world in which they are placed.

A world is defined in a text file that lists its elements and the plugins that enable its behaviour. Plugins are software modules that add functionality to the simulator, such as the calculation of buoyancy or the simulation of sensors.

The world used in this work represents an open water area, shown in Figure 4.1. It consists of a sea surface and a seabed below it, and the buoyancy is calculated from the density of the water, which allows the hull of the USV to float. The world also enables the simulation of the sensors that the vehicles can use to estimate their state, such as inertial, magnetic, pressure, and satellite positioning sensors.

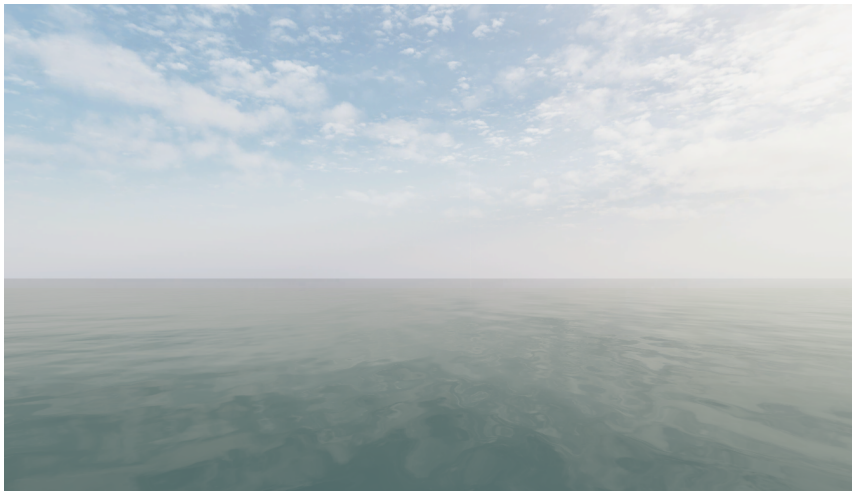


Figure 4.1: View of the simulated world

The vehicles operate at sea, where wind, waves, and currents act on them. The world allows these disturbances to be included, so that the controllers can be tested under conditions that their models do not predict. Each disturbance acts through a different physical path. Wind applies a force on the bodies exposed to the air, waves change the height of the water surface and therefore the buoyancy of the hull, and currents move the water itself, which changes the drag on the submerged bodies. These disturbances can be enabled, disabled, or adjusted for each test. In this work, a moderate sea, with a significant wave height of $0.6\,\mathrm{m}$, is present in every test unless stated otherwise.

4.1.2 Wave Adaptive Modular Vessel

The WAM-V is a catamaran with two inflatable hulls connected by a central payload deck, used in maritime robotics competitions. In this work it is the USV of the tethered system. It was chosen mainly because its payload deck is large enough to serve as a platform for the takeoff and landing of the UAV. In addition, a complete and publicly available model of the USV exists in the Virtual RobotX (VRX)

simulation framework [27], so that the work could focus on the control and not on building and validating a USV model. Figure 4.2 shows the vehicle in the simulated world.

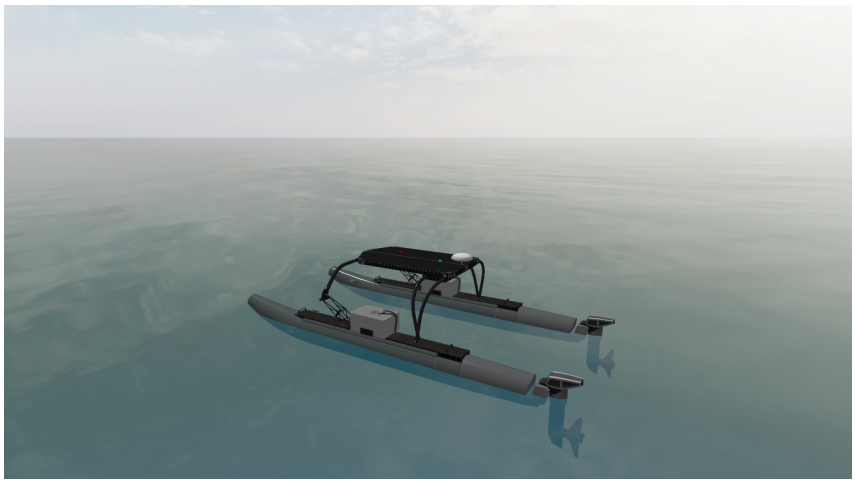
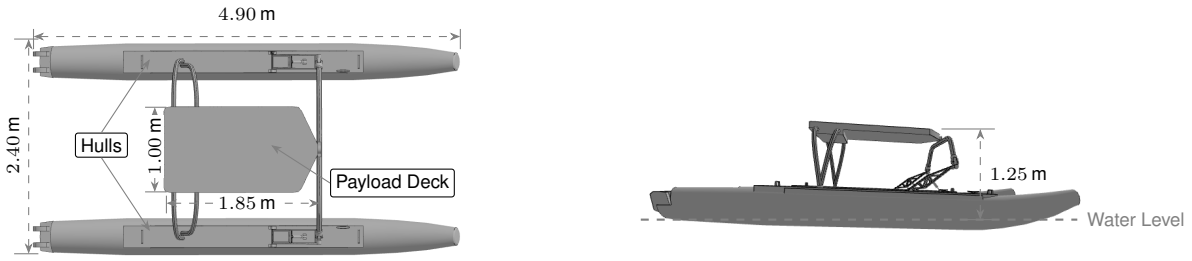


Figure 4.2: The WAM-V catamaran in the simulated world.

In the simulator, a vehicle is described by a model that lists its bodies, the joints that connect them, and for each body its mass, its inertia, and its geometry. The WAM-V is represented by a main rigid body, which includes the hulls, the deck, the batteries, and the mass of the thrusters, propellers, and sensors, with a total mass of $259.10\,\text{kg}$ and moments of inertia of 219, 568, and $683\,\text{kg}\cdot\text{m}^2$ about the longitudinal, lateral, and vertical axes. The thrusters, propellers, and sensors are also modelled as separate bodies attached to it through joints, with negligible mass. The thrusters can rotate to steer the USV, while the joints of the sensors are fixed. As the thrusters are mounted far aft, the centre of gravity of the whole vehicle lies $0.29\,\text{m}$ behind the centre of the hull geometry. The origin and the centre of mass of the main body are placed at this point, which is where the hydrodynamic forces described next are applied and which the odometry of the simulator refers to.

The catamaran has an overall length of $4.90\,\text{m}$ and a beam of $2.40\,\text{m}$. The payload deck, with $1.85\,\text{m}$ in length and $1.00\,\text{m}$ in width, is supported by the two hulls at about $1.25\,\text{m}$ above the water surface, as shown in Figure 4.3.



(a) Top view.

(b) Side view.

Figure 4.3: WAM-V catamaran layout.

The behaviour of the USV on the water is determined by how the simulator reproduces the forces that the water exerts on it. The hulls are kept afloat by a buoyancy force, which is calculated from the volume of each hull below the water surface. When waves are present, this volume changes along the

hulls and the catamaran pitches and rolls accordingly. As the USV moves, the water also resists its motion through drag forces and moments applied in the six degrees of freedom. For a velocity ν_i along or about each degree of freedom, the drag is

$$D_i = -\left(d_i^{\text{lin}} + d_i^{\text{quad}}\,|\nu_i|\right)\nu_i, \tag{4.1}$$

where d_i^{lin} and d_i^{quad} are the linear and quadratic coefficients. The negative sign makes the drag oppose the motion, and the linear term dominates at low speeds while the quadratic term dominates at higher speeds. This makes the vehicle slow down naturally when the propulsion is cut off. The coefficients of each degree of freedom, given in Table 4.1, are those provided by the VRX model.

Table 4.1: Hydrodynamic damping coefficients of the simulated WAM-V.

Damping Mode	Linear Coefficient	Quadratic Coefficient	Unit
Surge	100.00	150.00	N$/$(m/s) or N$/$(m/s)^2
Sway	100.00	100.00	N$/$(m/s) or N$/$(m/s)^2
Heave	500.00	–	N$/$(m/s)
Roll	300.00	600.00	N $\cdot$ m$/$(rad/s) or N $\cdot$ m$/$(rad/s)^2
Pitch	900.00	900.00	N $\cdot$ m$/$(rad/s) or N $\cdot$ m$/$(rad/s)^2
Yaw	800.00	800.00	N $\cdot$ m$/$(rad/s) or N $\cdot$ m$/$(rad/s)^2

The vehicle is propelled by two azimuth thrusters, one at the stern of each hull, and it steers by rotating them instead of using a rudder. To move, the model needs to receive, for each thruster, the thrust force that it must produce, in newtons, and its steering angle. The angle is limited to ± 30 degrees, to avoid twisting the power cables, and its rate of change to 10 degrees per second. The thrust is limited to 510.0 N in the forward direction, a value representative of the electric outboard motors used on the WAM-V in robotics competitions, such as the Torqeedo Cruise 2.0 [28]. In reverse, the propeller is less efficient, which is modelled with a multiplier of 0.746 that limits the thrust to -380.0 N.

4.1.3 x500 UAV

The Holybro X500 v2 [29] is a UAV, and in this work it is the aerial vehicle of the tethered system. It was chosen mainly because it is heavier and more stable than small UAVs, which makes it better suited to fly with a tether attached to it. Its size is also compatible with the payload deck of the WAM-V, leaving enough room for takeoff and landing, and a complete model of the vehicle is available in the simulation ecosystem of the PX4 autopilot [30], so that the work could focus on the control and not on building and validating a vehicle model. Figure 4.4 shows the vehicle in the simulated world.

The x500 is described in the simulator in the same way as the WAM-V, by a model that lists its bodies, the joints that connect them, and the mass and inertia of each one. It consists of a main rigid body, which includes the frame and the sensors, and of four rotors, which are separate bodies connected to it through rotating joints. Their masses, inertias, and positions are given in Table 4.2. The rotors are 0.348 m apart along the sides of the frame and about 0.49 m apart along the diagonal. The landing gear is 0.264 m wide, and the motor mounts are 0.26 m above the ground, as shown in Figure 4.5.

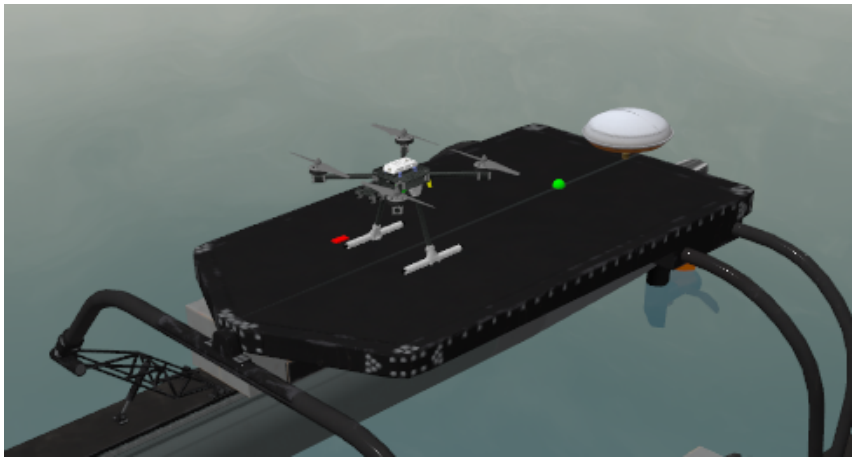
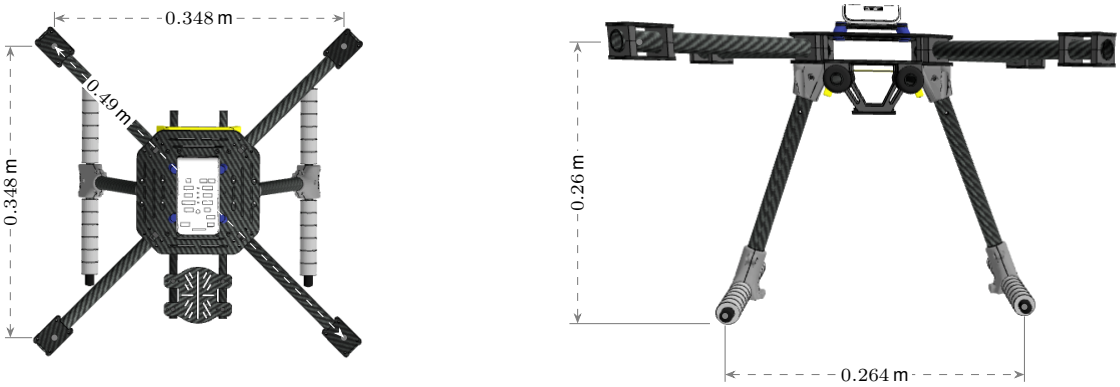


Figure 4.4: The x500 UAV in the simulated world.



(a) Top view.

(b) Front view.

Figure 4.5: x500 UAV layout.

Table 4.2: Parameters of the simulated x500, based on the PX4 model [31].

Parameter	**Value**	**Unit**
Main body mass	2.0	kg
Rotor mass (each)	0.016	kg
Total mass	2.06	kg
Moment of inertia, longitudinal and lateral axes	0.0217	kg · m^2
Moment of inertia, vertical axis	0.040	kg · m^2
Rotor position, along each horizontal axis	$\pm$0.174	m
Thrust coefficient k_F	1.11×10^{-5}	N/(rad/s)^2
Moment coefficient k_M	0.016	m
Rotor drag coefficient c_{drag}	8.06×10^{-5}	N · s/rad · m
Minimum rotor speed	150	rad/s
Maximum rotor speed	1000	rad/s
Rotor time constant, accelerating	12.5	ms
Rotor time constant, decelerating	25	ms

The vehicle is lifted by the thrust that its rotors produce by pushing air downwards. The model receives, for each rotor i, a command with its angular speed ω_i, in rad/s, and each rotor produces a thrust force F_i and a reaction moment M_i that depend on it [32],

$$F_i = k_F\,\omega_i^2, \qquad M_i = s_i\,k_M\,F_i, \tag{4.2}$$

where k_F is the thrust coefficient, k_M is the moment coefficient, and $s_i = \pm 1$ gives the sign of the moment, which depends on the direction of rotation. Rotors on the same diagonal rotate in the same direction and adjacent rotors in opposite directions, so that the reaction moments balance each other when the vehicle hovers. The model also includes the aerodynamic drag of the rotors.

The rotors have limits that the model reproduces. The speed of each rotor does not change instantly, and it follows the command with the time constants given in Table 4.2, up to the maximum speed given in the same table. At this speed, each rotor produces about $11.1\,\text{N}$, and the four together produce $44.5\,\text{N}$. As the weight of the vehicle is $20.2\,\text{N}$, the thrust-to-weight ratio is about 2.2. In hovering, the rotors produce about 45% of their maximum thrust, at a speed of about $675\,\text{rad/s}$, which leaves about $24\,\text{N}$ available to manoeuvre and to counter the force of the tether.

4.1.4 Tether

The tether couples the two vehicles, so the behaviour of the system depends on how it is represented. To evaluate the controllers as close as possible to real operation, they should be tested against a tether model that is more detailed than the one used in their design. The model used by the controllers, described in Section 3.1, approximates the tether by a catenary linearised around a constant slack. This simplification keeps the computational cost low, since the controllers evaluate the model at every step of the prediction horizon within each sampling interval. The simulated tether does not have this constraint, so a more detailed model can be used. Since this type of tether is not included in the vehicle models, it is simulated separately and coupled to them.

The tether is simulated with MoorDyn [10], a tool that represents a tether as a chain of lumped masses connected by segments that behave as springs and dampers. Each segment resists stretching and bending, while the weight and the drag of the tether are calculated at the nodes between segments. The slack and the tension of the tether result from the motion of these nodes, without assuming a shape for the tether. MoorDyn runs separately from Gazebo, which does not represent the tether itself but receives its effect on the vehicles through the forces that MoorDyn calculates. The tether is therefore displayed in a separate visualisation tool, Foxglove [33], as shown in Figure 4.6.

In this type of model, the mass of the tether is concentrated at the nodes, and the segments between them have no mass, as shown in Figure 4.7a. The motion of each free node i follows from Newton's second law, with the forces shown in Figure 4.7b,

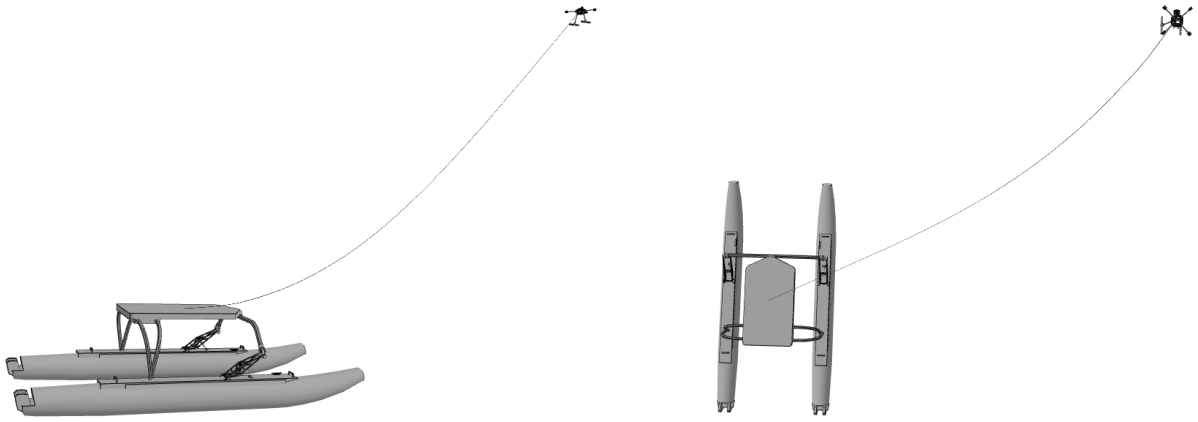
$$m_i\,\ddot{\mathbf{p}}_i = \mathbf{T}_{i-1} + \mathbf{T}_i + \mathbf{F}_i^{\text{bend}} + m_i\mathbf{g} + \mathbf{F}_i^{\text{drag}}, \tag{4.3}$$

where m_i and $\mathbf{p}_i$ are the mass and the position of the node, $\mathbf{T}_{i-1}$ and $\mathbf{T}_i$ are the tensions of the two adjacent segments, $\mathbf{F}_i^{\text{bend}}$ is the force due to the bending of the tether at the node, and $\mathbf{F}_i^{\text{drag}}$ is the drag, which opposes the velocity of the node. The magnitude of the tension of a segment depends on its elongation and on the rate at which it changes,

$$T = \begin{cases} EA\,\eta + BA\,\dot{\eta}, & \eta > 0, \\ 0, & \eta \le 0, \end{cases} \qquad \eta = \frac{\ell - \ell_0}{\ell_0}, \tag{4.4}$$

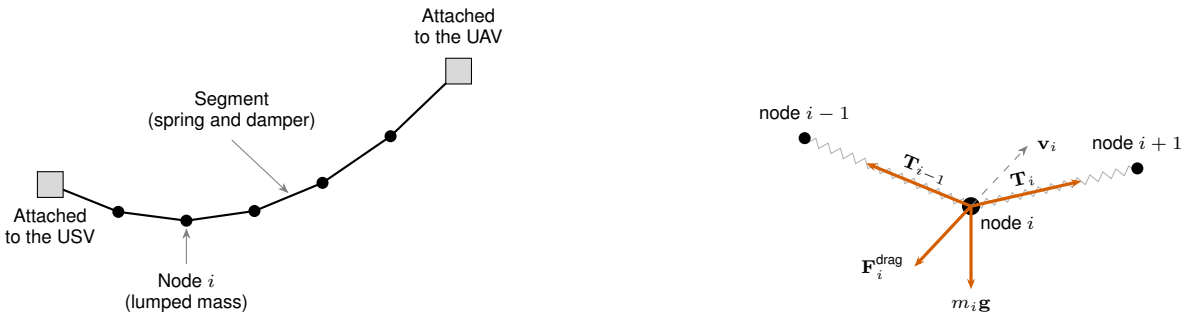
where $\ell = \|\mathbf{p}_i - \mathbf{p}_{i-1}\|$ is the current length of the segment, the distance between its two nodes, ℓ_0 is its unstretched length, EA is the axial stiffness, and BA is the damping coefficient, which is defined through the damping ratio given in Table 4.3. A segment that is shorter than its unstretched length carries no tension, so the tether can go slack. The accelerations are then integrated in time to obtain the new position and velocity of every node. The end nodes are not integrated, since their motion is imposed by the vehicles, and the forces at these nodes are the forces that the tether exerts on them.

The parameters of the model are the mass per unit length, which gives the mass m_i of each node in (4.3), the axial stiffness EA and the damping ratio, which define the tension in (4.4), the bending stiffness, and the drag coefficients. Their values are given in Table 4.3, together with the number of



(a) Side view. (b) Top view.

Figure 4.6: The tether between the USV and the UAV, displayed in Foxglove.



(a) Tether divided into nodes and segments. (b) Forces on a node.

Figure 4.7: Lumped-mass model of the tether.

segments and the time step used to integrate the equations. The mass per unit length and the diameter are based on commercial micro-tethers for UAVs that provide power and data [34, 35]. The axial stiffness is lower than that of a real tether, which would require an even smaller time step for the integration to remain stable, but under the tensions of a few newtons found in this work it already stretches the tether by less than $0.1\,\%$. The length of the tether is not fixed, and it is changed by the winch described in Section 4.1.5.

Table 4.3: Parameters of the simulated tether.

Parameter	**Value**	**Unit**
Diameter	4	mm
Mass per unit length	0.020	kg/m
Axial stiffness, EA	10^4	N
Damping ratio (relative to critical)	0.8	–
Bending stiffness	2×10^{-5}	$\mathrm{N \cdot m^2}$
Transverse drag coefficient	1.2	–
Axial drag coefficient	0.005	–
Density of the surrounding air	1.225	$\mathrm{kg/m^3}$
Number of segments	30	–
Integration time step	0.02	ms

The tether is coupled to the vehicles by a ROS node, as shown in Figure 4.8. At each step, the node reads the position and the velocity of the two ends of the tether, the winch on the deck of the USV and the centre of the body of the UAV, and gives them to MoorDyn. MoorDyn imposes them as the motion of the ends of the tether, integrates the tether over the step, and returns the force vector at each end, which is applied to the vehicles in Gazebo.

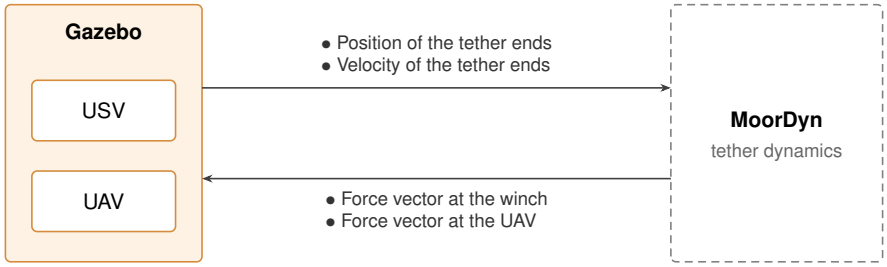


Figure 4.8: Data exchange between the simulation and the tether.

4.1.5 Tether Winch

The model of the tether used by the controllers in Section 3.1 assumes that its slack is kept close to a constant value ε_0, which requires a mechanism that adjusts the length of the tether as the vehicles move. In a real system, the winch spools the tether in and out, and in simulation it is represented by a virtual winch that varies the length of the tether.

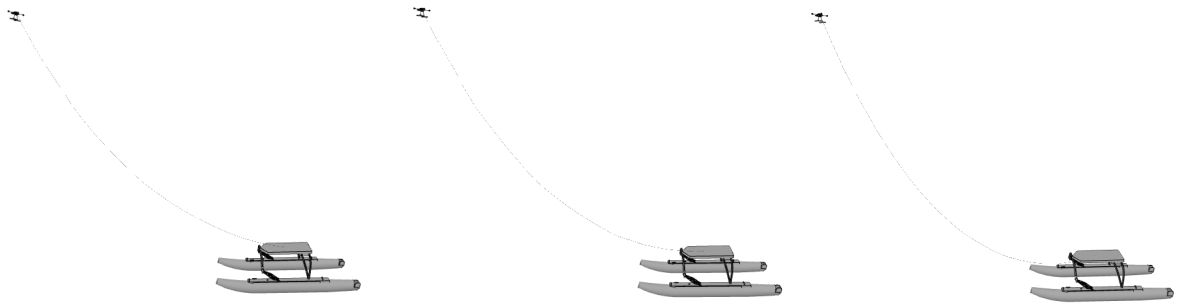
At each step, the virtual winch reads the distance d between the winch and the UAV and computes

the tether length L_{ref} that gives the desired slack,

$$L_{\text{ref}} = \frac{d}{1 - \varepsilon_0}, \tag{4.5}$$

which it then applies as the length of the tether in MoorDyn. Close to the deck, the winch can instead pull the tether with a constant force of $4\,\text{N}$, about a fifth of the weight of the UAV, raised gradually when the pull starts. A real system cannot follow the variations of the length instantly, because of the speed of the winch and the delay in the measurement of the distance. To account for this limitation, the rate of change of the length is limited to $v_{\text{winch}} = 2\,\text{m/s}$, which makes the slack deviate from ε_0 when the distance changes faster than the system can follow.

To evaluate the effect of the slack ratio, the UAV flew the same $60\,\text{s}$ trajectory with $\varepsilon_0 = 2\%$, 5%, and 10%, moving along x up to $10\,\text{m}$ to each side of the USV, which held its position, so that the distance between the winch and the UAV varied between about 5 and $14\,\text{m}$. Figure 4.9 shows the shape of the tether for each value, and a larger slack naturally makes the tether sag more between the vehicles. Figure 4.10 shows the tension at the UAV as a function of the distance between the winch and the UAV, averaged over intervals of $1\,\text{m}$, with the band showing its spread along the trajectory. For every slack ratio, the tension grows almost linearly with the distance, since the weight of the tether that the UAV supports increases with it. A larger slack lowers the whole curve, since a tether that sags more needs less tension to hold its own weight. Table 4.4 quantifies these results. From $\varepsilon_0 = 2\%$ to 10%, the mean tension decreases from $2.07\,\text{N}$ to $1.62\,\text{N}$, a reduction of 22%, and the maximum from $3.38\,\text{N}$ to $2.88\,\text{N}$.



(a) $\varepsilon_0 = 2\%$. (b) $\varepsilon_0 = 5\%$. (c) $\varepsilon_0 = 10\%$.

Figure 4.9: Shape of the tether for different slack ratios.

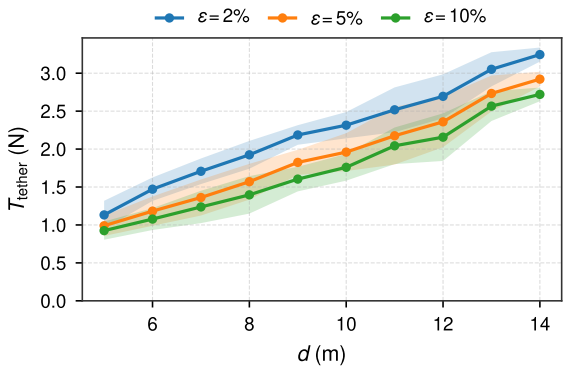


Figure 4.10: Tension at the UAV as a function of the distance, for different slack ratios.

Table 4.4: Tension at the UAV for different slack ratios.

ε_0	**Mean** (N)	**Maximum** (N)
2%	2.07	3.38
5%	1.78	3.08
10%	1.62	2.88

These tensions are small compared with the weight of the UAV, which is $20.2\,\mathrm{N}$, so the tension alone does not determine the choice of ε_0. The choice is instead a compromise on the amount of slack. Some slack is needed to absorb disturbances and avoid sudden tension spikes at the UAV, but too much makes the tether harder to manage and more likely to tangle. A value of $\varepsilon_0 = 5\%$ is used in the remainder of this work.

4.2 Models Validation

Before incorporating the vehicle and tether models into the real-time controllers developed in Chapter 3, their accuracy must be verified against the high-fidelity simulation environment described in Section 4.1 to ensure they capture the dominant dynamics while remaining computationally tractable. Each model is evaluated using the nominal simulator parameters, without fitting them to the validation data, driven by recorded inputs from representative manoeuvres. Model fidelity is quantified by the root-mean-square error (RMSE) in physical units and by the normalised root-mean-square error (NRMSE), defined as the RMSE normalised by the signal range.

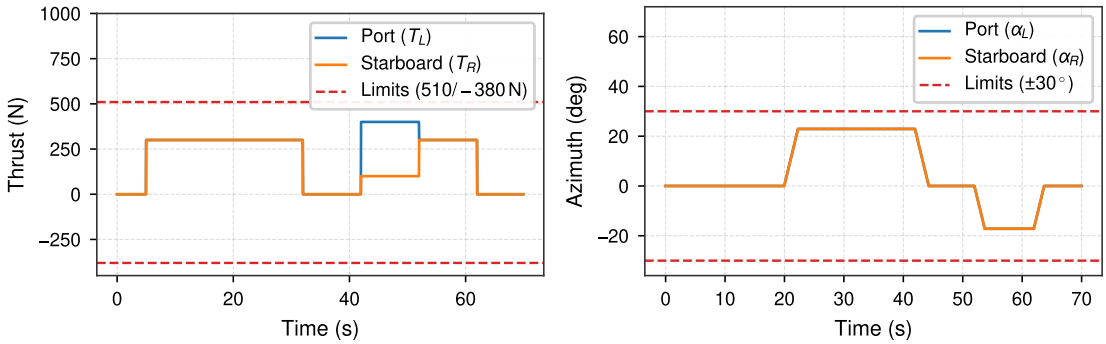
4.2.1 USV Model

The first model to be validated is the one used by the USV NMPC, presented in Section 3.2.3. It predicts how the USV moves from the commands sent to its thrusters, so its accuracy determines how well the controller can anticipate the motion of the USV. To compare it with the simulator, the parameters of the model are set to the values of the simulated USV, described in Section 4.1. These values are listed in Table 4.5.

Table 4.5: Parameters of the USV model.

Parameter	**Symbol**	**Value**
Mass	m	$259.10\,\mathrm{kg}$
Yaw moment of inertia	I_z	$683.22\,\mathrm{kg\cdot m^2}$
Added mass coefficients	$X_{\dot{u}},\ Y_{\dot{v}},\ N_{\dot{r}}$	0
Thruster longitudinal arm	x_{arm}	$2.0855\,\mathrm{m}$
Thruster lateral arm	y_{arm}	$1.0271\,\mathrm{m}$
Linear surge damping	X_u	$100.0\,\mathrm{N/(m/s)}$
Quadratic surge damping (forward, backward)	$X^+_{uu},\ X^-_{uu}$	$150.0\,\mathrm{N/(m/s)^2}$
Linear sway damping	Y_v	$100.0\,\mathrm{N/(m/s)}$
Quadratic sway damping	Y_{vv}	$100.0\,\mathrm{N/(m/s)^2}$
Linear yaw damping	N_r	$800.0\,\mathrm{N\cdot m/(rad/s)}$
Quadratic yaw damping	N_{rr}	$800.0\,\mathrm{N\cdot m/(rad/s)^2}$

To compare the model with the simulator, the USV was driven in open loop for $69.96\,\mathrm{s}$ with a sequence of thrust and azimuth commands chosen to excite the three velocities of the model, surge, sway, and yaw rate. Figure 4.11 shows the commands sent to the thrusters, which were recorded together with the resulting motion of the USV. The model is then given the same commands, starting from the initial state of the USV, and computes the velocities over the whole run without any correction from the simulator, to be compared with the recorded ones.

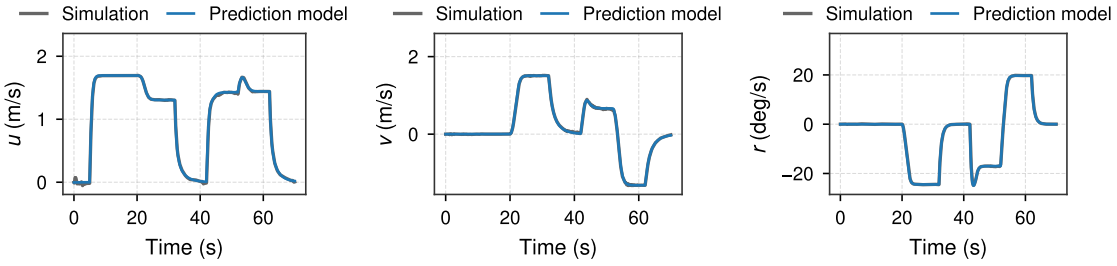


(a) Thrust commands.

(b) Azimuth angle commands.

Figure 4.11: USV thruster inputs.

The dynamics of the model are first tested over the whole run, in a single continuous test that covers all the conditions of the manoeuvre and in which the model never receives the state of the simulator. Figure 4.12 compares the surge velocity u, the sway velocity v, and the yaw rate r of the model with those of the simulator. Only the velocities are compared, because the damping force of the water grows with them and limits their error. The position and the heading are only the integral of the velocities, and would accumulate their small errors over the run. The errors of the three velocities, listed in Table 4.6, are below 1% of their range.



(a) Surge velocity u.

(b) Sway velocity v.

(c) Yaw rate r.

Figure 4.12: USV body-fixed velocities validation.

The NMPC does not use the model over the whole run, but over a prediction horizon of $T_h = 2.0\,\mathrm{s}$, chosen in Section 4.3.2, that always starts from the current state of the USV. To evaluate the model in this use, predictions of $2.0\,\mathrm{s}$ were computed along the trajectory of the run, each one starting from the state of the simulator and driven by the recorded commands. Figure 4.13 shows these predictions over the trajectory of the USV. Since each prediction starts from the state of the simulator, the errors of the position and the heading accumulate only over the horizon, which is the time span of the NMPC

prediction.

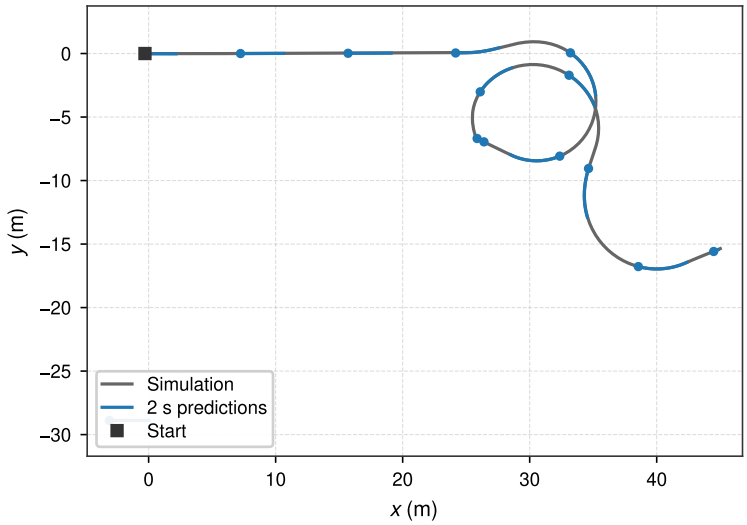
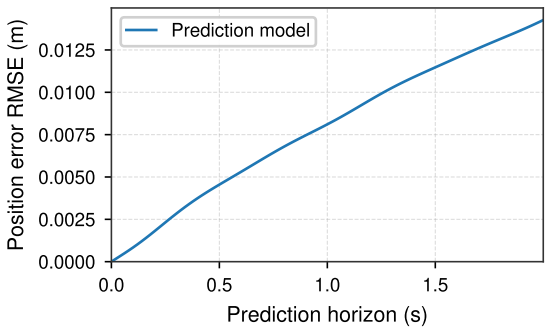
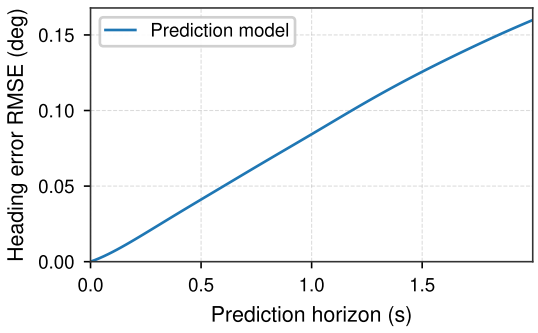


Figure 4.13: USV trajectory and $2.0\,$s predictions.

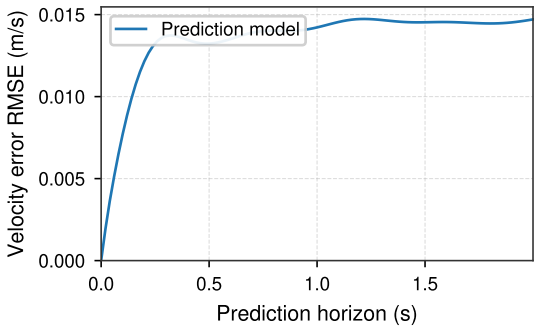
The error of these predictions grows with how far ahead they are made. Figure 4.14 presents the error of the position, the heading, the linear velocity, and the yaw rate as a function of the prediction horizon, up to $T_h = 2.0\,$s. The errors grow with the horizon, and at its end they are below 1% of the range of each state, as listed in Table 4.6, and the largest position error represents less than 2% of the length of the USV, which makes the model suitable for the NMPC.



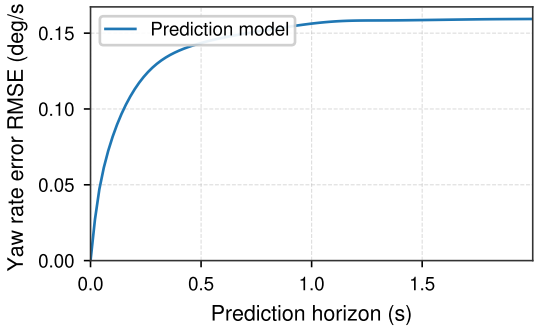
(a) Position error e_p.



(b) Heading error e_ψ.



(c) Linear velocity error e_v.



(d) Yaw rate error e_r.

Figure 4.14: USV prediction errors as a function of the prediction horizon.

Table 4.6: USV model validation errors.

State	Unit	RMSE	NRMSE (%)	Max Error
Velocities over the whole run (69.96 s)				
Surge velocity u	m/s	0.0129	0.74	0.0760
Sway velocity v	m/s	0.0092	0.32	0.0470
Yaw rate r	deg/s	0.1579	0.35	1.8787
Predictions over the horizon ($T_h = 2.0$ s)				
Position error e_p	m	0.0143	0.37	0.0933
Heading error e_ψ	deg	0.1599	0.33	0.8887
Linear velocity error e_v	m/s	0.0147	0.73	0.0756
Yaw rate error e_r	deg/s	0.1594	0.36	1.8723

4.2.2 UAV Model

The second model to be validated is the one used by the UAV NMPC, presented in Section 3.2.4. It predicts how the vehicle moves from the attitude and the thrust that are commanded, so its accuracy determines how well the controller can anticipate that motion. The tether force is left out of the model, so that the motion of the vehicle is validated on its own, and the tether is validated separately in Section 4.2.3. The parameters of the model are obtained from the simulated vehicle, described in Section 4.1.3, and are listed in Table 4.7.

Table 4.7: Parameters of the UAV model.

Parameter	Symbol	Value
Mass	m	$2.06\,\text{kg}$
Gravitational acceleration	g	$9.81\,\text{m/s}^2$
Rotor drag coefficient	d_r	$0.218\,\text{N}\cdot\text{s/m}$
Attitude time constant (roll and pitch)	τ	$0.154\,\text{s}$
Attitude time constant (yaw)	τ_ψ	$0.357\,\text{s}$
Transport delay (roll and pitch)	τ_d	$0.04\,\text{s}$
Transport delay (yaw)	$\tau_{d,\psi}$	$0.06\,\text{s}$

The mass and the gravitational acceleration are taken directly from the simulator, but the other parameters need to be calculated. The simulator applies to each rotor a drag force proportional to both its angular speed and the velocity of the vehicle, through the coefficient $c_{drag} = 8.06 \times 10^{-5}$ of Table 4.2. Around the hovering speed ω_0, at which the thrust of the four rotors balances the weight, the four rotors together give a linear drag on the horizontal velocity,

$$\omega_0 = \sqrt{\frac{mg}{4\,k_F}} \approx 675\,\text{rad/s}, \qquad d_r = 4\,c_{drag}\,\omega_0 \approx 0.218\,\text{N}\cdot\text{s/m}, \tag{4.6}$$

where k_F is the thrust coefficient of the rotors. The nominal attitude time constants are not parameters of the simulator, since they result from the attitude controller of the vehicle. This controller converts the angle error into an angular rate setpoint through a proportional gain, $P_\phi = 6.5\,\text{s}^{-1}$ for roll and pitch and $P_\psi = 2.8\,\text{s}^{-1}$ for yaw. Assuming that the inner loop that tracks the angular rate does so instantly, the

angular rate of the vehicle is equal to the setpoint, and the roll angle ϕ evolves as

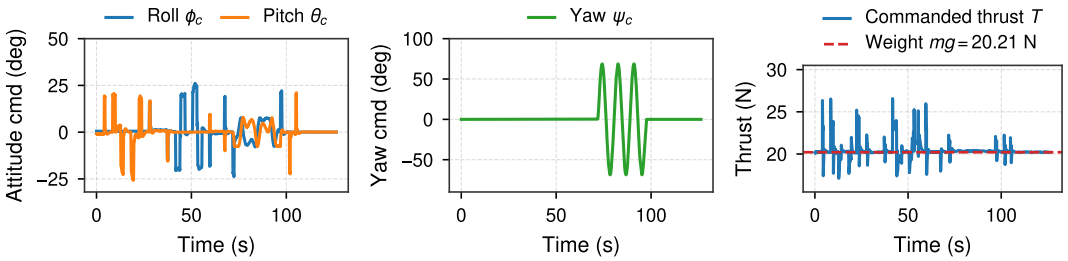
$$\dot{\phi} = P_\phi \, (\phi_{\mathrm{cmd}} - \phi), \tag{4.7}$$

where ϕ_{cmd} is the commanded roll angle, and the same relation holds for the pitch and, with the gain P_ψ, for the yaw. This is a first-order response, in which the attitude changes faster the further it is from the command, and its time constant is the inverse of the gain,

$$\tau = \frac{1}{P_\phi} \approx 0.154\,\mathrm{s}, \qquad \tau_\psi = \frac{1}{P_\psi} \approx 0.357\,\mathrm{s}. \tag{4.8}$$

The attitude controller does not, however, act on the roll and pitch angles separately. It rotates the vehicle so that the direction of the thrust approaches the commanded direction, and it controls the yaw independently, so the time constant τ applies to the direction of the thrust, as in the model of Section 3.2.4. The commands also reach the attitude controller only after a small transport delay. The delays are not parameters of the simulator either, and they were obtained from flights with the NMPC, different from the validation flight, as the values that minimise the error of the attitude predicted over $1\,\mathrm{s}$, giving $\tau_d = 0.04\,\mathrm{s}$ for roll and pitch and $\tau_{d,\psi} = 0.06\,\mathrm{s}$ for yaw.

To compare the model with the simulator, the UAV flew for $126\,\mathrm{s}$ following a reference that excites its motion in all directions, with inclined straight lines followed by a circle in which the yaw oscillates independently of the motion. The reference was tracked by the default position controller of PX4, so that the commands do not depend on the model being validated. Figure 4.15 shows the attitude and thrust commands generated by this controller, which were recorded together with the resulting motion of the vehicle. The model is given the same commands, and its response is compared with the recorded motion.

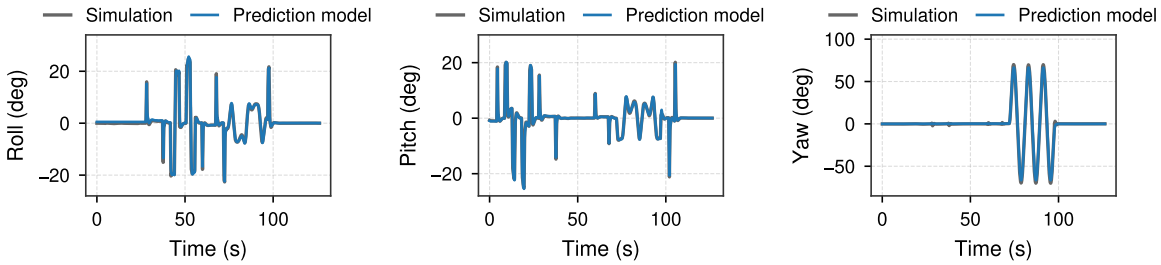


(a) Roll and pitch commands. (b) Yaw command. (c) Thrust command.

Figure 4.15: UAV attitude and thrust commands during the validation flight.

The attitude of the vehicle does not reach the commanded attitude instantly, so the model includes the response of the attitude controller, with the nominal time constants τ and τ_ψ and the transport delays τ_d and $\tau_{d,\psi}$. To test whether it predicts this response correctly, the recorded attitude commands were given to the response of Equation (3.74), and the resulting attitude was compared with the one recorded in the simulator. Figure 4.16 shows, for roll, pitch, and yaw, the attitude of the simulator and the attitude predicted by the model. The errors of the three angles, listed in Table 4.8, are below 2% of their range with the delays, against more than 4% if the commanded attitude were taken directly as the attitude of

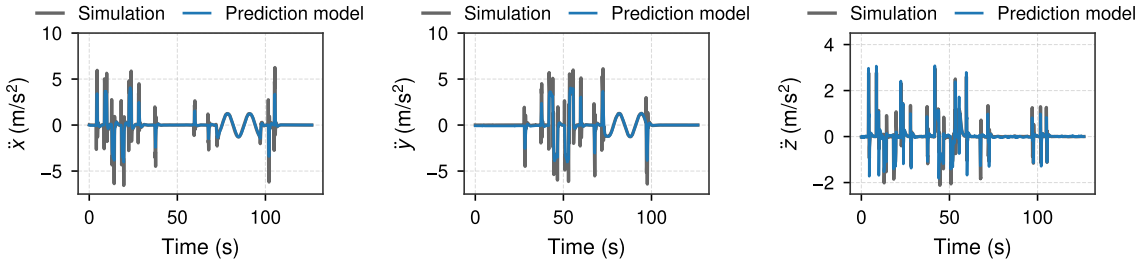
the vehicle.



(a) Roll. (b) Pitch. (c) Yaw.

Figure 4.16: UAV attitude tracking. Simulation and model with the nominal attitude response.

The model is first compared with the simulator on the inertial acceleration of the vehicle, which it computes directly from the attitude, the thrust, and the drag. Since this does not integrate any previous state, the error does not accumulate and can be evaluated over the whole flight. The model is given the recorded thrust and the attitude that it predicts with the delay, together with the velocity recorded in the simulator, which determines the drag. Figure 4.17 compares the acceleration along x, y, and z of the model with that of the simulator. The errors of the three accelerations, listed in Table 4.8, are about 4% of their range.



(a) Along x. (b) Along y. (c) Along z.

Figure 4.17: UAV inertial acceleration.

The NMPC does not use the model over the whole flight, but over a prediction horizon of $T_h = 1.0\,\text{s}$, chosen in Section 4.3.3, that always starts from the current state of the vehicle. To evaluate the model in this use, predictions of $1.0\,\text{s}$ were computed along the flight, each one starting from the state of the simulator and driven by the recorded commands. Figure 4.18 shows these predictions over the trajectory of the vehicle. Since the position and the velocity are obtained by integrating the acceleration, their errors accumulate only over the horizon, and they can be compared together with the acceleration.

The error of these predictions grows with how far ahead they are made. Figure 4.19 presents the error of the position and of the velocity as a function of the prediction horizon, up to $T_h = 1.0\,\text{s}$. At the end of the horizon, the errors are below 2.5% of the range of each state, as listed in Table 4.8. For the position, this corresponds to an RMSE of $11\,\text{cm}$ over the up to $8\,\text{m}$ that the vehicle travels in the horizon. The model is therefore suitable for the NMPC.

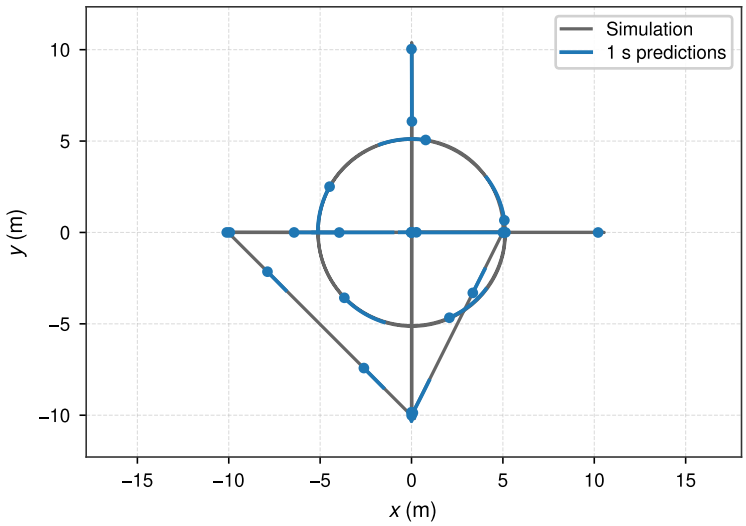
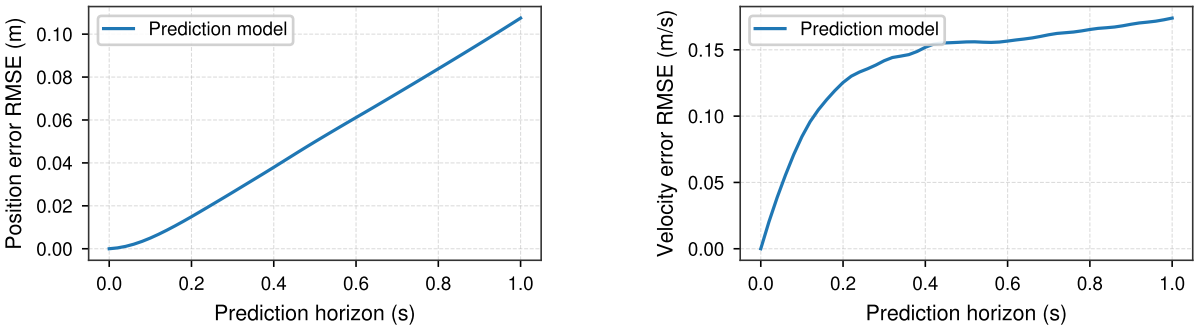


Figure 4.18: UAV trajectory and 1.0 s predictions.



(a) Position error.

(b) Velocity error.

Figure 4.19: UAV prediction errors as a function of the prediction horizon.

Table 4.8: UAV model validation errors.

State	Unit	RMSE	NRMSE (%)	Max Error
Attitude over the whole flight (126.32 s)				
Roll ϕ, with lag	deg	0.6316	1.32	3.8352
Roll ϕ, without lag	deg	2.4220	5.08	20.1176
Pitch θ, with lag	deg	0.4022	0.89	2.4640
Pitch θ, without lag	deg	2.2981	5.07	19.9585
Yaw ψ, with lag	deg	1.3520	0.96	6.3935
Yaw ψ, without lag	deg	5.7076	4.07	18.7711
Acceleration over the whole flight (126.32 s)				
Acceleration $\ddot{x}$	m/s^2	0.4551	3.55	3.3270
Acceleration $\ddot{y}$	m/s^2	0.4969	3.97	3.2636
Acceleration $\ddot{z}$	m/s^2	0.1913	3.77	1.5908
Predictions over the horizon ($T_h = 1.0$ s)				
Position error e_p	m	0.1075	1.34	0.7177
Horizontal position error $e_{p,xy}$	m	0.1024	1.31	0.7078
Vertical position error $e_{p,z}$	m	0.0326	1.34	0.2380
Velocity error e_v	m/s	0.1739	2.02	0.8724

4.2.3 Tether Model

The last model to be validated is the tether force model of Section 3.1.3, which gives the force that the tether applies on the UAV from the positions of the two vehicles. The force has a tension, proportional to the distance d between the vehicles, and a direction. The chord, the straight line between the vehicles, would be the direction of the force if the tether were weightless, but the weight of the tether bends it and rotates the force from the chord towards the vertical. The tension is approximately the weight of the tether hanging between the two vehicles, $T_{\text{tether}} = \mu g L$, where μ is the linear mass and L the length of the tether. Since the winch keeps $L = d/(1-\varepsilon_0)$, the tension is proportional to the distance between the vehicles, $T_{\text{tether}} = c_d\, d$, with

$$c_d = \frac{\mu g}{1-\varepsilon_0} = \frac{0.020 \times 9.81}{1-0.05} = 0.2065\,\text{N/m}. \tag{4.9}$$

The deflection of the force is largest when the chord is horizontal and decreases as it approaches the vertical. For a horizontal chord, the weight bends the tether into a catenary [36], and the deflection γ_0 is the angle that the tether makes with the chord at the UAV. For a tether of length L between two points at distance d, it is given by

$$\gamma_0 = \arctan\left(\sinh\kappa\right), \qquad \frac{\sinh\kappa}{\kappa} = \frac{L}{d} = \frac{1}{1-\varepsilon_0}, \tag{4.10}$$

where κ is the positive solution of the second equation, obtained iteratively. For $\varepsilon_0 = 5\%$, this gives $\kappa = 0.558$ and $\gamma_0 = 30.4°$. The parameters of the model are listed in Table 4.9.

Table 4.9: Parameters of the tether model.

Parameter	Symbol	Value
Linear mass	μ	0.020 kg/m
Slack ratio	ε_0	5%
Tension per unit of distance	c_d	0.2065 N/m
Deflection for a horizontal chord	γ_0	$30.4°$

The tether force was recorded in the simulator during a flight of 60.5 s, in which the UAV moved along x, and the distance between the two vehicles varied between 4.6 m and 14.0 m. The winch regulated the slack to ε_0. The reference is the force computed by MoorDyn and applied on the UAV in Gazebo described in Section 4.1. At each instant, the model receives the positions of the two vehicles and computes the force, which is compared with the reference in tension and in its components in the inertial frame. The comparison is also made for two simpler models of the force, which differ from the catenary model only in its direction: the taut model, in which the tether is taut and the force points along the chord, and the vertical model, in which the tether hangs from the UAV as a vertical load. Since the models give the force directly from the positions, there is no initial state and no error accumulates, so the whole flight is compared instant by instant.

Figure 4.20 compares the tension of the model with that of the simulator along the flight, and Figure 4.21 compares the components of the force on the UAV along x and z. The catenary model follows

the tension and the components, with errors between 4% and 11% of their range, as listed in Table 4.10. The two simpler models fail in opposite ways. The vertical model gives no force along x and places the whole tension along z, while the taut model, which keeps the force along the chord, overestimates it along x and underestimates it along z.

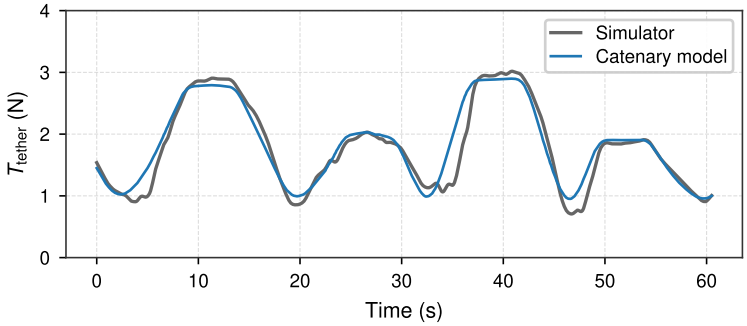
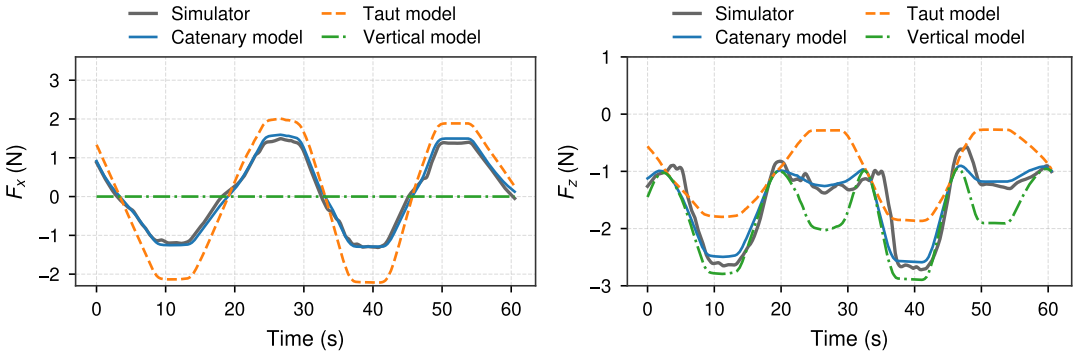


Figure 4.20: Tether tension of the model and of the simulator.



(a) Force along x.

(b) Force along z.

Figure 4.21: Tether force components at the UAV.

Figure 4.22 shows, for each model, the error of the force vector along the flight, combining its components, and Table 4.10 lists the corresponding errors. Since the three models use the same tension, their errors differ only through the direction of the force. The catenary model reduces the error of the force by 72% relative to the taut model and by 76% relative to the vertical model, and its error of $0.25\,\text{N}$ represents 1.2% of the weight of the UAV, which is $20.2\,\text{N}$.

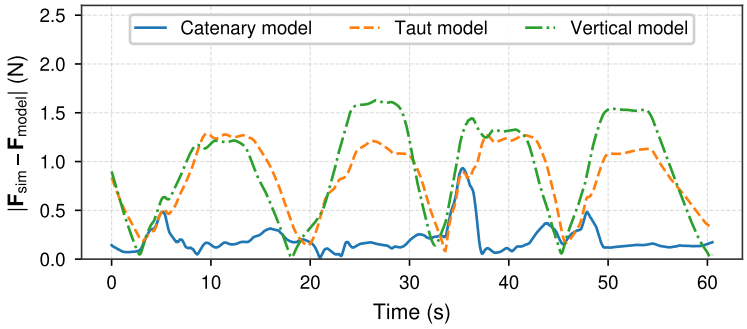


Figure 4.22: Error of the force vector of the tether models.

Table 4.10: Tether force model errors.

Quantity	Unit	RMSE	NRMSE (%)
Catenary model			
Tension T_{tether}	N	0.2031	8.79
Force F_x	N	0.1080	3.85
Force F_z	N	0.2259	10.50
Error of the force vector			
Catenary model	N	0.2505	10.84
Taut model	N	0.8991	38.91
Vertical model	N	1.0332	44.72

4.3 Control Implementation

With the simulation environment of Section 4.1 developed and the models validated against it in Section 4.2, the next step is to implement the controllers of Chapter 3, which use these models to drive both vehicles.

4.3.1 Optimisation Framework

The controllers of Chapter 3 must solve an optimisation problem at every sampling instant, within the time available before the next one. To make this possible, they are implemented with two complementary tools. The problem is first written in CasADi [37], which describes the models, the cost, and the constraints as symbolic expressions and obtains their derivatives automatically. From this description, acados [38] generates a solver in C dedicated to that specific problem, which is compiled once, before operation. During operation, each controller only runs this compiled solver, which is much faster than a general-purpose optimiser.

The prediction model requires integrating the vehicle models over each sampling interval, which is done with the fourth-order Runge–Kutta method [39]. Since the models are non-linear, the resulting problem cannot be solved directly, and acados approaches the solution iteratively, solving at each iteration a simpler, quadratic approximation of the problem [40]. More iterations bring the result closer to the solution, but take more time.

4.3.2 USV Control

Most limits of the controller are not design choices, but the limits of the thrusters of the simulated WAM-V, described in Section 4.1.2, on the thrust, on the azimuth angle, and on how fast each can change. The remaining two come from how the system is meant to operate. The minimum surge speed keeps the USV sailing bow first, while still allowing the slow backing needed to hold a position or to brake, and the maximum separation d_{max} defines the working radius of the UAV around the USV, with a value of 15 m that is raised in the missions requiring a longer reach. These limits are listed in Table 4.11.

The sampling time, which sets how often the controller computes a new command, and the prediction horizon, which sets how far ahead it plans, are both chosen from how fast the USV responds to its

commands. At the operating speeds, the hull takes between half a second and a second to react to a change of force. The controller runs at 10 Hz, so it computes a new command every 0.1 s, five to ten while the USV responds to a single one, and can correct this response as it develops. The horizon must be long enough for the controller to see most of this response, and is set to 2 s, or 20 steps, which covers two to four times the response time. Since the UAV predicts only 1 s ahead, its trajectory covers the first $N_c = 10$ steps of this horizon, over which the tether constraint is applied.

The weights of the cost set what the controller values most, and were tuned in simulation until the USV followed its references smoothly. They form a baseline for all the missions of this work, which could be adapted to each manoeuvre, favouring precision when holding a station or smoothness on a long transit. The position and the heading dominate the cost, since following the reference is the purpose of the controller. The sway velocity is also penalised, so that the USV does not drift sideways, and the thrust and its rate of change receive a light penalty that avoids unnecessary effort without compromising the tracking. The rate of change of the azimuth receives a larger weight, which keeps the steering smooth. The weights are listed in Table 4.12.

Since the solver approaches the solution iteratively, its settings trade the quality of the solution for computation time. To choose them, three settings were compared on a recorded run, on a laptop with an Intel i5-1135G7 processor (4 cores, 2.4 GHz), through their iterations, computation time, and optimality residual, which measures how far the command is from satisfying the conditions that define a solution. The results are shown in Figure 4.23, and summarised in Table 4.13 after the first 2 s, once the solver has stabilised. A strict tolerance gives the smallest residual but needs the most iterations, a single iteration per instant, known as RTI [41], is the fastest but leaves the command furthest from the solution, and a moderate tolerance lies between the two. The controller uses this moderate tolerance, of 10^{-4},

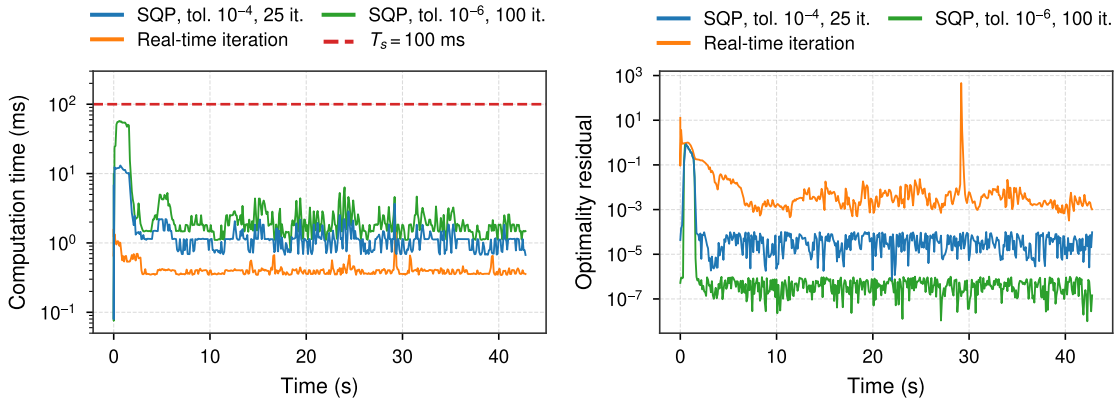
Table 4.11: Limits of the USV NMPC.

Parameter	**Symbol**	**Value**
Thrust limits	$T_{L,max} = T_{R,max}$	510.0 N
	$T_{L,min} = T_{R,min}$	-380.0 N
Azimuth limits	$\alpha_{L,max} = \alpha_{R,max}$	$30°$
	$\alpha_{L,min} = \alpha_{R,min}$	$-30°$
Thrust rate limit	$\dot{T}_{L,max} = \dot{T}_{R,max}$	500 N/s
Azimuth rate limit	$\dot{\alpha}_{L,max} = \dot{\alpha}_{R,max}$	$10°/$s
Minimum surge speed	u_{min}	-0.3 m/s
Surge constraint penalty	$\mu_{u,1},\ \mu_{u,2}$	$10^5,\ 10^5$
Maximum tether separation	d_{max}	15.0 m
Tether constraint penalty	$\mu_1,\ \mu_2$	$500,\ 5000$

Table 4.12: Cost weights of the USV NMPC.

Parameter	**Symbol**	**Value**
Position weight	Q_p	$\mathrm{diag}(150,\ 150)$
Heading weight	q_ψ	150
Velocity weight	Q_ν	$\mathrm{diag}(2,\ 120,\ 1)$
Thruster command weight	Q_τ	$\mathrm{diag}(10^{-5},\ 10^{-5},\ 0.1,\ 0.1)$
Thruster rate weight	R	$\mathrm{diag}(10^{-5},\ 10^{-5},\ 100,\ 100)$

since a stricter one changes the applied thrust by less than $0.1\,\mathrm{N}$ and the azimuth by less than $0.01°$, with a limit of 25 iterations that keeps even the first instants of the run well within the $100\,\mathrm{ms}$ available.



(a) Computation time. (b) Optimality residual.

Figure 4.23: USV NMPC computation time and optimality residual for different solver settings.

Table 4.13: USV NMPC solver statistics for different solver settings.

	Iterations		**Time (ms)**		**Residual**	
Settings	Mean	Max	Mean	Max	Mean	Max
Tolerance 10^{-6}, 100 iterations	5.3	14	2.0	6.3	4×10^{-7}	1×10^{-6}
Tolerance 10^{-4}, 25 iterations	3.0	7	1.1	3.7	4×10^{-5}	1×10^{-4}
Single iteration	1	1	0.4	1.4	1.1	5×10^{2}

The solver may also fail to return a usable command, for instance when the solution of the previous instant is a poor starting point for the current one. In that case, the solver is restarted from the current state of the USV and solves the problem once more. If it fails again, the thrusters keep the last command, and the UAV keeps the last predicted trajectory of the USV, until a new solution is found. Reaching the iteration limit is not treated as a failure, and the command reached is still used. In the recorded run, the solver never failed, and the limit was reached only in the first instants of the run.

4.3.3 UAV Control

Unlike the USV, the UAV is not driven directly by its controller, but through the PX4 autopilot [30]. PX4 can follow commands given at different levels, from positions to the speeds of the rotors, and the controller uses it at the attitude level. At every sampling instant, it sends the commanded attitude and thrust, and PX4 turns them into rotor speeds through its own attitude controller. The NMPC does not need to know how this is done, only how quickly the attitude follows the command, and these dynamics are already included in its prediction model, as described in Section 3.2.4.

The NMPC computes the thrust in newtons, while PX4 expects a throttle δ_T between zero and one, which sets the speed of the rotors. This conversion is usually based on the hover throttle that PX4 estimates in flight, but here that estimate also absorbs the pull of the tether, which the NMPC already accounts for. Since the vehicle is powered through the tether, its supply voltage does not drop during

the flight as that of a battery would, which is one of the main causes of variation of this relation, and any remaining deviation is compensated by the disturbance observer. The conversion therefore uses the theoretical relation of the rotors. The total thrust of the four rotors grows with the square of their speed ω, and the throttle sets this speed linearly between ω_{min} and ω_{max},

$$T = 4\,k_F\,\omega^2, \qquad \delta_T = \frac{\omega - \omega_{min}}{\omega_{max} - \omega_{min}}, \tag{4.11}$$

where k_F, ω_{min}, and ω_{max} are given in Table 4.2. Solving the first relation for the speed and replacing it in the second gives the throttle sent to PX4 for the thrust T computed by the NMPC,

$$\delta_T = \frac{\sqrt{T/(4\,k_F)} - \omega_{min}}{\omega_{max} - \omega_{min}}, \tag{4.12}$$

which is limited to the interval $[0, 1]$.

As for the USV, some limits of the controller come from the vehicle itself. The thrust is kept above a small minimum, so that the rotors do not stop during descents, and its maximum is set slightly below what the rotors produce at full speed, since rotating the vehicle requires some rotors to spin faster than others, and this margin leaves them room to do so. When the margin is not enough, the mixer of PX4, which distributes the commands among the rotors, gives priority to the attitude and reduces the total thrust instead. Representing each rotor in the model would let the NMPC account for this directly, but it would have to command the rotors itself, bypassing the attitude controller of PX4. The rate at which the separation between the UAV and the USV can change is limited to the speed of the winch, described in Section 4.1.5, and the maximum separation is the same as in the USV controller. The remaining bounds, on the altitude, the speed, the tilt, and the rates of the commands, define the operating envelope of the vehicle, and were chosen for the intended operation, within the capabilities of the vehicle. These limits are listed in Table 4.14.

Table 4.14: Limits of the UAV NMPC.

Parameter	**Symbol**	**Value**
Thrust limits	$T_{min},\ T_{max}$	$2.02\,$N, $39.96\,$N
Thrust rate limit	$\dot{T}_{max}$	$100\,$N/s
Tilt limit	ϑ_{max}	$0.5\,$rad
Attitude command rate limits	$\dot{\phi}_{max},\ \dot{\theta}_{max},\ \dot{\psi}_{max}$	$0.8,\ 0.8,\ 1.0\,$rad/s
Horizontal and vertical speed limits	$v_{xy,max},\ v_{z,max}$	$10.0,\ 3.0\,$m/s
Minimum altitude	z_{min}	$2.0\,$m
Penalty, height and vertical speed h_1, h_3	$\mu_{1,j},\ \mu_{2,j}$	$10^3,\ 10^3$
Penalty, horizontal speed h_2	$\mu_{1,2},\ \mu_{2,2}$	$300,\ 300$
Penalty, tilt h_4, h_5	$\mu_{1,j},\ \mu_{2,j}$	$10^4,\ 10^4$
Maximum tether separation	d_{max}	$15.0\,$m
Maximum separation rate	$\dot{d}_{max}$	$2.0\,$m/s
Penalty, tether h_6, h_7	$\mu_{1,j},\ \mu_{2,j}$	$10^5,\ 10^5$

The sampling time and the prediction horizon of the UAV controller follow from how fast the vehicle responds, which is much faster than the USV. Its roll and pitch respond to a new command within about $0.2\,$s, the delay and the time constant of Table 4.7 together. The controller runs at $50\,$Hz, the rate at

which the state estimate of the vehicle is received, so it computes a new command every 0.02 s, about ten while the vehicle responds to a single one, and can correct this response as it develops. The horizon is set to 1 s, or 50 steps, which covers about five times this response and more than twice the slower response of the yaw.

The weights of the cost were also tuned in simulation, until the vehicle followed its references closely with smooth commands, and are likewise a baseline that could favour precision when landing or smoothness when following a target. The position dominates the cost, with a larger weight on the height, since the thrust corrects it directly, whereas a horizontal correction requires the vehicle to tilt first and a lighter weight keeps this tilting gradual. The yaw is tracked as well, independently of the position, since the vehicle can turn without moving. Among the inputs, the rates of the attitude commands are penalised the most, while the thrust may change quickly within its rate limit. The tilt and the deviation of the thrust from the hover value are lightly penalised to discourage values larger than the tracking requires. The weights are listed in Table 4.15.

Table 4.15: Cost weights of the UAV NMPC.

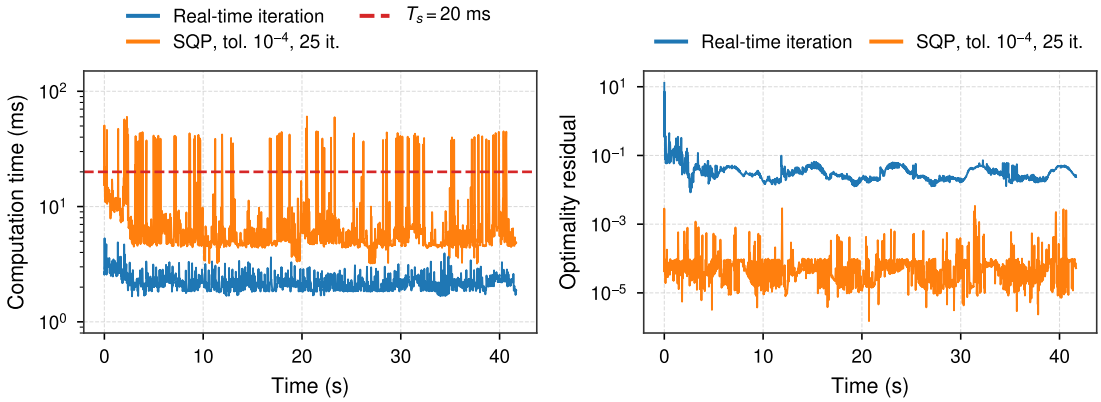
Parameter	**Symbol**	**Value**
Position weight	Q_p	$\mathrm{diag}(60,\, 60,\, 150)$
Velocity weight	Q_v	$\mathrm{diag}(5,\, 5,\, 7.5)$
Tilt weight	Q_{tilt}	$\mathrm{diag}(5,\, 5)$
Yaw weight	q_ψ	20
Thrust weight	Q_T	0.118
Input weight	R	$\mathrm{diag}(30,\, 30,\, 30,\, 0.01)$

The disturbance observer of Section 3.2.4 is set by its gain and its bound. With $\beta = 0.03$, the estimate converges with a time constant of about $T_s/\beta \approx 0.7$ s, slower than the response of the attitude, so that it does not react to the short prediction errors of a manoeuvre, but fast enough to follow the gusts of the wind, which vary over periods of several seconds. The bound of 8 N lies well above the force of the strongest wind tested, of about 2.3 N at 8 m/s.

The same comparison was made for the UAV, close to its limits, between iterating until a tolerance of 10^{-4}, with at most 25 iterations, as in the USV, and the RTI, which relies on the solution changing little between consecutive instants, so that refining the previous one once is enough to follow it. Iterating towards convergence leaves a residual several hundred times smaller, but it exceeded the 20 ms available in about 6% of the instants, most of them ending at the iteration limit without converging, as shown in Figure 4.24 and Table 4.16. The single iteration never took more than 5 ms, and its commands differed from the converged ones by less than 0.7° in roll and pitch and 0.6 N in thrust, making the RTI the strategy used by the controller.

Table 4.16: UAV NMPC solver statistics for different solver settings.

	Iterations		**Time (ms)**		**Residual**	
Settings	Mean	Max	Mean	Max	Mean	Max
Tolerance 10^{-4}, 25 iterations	4.6	25	7.7	60.4	8×10^{-5}	3×10^{-3}
RTI	1	1	2.2	4.7	3×10^{-2}	2×10^{-1}



(a) Computation time.

(b) Optimality residual.

Figure 4.24: UAV NMPC computation time and optimality residual for different solver settings.

If the solver fails to return a usable command, the vehicle keeps the last attitude and thrust commanded until a new solution is found, while the disturbance estimate is held, since there is no prediction to compare with the measured velocity. With a single iteration there is no iteration limit to reach, and in the recorded run the RTI never failed.

4.4 Manoeuvres and Missions

The controllers of Section 4.3 follow the references they are given, but do not decide where each vehicle should go. This is the role of the manoeuvres, which turn a simple motion, such as an orbit around the USV or a landing on its deck, into the reference of each controller at every instant. The missions then chain these manoeuvres, for both vehicles, into a complete operation.

4.4.1 Manoeuvres

A manoeuvre gives the reference of one vehicle as a function of time, sampled over the horizon of the controller at every instant. Five manoeuvres were defined for the UAV, takeoff, formation, orbit, follow and landing, and four for the USV, which holds its position, sails a route of waypoints, follows a target, keeping a safe distance from it, or supports the UAV. Each is set by a few parameters, such as a destination or a speed. The references are simple geometric curves in time, such as lines and circles, since the controllers keep the vehicles within their limits whatever reference they are given, even a single point.

Most manoeuvres of the UAV are given relative to the winch, since the vehicle works around the USV and must return to it. Their references move with the USV along the trajectory predicted by its controller, either in fixed axes or in the axes of the USV, so that they turn with it, as when the UAV keeps ahead of the bow. The height is taken above the height z_t of the winch, so that the UAV does not follow the heave of the deck. The point of the formation and the centre of the orbit can also be fixed in the world, to reach or circle a structure, and the reference of the follow manoeuvre moves with the target.

When the UAV circles a structure, a tether anchored at a fixed point would wrap around it. The USV

then supports the UAV, taking as its reference the point on the water below the UAV, pushed away from the structure to a safe distance, and sailing along with it as it moves around the structure. This is only possible when the reference of the UAV is fixed in the world, since a reference given relative to the winch would move with the USV, and each vehicle would then chase the other.

Each manoeuvre also sets the heading of the UAV, through the point that its controller points it at, described in Section 3.2.4, such as the winch, a structure or a target, so that a camera on the vehicle keeps it in view. To hold a given heading instead, this point is placed ahead of the winch along that heading, which for the takeoff and the landing is the heading of the USV, so that the vehicle is aligned with the deck.

Figure 4.25 shows the manoeuvres of the UAV as performed in simulation, seen from above in fixed axes, with the two vehicles drawn at the same instants, the older ones more transparent, so that their motion can be seen together. The formation (b) keeps the UAV ahead of the bow, in the axes of the USV, while it follows a route, and the orbit (c) circles the USV as it sails, shown in its axes, where the orbit is a circle, while in (d) the UAV orbits the USV on patrol until it detects a target, which it then follows (e) together with the USV. The takeoff and the landing (a, f) are shown instead through the height of the UAV above z_t, which shows the heave of the deck.

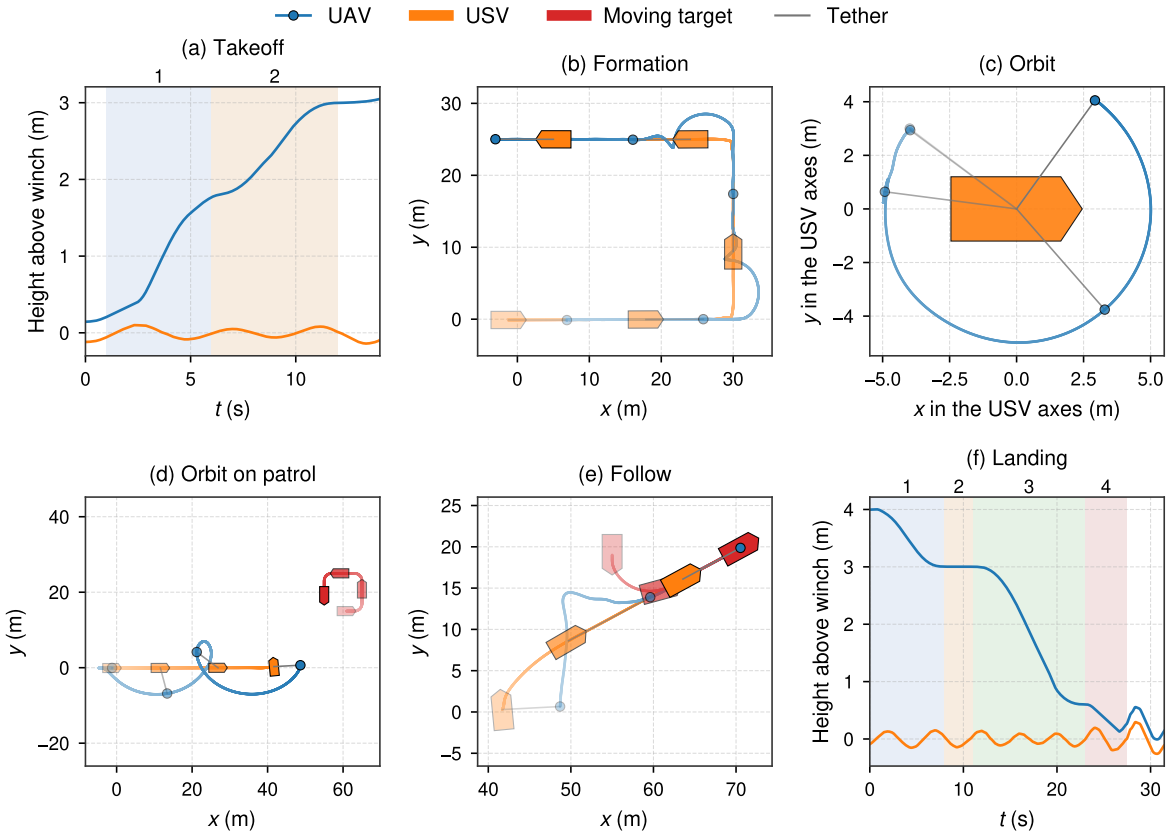
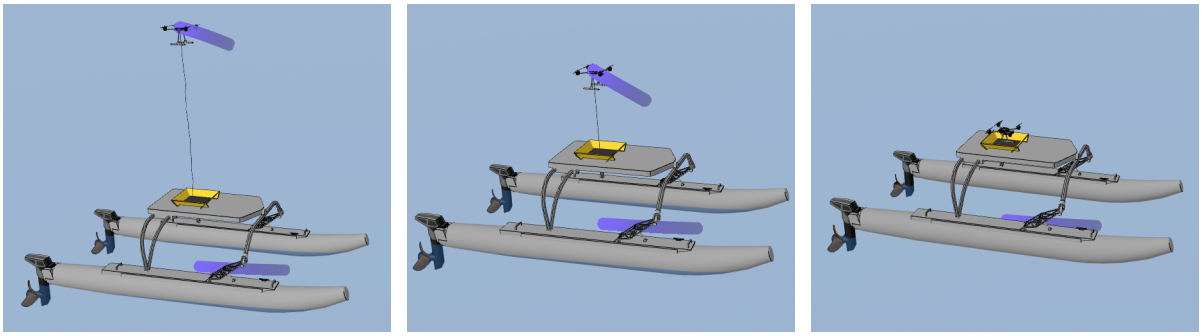


Figure 4.25: Manoeuvres performed in simulation.

The takeoff and the landing are the manoeuvres in which the UAV is closest to the deck and moves with it. Both start or end on a landing pad of $0.40\,\mathrm{m}$ by $0.40\,\mathrm{m}$ at the centre of the deck, bordered by inclined walls and placed directly above the winch. To hold the vehicle against the deck, so that the

waves do not tip it over and throw it into the sea, the winch pulls the tether, as described in Section 4.1.5, from the start of the takeoff until the UAV is clear of the deck. The controller of the UAV does not include this pull in its model, and the vertical component of its disturbance estimate is held constant while the pull acts, so that the estimate does not cancel the pull, which keeps acting on the UAV and holds the tether taut. The takeoff therefore begins with a vertical climb off the pad, phase 1 in Figure 4.25a, and only then moves the vehicle to its first position, phase 2, so that it does not drift over the edge of the deck at low height.

The landing is divided into four phases, shown in Figure 4.25f and Figure 4.26. The UAV first moves above the winch (1) and waits until it is centred above the pad and moving with it (2). If it is not centred within 40 s, the landing on the moving deck is aborted and the USV starts the recovery of the UAV, described in Section 4.4.2. It then descends with the winch pulling it towards the deck (3), and ends with a slow final approach towards a point below the deck (4), so that it touches down even if the deck falls away, and the speed of the contact is set by the heave rather than by the descent. Once it rests on the pad, the vehicle is disarmed, and the winch keeps it against the deck.



(a) Waiting above the pad (2). (b) Descent pulled by the winch (3). (c) Touchdown on the pad (4).

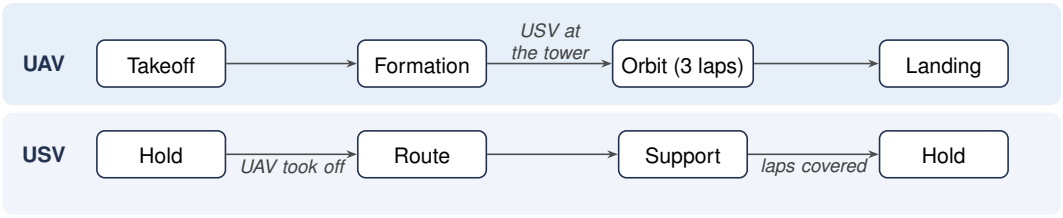
Figure 4.26: Phases of the landing, with the USV sailing at 0.75 m/s.

4.4.2 Missions

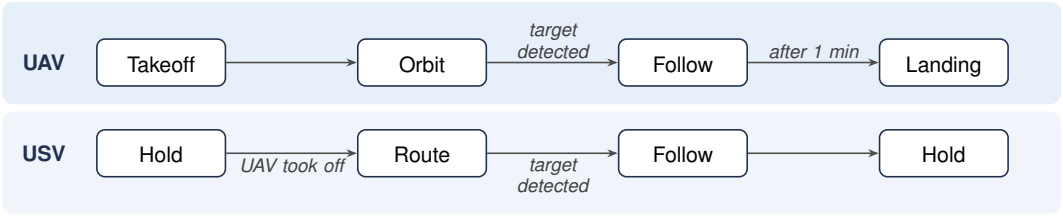
A mission is a file with two sequences of manoeuvres, one for each vehicle. Each manoeuvre ends when a condition is met, either an event, such as the detection of a target, or the end of a manoeuvre of the other vehicle.

A mission is aborted when a manoeuvre fails, when it takes too long, or when the operator requests it. The USV then holds its position and the UAV lands.

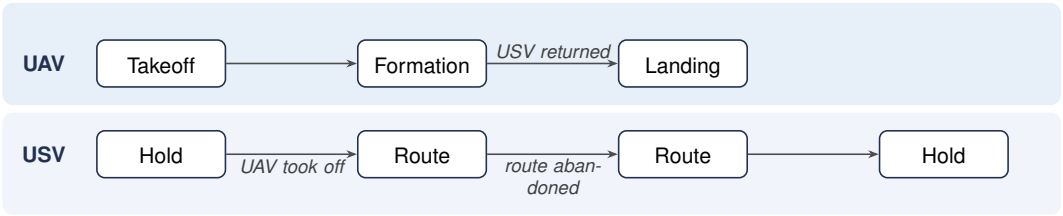
Three missions were tested in simulation, shown in Figure 4.27. In the inspection (a), the UAV circles a tower standing in the water, 43 m high, at three heights while the USV supports it. In the patrol (b), the UAV orbits the USV until it detects a moving target, which both vehicles then follow. The target is a second WAM-V that patrols a forbidden zone at 0.6 m/s and flees at 2 m/s when either vehicle comes within 15 m, and it is detected when it enters the view of a camera on the UAV, pointed downwards and assumed to cover $120°$. In the recovery (c), the UAV is held at a point it cannot leave, while the USV sails away, waits at the reach of the tether and returns to collect it.



(a) Inspection.



(b) Patrol.



(c) Recovery.

Figure 4.27: Missions tested in simulation.

4.5 Software Architecture and Mission Control

The models, controllers and manoeuvres of the previous sections are implemented as independent software components, integrated with the simulator and with the autopilot of the UAV. This section describes the architecture that connects these components and the interface through which an operator executes the missions. The code of the system is publicly available at `https://github.com/tomasgesteves1/tethered-usv-uav-nmpc`.

4.5.1 Architecture

The software is built on ROS 2 [42], in which each component runs as an independent node, and the nodes exchange messages through named topics. The controllers therefore do not depend on whether their data comes from the simulation or from the real vehicles.

Figure 4.28 shows the components and the main data exchanged between them. Gazebo simulates the vehicles and the waves, while PX4 runs as the autopilot of the UAV, in lockstep with the simulation. Each is connected to ROS 2 by a bridge, the ROS–Gazebo bridge [43] and the Micro XRCE-DDS agent [44]. The tether node applies in Gazebo the force of the tether, computed by MoorDyn [10] for the length set by the winch. The two NMPC controllers send their commands to the thrusters of the USV and to the autopilot of the UAV while exchanging their predicted trajectories, and the mission executor produces the references of both. A web server connects ROS 2 to a Mission Control Centre.

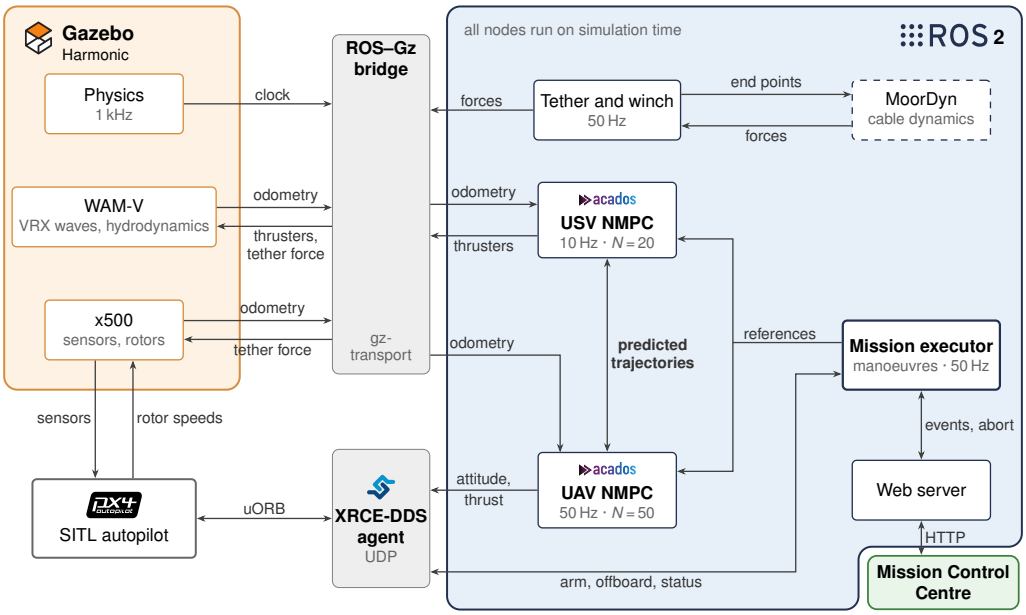


Figure 4.28: Software architecture.

4.5.2 Mission Control Centre

The Mission Control Centre is a web page, shown in Figure 4.29, from which an operator runs the whole system without the command line. Since every mission is launched and recorded in the same way, it also made the tests in simulation repeatable. The bar at the top starts and stops the simulation, the panel on the left lists the missions with their manoeuvres and starts, records or aborts them, the map in the centre shows the two vehicles, the tether and their references from above, and the panels on the right show the state of each vehicle and of the tether and the progress of the mission.



Figure 4.29: Mission Control Centre.

Chapter 5

Results

The previous chapters developed models of the two vehicles and of the tether, and a controller for each vehicle that relies on them. Whether these controllers can operate the tethered system at sea can only be judged by operating it, where the limits of each vehicle, the coupling through the tether and the disturbances of the sea all act at once.

Evaluated all together, these effects would not reveal which of them is responsible for a given behaviour. The evaluation therefore introduces them one at a time, moving gradually closer to a real operation. It begins with each vehicle alone, where only its own dynamics and limits are involved, then exposes the landing on the moving deck, the manoeuvre most sensitive to the sea, to higher waves and to wind that the models do not describe, and finally couples the two vehicles through the tether in complete missions.

5.1 Trajectory Tracking

Evaluated on its own vehicle, each controller faces two distinct questions. The first is how closely it tracks a trajectory that the vehicle can follow, which depends on how well it predicts the motion of the vehicle. The second is how it behaves when the trajectory demands more than the vehicle can deliver, since the controller must then give up following the reference in order to keep the vehicle within its limits.

5.1.1 USV Controller

The USV controller is evaluated with the UAV resting on the deck, so that the USV moves alone and its response depends only on its own controller.

Tracking Accuracy

The reference is a figure-eight sailed twice, as shown in Figure 5.1. Its two circles, of $12\,\mathrm{m}$ radius, are turned in opposite directions and joined by straight segments, so that the USV turns both ways and alternates between turning and sailing straight.

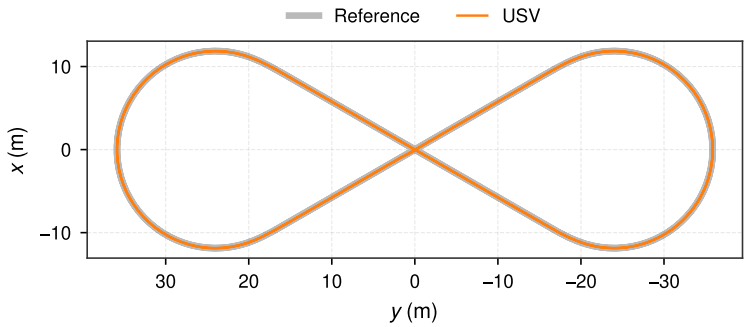
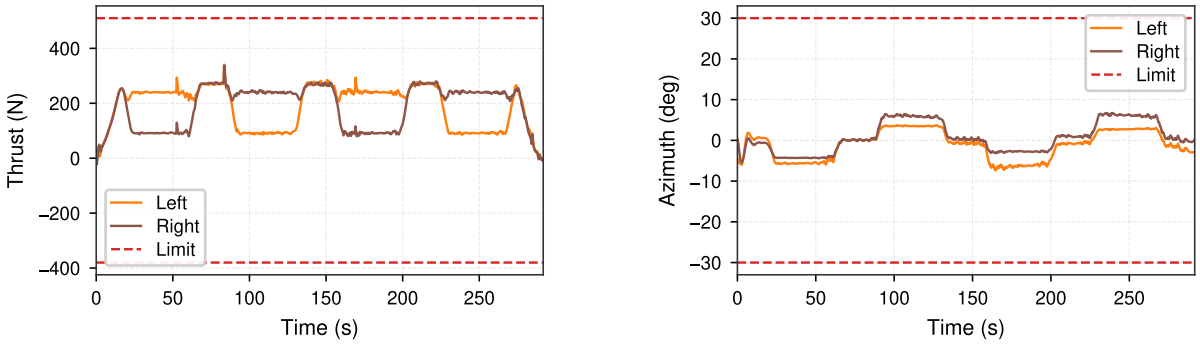


Figure 5.1: Reference figure-eight of the USV and path sailed, seen from above.

The reference lies within the capability of the USV. The thrust and the azimuth computed by the controller remain far from the limits of the thrusters throughout, as shown in Figure 5.2, and neither their rates nor the surge speed approach their own limits. The USV turns mainly through the difference between the thrusts of its two thrusters, which reverses between the two circles, while the azimuth changes by only a few degrees. The tracking error therefore reflects the controller alone, and not the limits of the USV.



(a) Thrust.

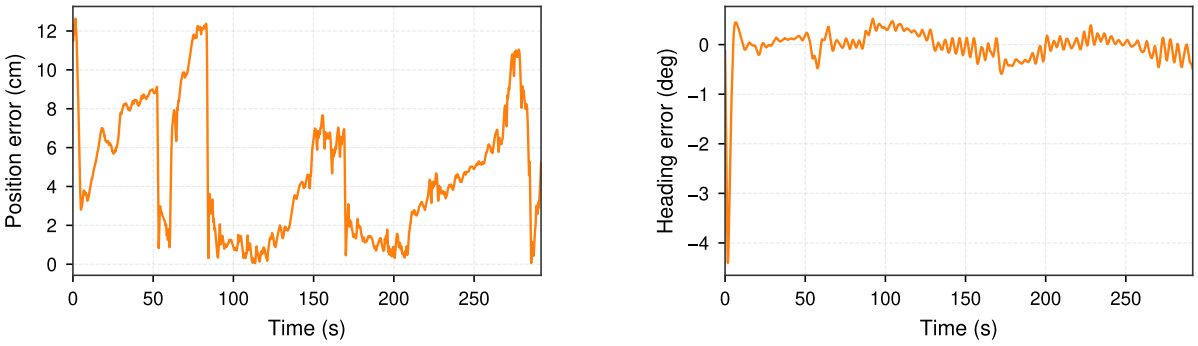
(b) Azimuth.

Figure 5.2: Thrust and azimuth of both thrusters of the USV on the figure-eight, against their limits.

The heading error of the USV reaches about 4° at the start of the run, while the USV accelerates from rest, and decays within the first seconds, as shown in Figure 5.3. Thereafter, the heading error remains below one degree, and the position error is of the order of one percent of the length of the USV, as listed in Table 5.1. At the scale of the manoeuvre, the path sailed is therefore indistinguishable from the reference, with which it coincides in Figure 5.1.

Table 5.1: Tracking error of the USV.

	RMSE	Maximum
Position (cm)	5.5	12.4
Heading (deg)	0.2	0.6



(a) Position error.

(b) Heading error.

Figure 5.3: Position and heading errors of the USV.

Operating Limits

The second reference asks the USV for motions it cannot perform, as shown in Figure 5.4, where the USV is drawn at a few instants and the dashed outlines mark the poses held by the reference. It begins with a straight segment at $3\,\text{m/s}$, beyond the top speed of the USV, which ends in a sharp turn of 90° at that speed. It then comes to rest at a point, which is moved first $15\,\text{m}$ backwards and then $15\,\text{m}$ to the left of the USV, keeping the same heading.

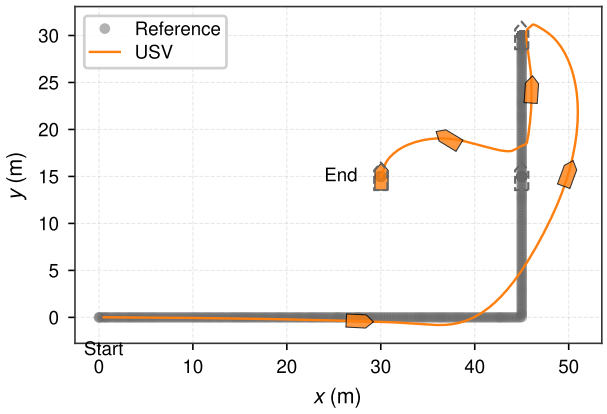


Figure 5.4: Reference beyond the limits of the USV and path sailed, seen from above.

On the straight segment, both thrusters reach their maximum thrust, as shown in Figure 5.5, and the USV settles at its top speed of $2.3\,\text{m/s}$, falling behind the reference. The reference has therefore already turned when the USV approaches the corner, and the NMPC cuts the corner towards it. The USV turns mainly through the difference between the thrusts, with the outer thruster at its maximum and the azimuth below 10°, and, still close to its top speed, crosses the new segment and goes beyond it by about $6\,\text{m}$. To return to the point held by the reference, the NMPC then turns both azimuths towards their limit at the maximum rate and reverses one thruster at its maximum rate, so that the USV slows down and turns at the same time.

When the point moves backwards, the speed is bounded by the soft lower limit on the surge speed, and the USV reverses in a straight line at exactly this limit, as shown in the last panel of Figure 5.5. While the surge speed rests on this limit, the thrust oscillates by a few newtons about the value that holds it

there, without any visible effect on the motion of the USV. To reach the point on its left, the NMPC turns the USV towards it, sails to it and turns back to the heading of the reference.

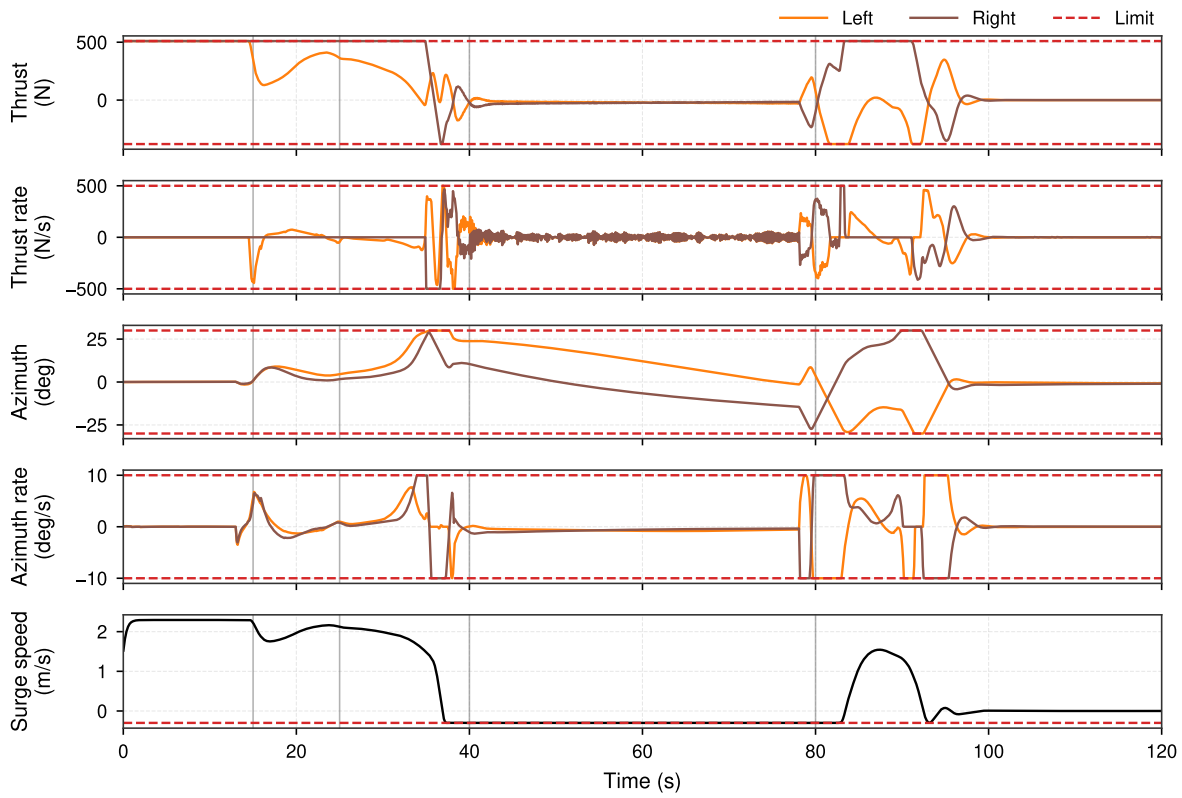


Figure 5.5: Thrust and azimuth of both thrusters of the USV, their rates, and its surge speed, against their limits.

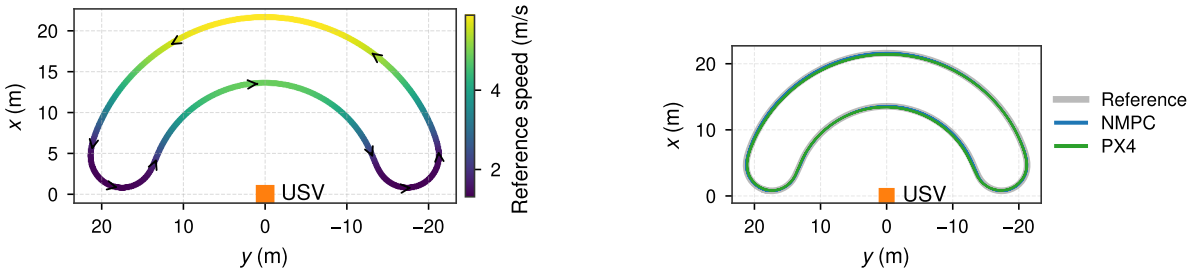
Throughout the test, the limits of the USV are reached but never exceeded, including the soft limit on the surge speed. The controller therefore gives up following the reference only as far as the limits of the USV require, and uses its full capability in doing so.

5.1.2 UAV Controller

The UAV controller is evaluated with the UAV tethered to the USV, which holds its position at the centre of each reference, so that the response of the UAV depends only on its own controller and on the force of the tether.

Tracking Accuracy

The reference is a closed trajectory around the USV, as shown in Figure 5.6a. It consists of two fast arcs, along which the separation between the vehicles is constant, joined by two slow turns, along which this separation varies. The UAV therefore flies at both high and low speed, under a tether force that changes along each lap, since this force grows with the length of tether suspended between the vehicles.



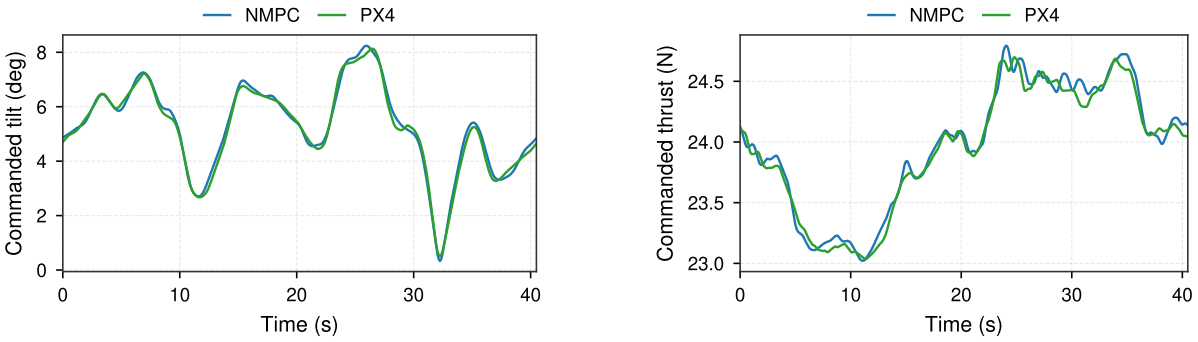
(a) Reference, coloured by its speed.

(b) Paths with the NMPC and with PX4 over one lap.

Figure 5.6: Reference trajectory of the UAV around the USV and paths flown, seen from above.

As a baseline, the same trajectory is tracked by the position controller supplied with PX4. At each instant, this controller receives the current point of the trajectory and acts on the error between this point and the current position of the UAV, whereas the NMPC receives the trajectory over a one-second horizon and predicts the response of the UAV and of the tether. The comparison therefore quantifies the benefit of prediction.

The trajectory keeps both controllers well within the limits of the UAV, as shown in Figure 5.7. The commanded tilt stays below $9°$, against a limit of $28.6°$, and the commanded thrust varies by less than $2\,\mathrm{N}$ about its value in hover. Both controllers command nearly the same tilt and thrust along the lap, since both must produce the accelerations that the trajectory requires. The difference between the two results therefore reflects only the tracking performance of each controller, and not a larger control effort or the limits of the UAV.



(a) Commanded tilt.

(b) Commanded thrust.

Figure 5.7: Commands sent to the autopilot by the NMPC and by PX4 over one lap.

Over a full lap, the paths of both controllers are nearly indistinguishable from the reference, as shown in Figure 5.6b. Their errors are of the order of centimetres, and over three laps the error of the NMPC is more than four times smaller than that of PX4, both on average and at its maximum, as listed in Table 5.2.

Table 5.2: Position error of the UAV (cm).

Controller	RMSE	Maximum	Horizontal	Vertical
NMPC	6.5	12.8	6.4	1.0
PX4	29.9	53.8	29.9	1.8

The error of PX4 is not uniform along the lap. It increases on each fast arc and decreases on the slow turns, as shown in Figure 5.8, and lies mostly along the path. On the fast arcs, PX4 trails the reference by about 80 ms, and a constant lag produces an error proportional to the speed. The NMPC, which receives the trajectory ahead and predicts the response of the UAV, shows no such lag.

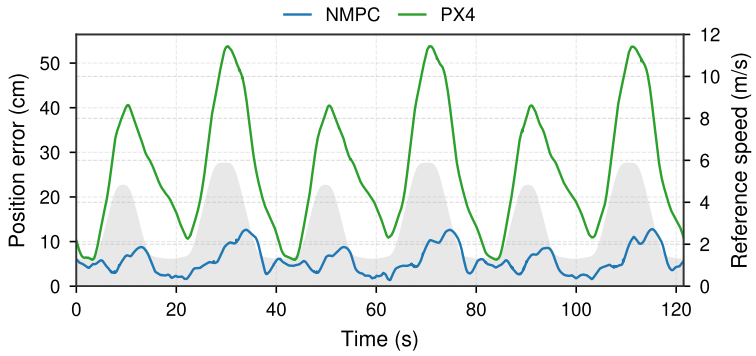


Figure 5.8: Position error of the UAV, with the speed of the reference shaded.

Operating Limits

As for the USV, the second reference asks the UAV for motions it cannot perform, with each manoeuvre designed to exceed a different limit, as shown in Figure 5.9. The first arc, of 15 m radius at 10.5 m/s, requires a tilt of about $37°$, beyond the limit of $28.6°$, and the second requires a horizontal speed of 11 m/s, beyond the limit of 10 m/s. After moving away from the USV and down, the reference performs two steps in height, the first up to 10 m and the second down to the surface of the water, below the minimum height of 2 m.

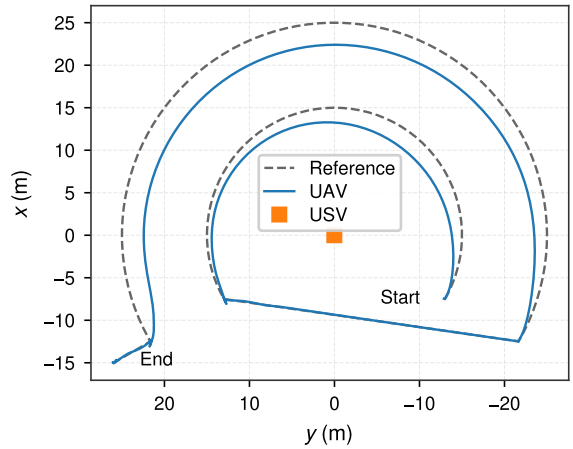


Figure 5.9: Reference beyond the limits of the UAV and path flown, seen from above.

On each arc, the NMPC drives the UAV to the limits that make the reference unattainable and holds it there until the end of the arc, as shown in Figure 5.10. On the first arc, the tilt reaches its limit at the maximum rate of the commanded roll and pitch, and the horizontal speed soon reaches its own limit as well. The tether, attached to the USV at the centre of the arc, pulls the UAV inwards and adds to the horizontal force of the rotors. The centripetal acceleration therefore reaches about 7 m/s^2, against the

$5.4\,\mathrm{m/s^2}$ that the rotors provide with the tilt on its limit, so that the UAV holds $10\,\mathrm{m/s}$ on this arc. On the second arc, the tilt reaches its limit only while the UAV accelerates, and the horizontal speed then settles on its limit with the tilt well below its own. In both cases the horizontal speed stays on its limit without oscillation, so that each arc is flown as fast as the limits allow.

The UAV therefore falls behind the reference, by up to about $5\,\mathrm{m}$ at the end of the second arc. Since the reference lies ahead along the arc, the NMPC steers the UAV towards it across the inside of the arc, so that the UAV flies inside the reference, as shown in Figure 5.9. When the reference comes to rest, the UAV recovers this lag within about two seconds.

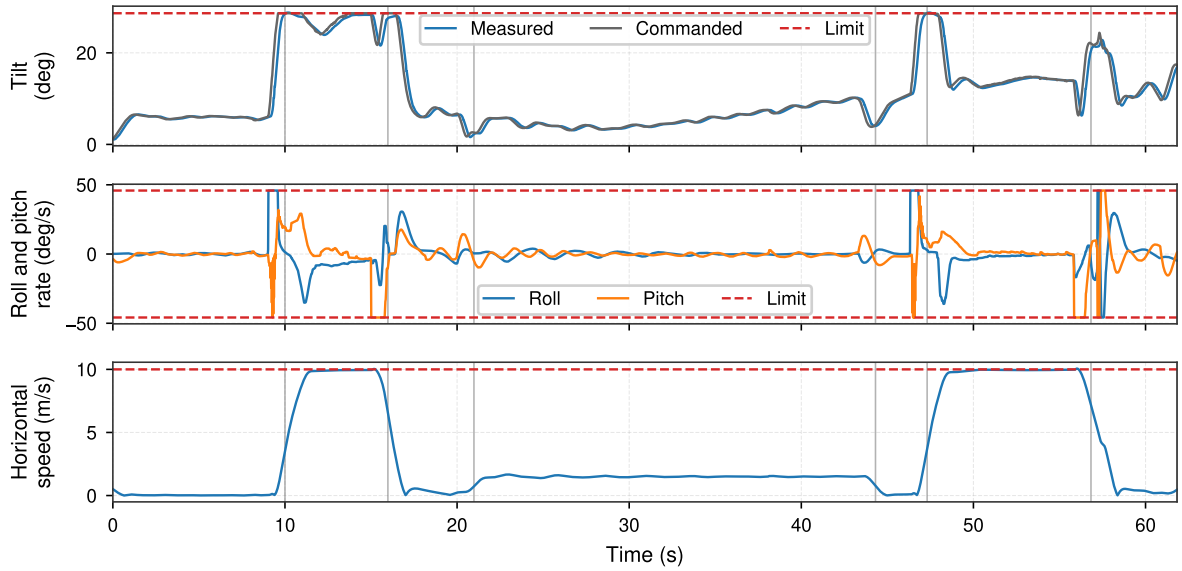


Figure 5.10: Tilt, its rate and horizontal speed of the UAV against their limits.

On the steps in height, the UAV climbs and descends at the maximum vertical speed of $3\,\mathrm{m/s}$, as shown in Figure 5.11, which it reaches through a rapid change of the thrust, with the thrust and its rate within their limits. During the descent, the minimum height enters the prediction horizon before the UAV reaches it, and the NMPC begins to decelerate at about $3.3\,\mathrm{m}$, so that the UAV comes to rest at $2\,\mathrm{m}$ even though the reference lies at the surface of the water. Since stopping from $3\,\mathrm{m/s}$ takes about $1.3\,\mathrm{m}$, a controller that reacts to this limit only upon reaching it would cross it.

Throughout the test, the limits on the inputs, which are hard constraints, are reached but never exceeded. The limits on the states targeted by the reference are soft constraints, which the NMPC may exceed when no feasible solution exists, and are exceeded by less than $1\,\%$ and only in brief transients. As for the USV, the controller therefore gives up following the reference only as far as the limits of the UAV require, and uses its full capability in doing so.

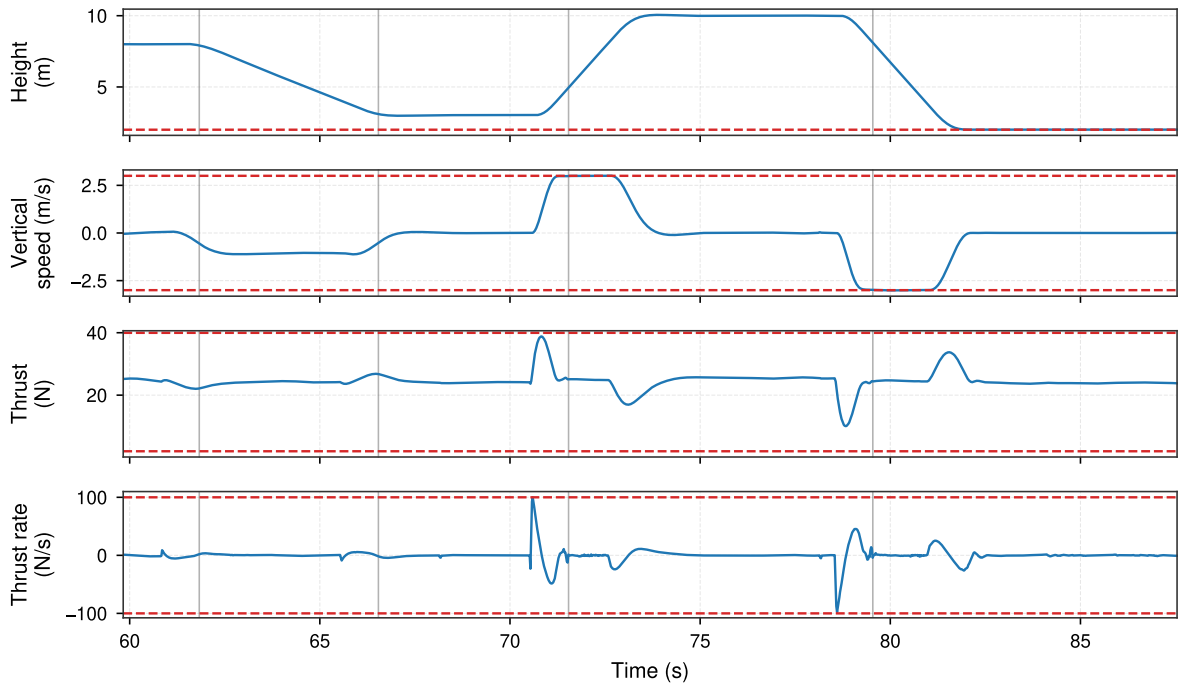


Figure 5.11: Height, vertical speed, thrust and thrust rate of the UAV against their limits.

5.2 Landing on the Moving Deck

With the tracking of each controller established, the evaluation turns to the landing on the deck of the moving USV, the manoeuvre that closes every mission. Unlike the other manoeuvres, which reduce to following a reference, the landing brings the UAV into contact with a deck that moves with the waves, and it is therefore the manoeuvre that depends most on the sea.

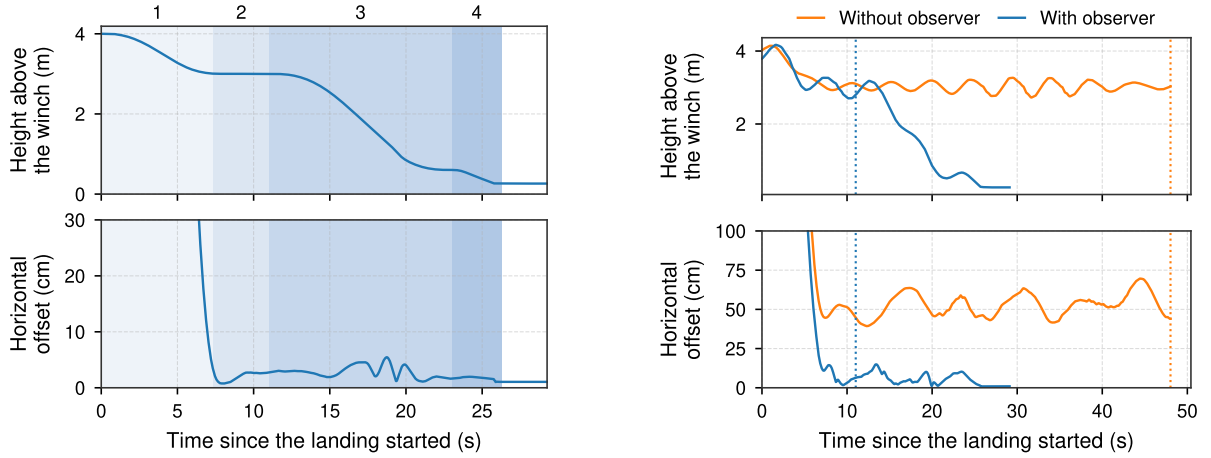
In each run, the UAV takes off while the USV starts a route at 0.75 m/s, moves away from it and then lands back on the pad, in the phases described in Section 4.4.1. A landing is judged by four criteria. The UAV must be centred above the pad when the descent starts, touch down close to the centre of the pad, which is 0.40 m wide, meet the deck at a low speed, and complete the landing without delay.

Without waves and wind, each phase of the landing completes as intended, as shown in Figure 5.12a. The UAV moves above the winch in about 7 s and, once centred within 3 cm of it, descends with the winch pulling it towards the deck. The final approach then brings it into contact with the pad at 0.15 m/s, less than 1 cm from its centre, 26 s after the landing started.

The landing is then repeated in two seas, with significant wave heights of 0.6 and 1.2 m. Since the descent is referenced to the height z_t of the winch, and not to the heaving deck, the UAV descends as it does without waves, and is centred, touches down within 4 cm of the centre of the pad and completes the landing in the same time, as listed in Table 5.3. What the waves change is the contact. Since the final approach descends slowly towards a point below the deck, the speed of the contact is set by the motion of the deck, and grows from 0.15 m/s without waves to 0.39 m/s in the higher sea, still below the default landing speed of PX4, of 0.7 m/s [45].

The wind acts on the UAV as a horizontal drag force along a fixed direction, with gusts of $\pm 30\,\%$, and is applied with the waves of 0.6 m at 5 and 8 m/s, the stronger producing a mean force of 2.3 N, about

a ninth of the weight of the UAV. Without the disturbance observer of Section 3.2.4, the wind holds the UAV away from the pad, and it is never centred above it, as shown in Figure 5.12b for the stronger wind. When the wait for the centring times out, the landing on the moving deck is therefore aborted, and the USV starts the recovery of the UAV.



(a) Without waves and without wind, with the four phases shaded.

(b) Wind of 8 m/s, with and without the disturbance observer.

Figure 5.12: Landing on the moving deck.

With the observer, the UAV is centred soon after it arrives, and the landing takes the same time as without wind, as listed in Table 5.3. In both winds, it touches down within 2 cm of the centre of the pad.

Table 5.3: Landing under waves and wind.

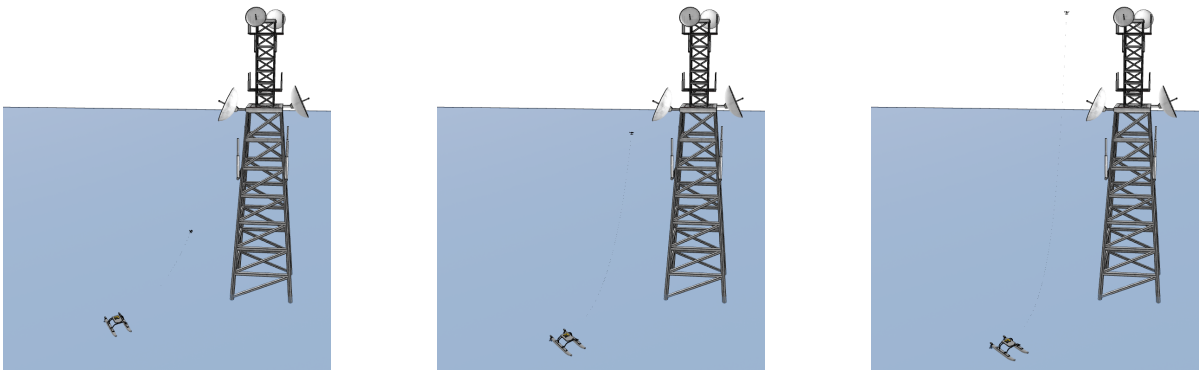
	Centring (cm)	**Touchdown** (cm)	**Contact** (m/s)	**Duration** (s)
Without waves	2.7	0	0.15	26.3
Waves of 0.6 m	3.8	2	0.26	25.2
Waves of 1.2 m	1.3	4	0.39	26.9
Wind, without / with the disturbance observer				
5 m/s	18.3 / 4.3	– / 1	– / 0.26	– / 27.7
8 m/s	44.1 / 6.4	– / 2	– / 0.25	– / 26.1

5.3 Missions

With the tracking of each controller and the landing on the moving deck established, the evaluation proceeds to the system as a whole. In the missions defined in Section 4.4.2, both controllers operate simultaneously, coupled by the tether, and each executes the sequence of manoeuvres assigned to its vehicle. Three missions are evaluated, each forming a complete operation, from the takeoff of the UAV to its landing on the deck.

5.3.1 Inspection

The inspection represents the survey of a structure standing in the water, here the tower. The UAV circles the tower at $13\,\text{m}$ from its centre, in three laps at heights of 12, 27 and $42\,\text{m}$, with its heading pointed at the tower, so that a camera on board covers it from bottom to top, as shown in Figure 5.13. To reach the top of the tower, the maximum separation is raised to $d_{max} = 45\,\text{m}$.

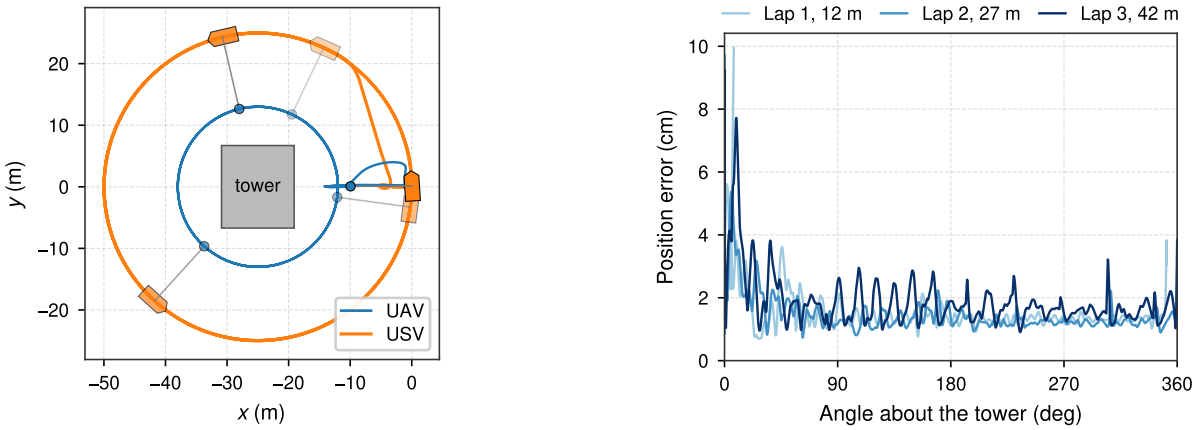


(a) First lap, $12\,\text{m}$. (b) Second lap, $27\,\text{m}$. (c) Third lap, $42\,\text{m}$.

Figure 5.13: Inspection of the tower on each lap.

The USV supports the UAV from the water, following it around each lap at the same angle about the tower and $12\,\text{m}$ further out, as shown in Figure 5.14a. The tether therefore runs outwards from the tower on every lap, without crossing it, and the separation between the vehicles depends only on the height of the lap, remaining within the reach of the tether even on the highest one.

With the longer tether, the force on the UAV grows with the height of the lap, and its tension more than doubles from the lowest to the highest one. The position error of the UAV nevertheless remains of about $2\,\text{cm}$ on all three laps, as shown in Figure 5.14b, so that the two vehicles survey the tower from bottom to top at a constant distance, which neither could achieve alone. After the last lap, the UAV returns to the USV and lands on the centre of the pad.

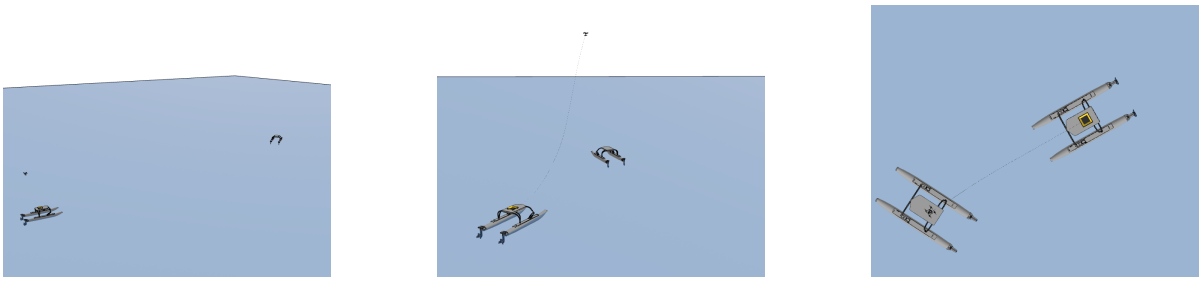


(a) Paths of both vehicles, seen from above. (b) Position error of the UAV along each lap.

Figure 5.14: Inspection of the tower.

5.3.2 Patrol

The patrol represents the surveillance of an area, in which the UAV acts as a lookout for the USV. The USV sails a route near the forbidden zone while the UAV orbits it, $15\,\text{m}$ above the winch. Once the UAV detects the target, both vehicles pursue it for one minute. The maximum separation is $d_{max} = 20\,\text{m}$. Figure 5.15 shows the system at three moments of the pursuit.

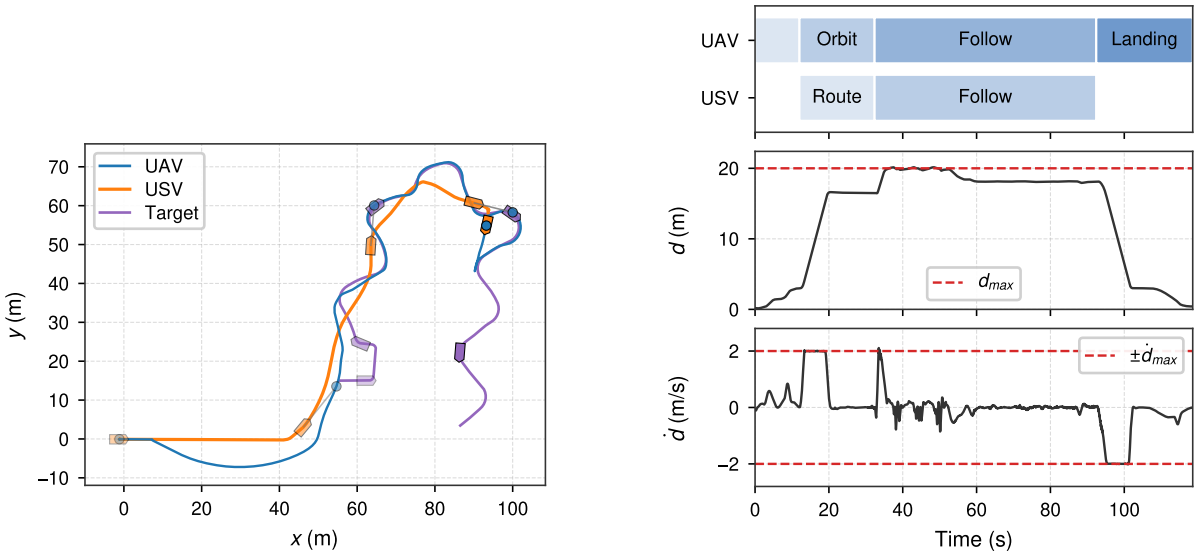


(a) Detection of the target. (b) UAV at the limit of the tether. (c) UAV above the target.

Figure 5.15: Patrol at three moments of the pursuit.

The target is detected at the edge of the view of the camera, beyond the reach of the tether, as shown in Figure 5.16a. The UAV flies towards it until the separation reaches d_{max}, and then remains at the limit of the tether while the USV closes the distance. Once the target is within reach, the UAV keeps above it until the end of the pursuit, so that the target never leaves the view of the camera.

Throughout the mission, the UAV keeps within both limits of the tether, as shown in Figure 5.16b. Besides the separation at d_{max}, its rate of change reaches the speed of the winch, $\pm\dot{d}_{max}$, as the UAV climbs to its orbit, sets off after the target and descends to land, and the two exceed these soft limits by at most $0.1\,\%$ and $6\,\%$, respectively. The UAV then lands $1\,\text{cm}$ from the centre of the pad.



(a) Paths of both vehicles and of the target. (b) Manoeuvres, separation d and its rate of change $\dot{d}$.

Figure 5.16: Patrol mission.

5.3.3 Recovery

The recovery represents the rescue of a UAV that cannot follow the USV, for instance against a wind it cannot overcome. The UAV is held at a point fixed in the world, without giving way to the tether, as it could not against such a wind, while the USV is sent along a route of $60\,\mathrm{m}$, about three times the maximum separation of $d_{max} = 20.5\,\mathrm{m}$. After a wait, the USV abandons the route and returns below the UAV, which then lands.

The USV sets off at its top speed and brakes as the separation predicted over its horizon approaches d_{max}, and then waits at the reach of the tether, as shown in Figure 5.17. While braking, the separation briefly exceeds d_{max} by less than $0.5\,\mathrm{m}$, since the USV sees the UAV only over the horizon of the UAV, which is shorter than the time the USV needs to stop from its top speed.

The UAV holds its point throughout the wait. Once the wait is over, the USV turns back and sails below the UAV, which lands $3\,\mathrm{cm}$ from the centre of the pad. The mission thus shows that, when the UAV cannot follow, the USV moves no further than the tether allows, and can return to rescue it.

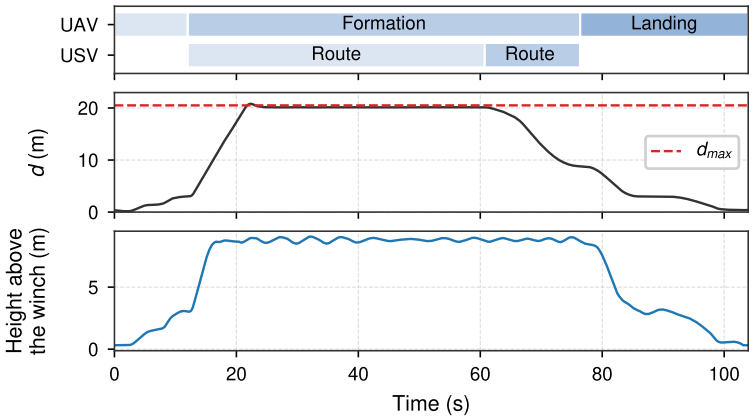


Figure 5.17: Manoeuvres, separation d and height of the UAV in the recovery mission.

Chapter 6

Conclusions

6.1 Achievements

This work set out to control a USV and a UAV connected by a tether, so that they carry out maritime missions autonomously as a single system, each within its own limits and both within the reach of the tether, also under waves and wind. It pursued the four objectives of Section 1.2: models of both vehicles and of the tether, a controller for each vehicle that anticipates the motion of the other, the manoeuvres and missions that put the two vehicles to work together, and their evaluation in a simulation of the tethered system at sea. Each is reviewed in turn.

The models were derived from the physics of each vehicle and evaluated against the simulator with its nominal parameters, without fitting them to the validation data. Over the horizons of the controllers, the model of the WAM-V predicts its motion with errors below $1\,\%$ of the range of each state, and that of the x500 with errors below $2.5\,\%$, which for its position is 11 cm over the up to 8 m it travels in one second. For the tether, a closed-form model of its force, which accounts for its weight through the deflection of the force from the chord, reduces the error of the force by more than $70\,\%$ relative to a taut tether or a hanging load, to $1.2\,\%$ of the weight of the UAV. Unlike the massless and taut tethers or the lumped disturbances common in the literature, the model thus describes the force of a slack tether at little computational cost, and a disturbance observer compensates what it leaves out.

Each vehicle is driven by its own NMPC, which respects the limits of the vehicle and plans its motion over a horizon of 2 s for the USV and 1 s for the UAV. The two controllers exchange the trajectories they predict, so that both keep the separation within the reach of the tether, and the UAV also keeps the rate at which it changes within the speed of the winch, taking into account where the other vehicle is going rather than where it is. Both solve their problem well within their sampling periods, in 4.8 ms of the 20 ms of the UAV and 1.9 ms of the 100 ms of the USV on average, so the cooperation does not require a single, centralised problem.

On a reference both vehicles can follow, the NMPC of the UAV tracks a circle with a varying tether force with an error of 4.0 cm, less than half of that of the position controller of PX4 on the same circle, since it anticipates the motion instead of reacting to the error. More importantly, the controllers do not

need feasible references. Given single points that jump from one place to another, or that lie out of reach of the tether, they plan a motion within the limits of the vehicles and of the tether, with the separation never beyond 14.5 m of its 15 m, the commands at their limits without exceeding them, and the tilt of the UAV at most $1.7°$ above its limit. The manoeuvres can therefore remain simple, and give where each vehicle should be rather than how to get there.

The three missions show the two vehicles working as one system. In the inspection, the UAV surveys a tower from bottom to top at a constant distance, with a tether of 45 m that the USV carries around the tower from the water, which neither vehicle could do alone. In the patrol, the UAV keeps a fleeing target in view of its camera, waiting at the limit of the tether while the USV closes the distance, and within the limits of the separation and of its rate of change. In the recovery, when the UAV cannot follow, the USV moves no further than the tether allows and returns to rescue it. In all three, the UAV lands back on the moving deck within 3 cm of the centre of the pad.

The landing on the moving deck holds under waves and wind. In seas with significant wave heights of up to 1.2 m, the UAV lands within 4 cm of the centre of the pad, and only the speed of the contact grows with the waves, up to 0.39 m/s. In winds of up to 8 m/s, the disturbance observer keeps the UAV centred above the pad, and it lands within 2 cm of its centre in the same time as without wind, while without it the UAV is never centred and the landing takes more than twice as long.

Together, these results show that two cooperating NMPC controllers, with models of both vehicles and of the tether, can operate a tethered USV and UAV as a single system at sea, in simulation. The tether, instead of being only a constraint to be avoided, is used through the cooperation of the two vehicles to extend what each can do alone, as the inspection of the tower shows. The planning carried out by the controllers keeps the missions simple, and the disturbance observer keeps the landing accurate under wind.

6.2 Future Work

The system was evaluated in simulation only, with the state of both vehicles taken from the simulator. The next step is to test it with the real vehicles, where the state is estimated from noisy sensors, the communication between the vehicles has delays, and the tether is stiffer than the one simulated, so that its tension changes faster with the distance. The models were validated against the simulator, and would first need to be identified again against the real vehicles, with the same procedure.

The takeoff in strong wind is the moment most exposed to disturbances, since the UAV can be tipped against the pad while the winch still holds it, before the controller has any effect. Releasing the pull of the winch progressively, turning the USV so that the UAV faces the wind, or starting the estimate of the disturbance observer before the UAV leaves the deck could make it more robust. In the landing, a prediction of the motion of the deck could time the touchdown to when the deck is moving down, and reduce the speed of the contact, which grows with the waves.

The missions are fixed sequences of manoeuvres, in which every manoeuvre ends with an event or with the end of a manoeuvre of the other vehicle. Real operations also need missions that branch, for

instance to return to the search when the target is lost, or to change the route when the sea is too rough to land. A reactive mission layer, such as a behaviour tree, could choose the next manoeuvre from the state of both vehicles, and a higher-level interface could let an operator describe the mission in more general terms, leaving its decomposition into manoeuvres to the system.

Since the controllers plan the motion within all their limits from simple references, the manoeuvres could be replaced by references planned over a longer horizon, for instance around obstacles, without changing the controllers. The controllers do not avoid obstacles themselves. Obstacles described by simple shapes could be added to both NMPCs as further constraints, on the positions of the vehicles and on points along the tether, which would let the UAV inspect the far side of a structure such as the tower, which the tether would otherwise cross.

In the cooperation, it is the UAV that gives way when the reference leads beyond the reach of the tether, since its trajectory already respects the separation and the constraint of the USV never becomes active. Sharing this between the vehicles, so that the USV also moves to give the UAV its reach, would let the cooperation emerge from the controllers rather than from a manoeuvre such as the support of the inspection.

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